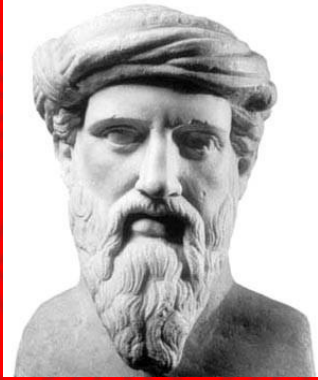
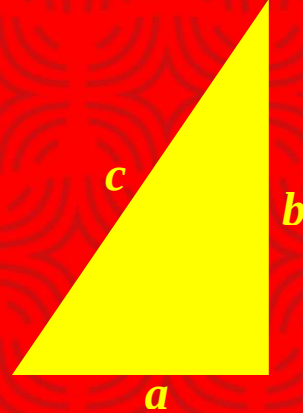




Pythagoras almost 2500 years ago formulated the mathematical theorem that bears his name. *



The Theorem of Pythagoras

$$a^2 + b^2 = c^2$$

***A Mathematical Contribution in the
21st Century: Factorization of $a^2 + b^2$,
in Real Numbers.***

Bienvenido Hernaldo Acevedo González

**A Mathematical Contribution in the 21st
Century: Factorization of $a^2 + b^2$, in
Real Numbers.**

First Edition

By: Bienvenido Hernaldo Acevedo González.

First Original Edition in Spanish, consisting of 1000 copies.

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Dedicated to

JEHOVAH

The Creator of the Universe, source of life
and of the Mathematical Laws of the Cosmos.

JESUS CHRIST

The King of the next earthly paradise.

My Mother: Ms. Paulita

My Family

My Students

Friends: Fausto, Gerald, Antonio, Arturo, Selmira,

Mar Ekaterina y Kenia Melissa.

To the directors, and all the personnel who work in the
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Valle,

Nurse Renel Moses Levy, and all those who collaborate with him in the
Intensive Care Unit (ICU).

Thanks to:

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and for the miracle of sparing my life, which made this book possible.

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shed His blood for us. The King who said: “[...] you will be with me in
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To all those people who made the publication of this book possible.

**A Mathematical Contribution in the 21st
Century: Factorization of $a^2 + b^2$, in
Real Numbers.**

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Why was this book written?

This book is about factoring a sum of squares in real numbers. This sum refers to the Pythagorean Theorem *: $a^2 + b^2 = c^2$. Here, a , b , and c are natural numbers and are the measures of the sides of a rectangle triangle. It is called a rectangle triangle, because two sides form a straight angle to each other (a 90-degree opening). The side c opposite the straight angle is called the hypotenuse. * The image of Pythagoras on the cover was obtained from (www.astronomia.com, 2021).

The formula is true for every rectangle triangle and all three numbers; a , b and c , natural numbers that belong to the set of real numbers, fulfill the formula and are called Pythagorean triple. The Hindus, the Chinese, the Babylonians and the Egyptians in many activities already used the relationship between these three numbers, perhaps a thousand years before Pythagoras.

The point is that, up to the present, $a^2 + b^2 = c^2$, it is said to be irreducible, that is, it cannot be factored in real numbers. Here a , b and c are integers with a less than b , c being the longest side.

This book was written to present a contribution in which $a^2 + b^2 = c^2$ is factored as the product of two distinct rational factors with coefficients on the set of real numbers. The contribution consists of two formulas: "**bienvé₁**" and "**bienvé₂**", in which the rational factors f_1 y f_2 of "**bienvé₁**" differ between an addition and an addition with a rational coefficient, while in "**bienvé₂**" its two rational factors f_3 y f_4 contain an addition and a subtraction.

Bienvenido Hernaldo Acevedo González

Managua, Nicaragua, Centroamérica.

agosto 2021

Summary of Contents

The book consists of eight chapters. We describe the central idea of each chapter below:

In Chapter I: Deals with the origin of mathematical laws, their definition, their characteristics, their sources and their importance.

Chapter II: Some brief accounts of the life and work of Pythagoras and his influence to this day are presented.

Chapter III: Refers to the topic of the Pythagorean Theorem, a geometric and a numerical proof.

Chapter IV: Deals with concepts and terms in basic geometry, common geometric figures, and number sets.

Chapter V: The Topic of Notable Products and Factorization is presented; different sources show notable products factored and in which the sum of two squares does not appear.

Chapter VI: Relates what has been done so far to factor $a^2 + b^2 = c^2$.

Chapter VII: The factorization formula "**bienvé₁**" is presented in which the factors f_1 and f_2 contain each a sum of terms; one of them has a rational coefficient.

Chapter VIII: The "**bienvé₂**" factoring formula is presented here, the rational factors f_3 and f_4 ; contain an addition and a difference between their terms. In both formulas, c^2 is the product of two distinct rational factors, which *belong to the set of real numbers*. Only the four basic operations of arithmetic are involved in these rational factors.

Chapter 1: "Something that is said about Mathematics."

1.1 Mathematical laws govern the material and immaterial that cannot be seen.

In my opinion: since man was created, he received the assignment to reproduce, take care of the environment, and name the animals and objects around him. In addition, I think that man was endowed with language, free will, reasoning ability, use of common sense, and that he was designed with a mathematical neural program installed in his brain that allows him to see patterns of all kinds in his surroundings. These later are capable of being reproduced by formulas or mathematical laws. (TNMSE, 1987). Man has an immense neural network in his brain, which makes it act as a biological supercomputer sustained by a biological-electro-chemical energy, with an expandable memory, each time it stores new information.

This is where the mysterious human mind is housed, with its neuro-mathematical program, which allows it, after seeing and observing the structures and patterns of its environment: to think, reason, analyze, compare, use common sense, extract, process, and discover information, which the human mind later formulates and makes known through mathematical laws pre-established by the Creator of the Cosmos.

Regarding the classification of things, Nicomachus (60-120 AD) is mentioned and it is said that, "Nicomachus wrote a treatise on arithmetic [...]", the content of Book I, Chapter I. *Definition of wisdom and "classification of things in material and immaterial"*. » (Jiménez, 2016). Why do I mention and refer to this Nicomachean mathematician? Well, I do it to emphasize that, I am neither the first nor the last to think that there is a classification of things in, material and immaterial, that is to say that, in the "all", there are two universes.

That the totality of the Cosmos (Order) seen as a "whole", *has two components:*

- a) An immaterial “spiritual” universe impenetrable, inscrutable, unintelligible and un-researchable for the human mind and
- b) A material universe, penetrable, scrutable, understandable and researchable up to a certain limit. This is the macro universe, that of galaxies and the micro, the atomic, subatomic and quantum.

For living in the era of computing, artificial intelligence, information, knowledge, discoveries, the world of quantum physics and the advances in science and technology, I think that the Universe is made up of these two components: an immaterial world of unimaginable entities and the material world of which we are part.

I think that different mathematical laws govern these two components, and that the material has arisen from the immaterial that cannot be seen and that man only observes, analyzes, discovers and gives existence only to the mathematical laws of the material world, to as far as what your mind allows. (DHH, 2021)

The Creator of the Material and Immaterial Universe, established laws and patterns that can be expressed through equations and formulas for each of them, but that human beings only have access to the material world and up to certain limits.

1.2 Some mathematicians share opinions about mathematical laws.

Regarding the issue of the origin of mathematical laws, it is stated that, «geometry, as a sovereign science, is the key to understanding the entire universe. Moreover, that the great Galileo Galilei (1564-1642), having a great cult for this science, wrote, “This great book that we continually have open before our eyes (I speak of the universe) [...] cannot be understood if one does not first learn to understand the language, and to know the characters in which it is written. It is written in mathematical language and the characters are triangles, circles and other geometric figures ...,” When Plato said that “God himself geometrizes”, [...] he wanted to affirm that the universe is constituted according to geometric forms and laws. » (Radice, 1976).

It is also said that, «Until today, Mathematics is debated in the Aristotle-Plato polarity on its origin and nature. According to Aristotle, they are the creation of the human mind. According to Plato, they have their own nature, independent of the human being. Platonism connects with the tradition of the Egyptians. (Garcia Cuerva, 2009).

Regarding the laws of the universe, he states:

« [...] the universe is not arbitrary, but it is governed by certain well-defined laws, [...], we are rational beings, free to observe the universe as we please and to extract logical deductions from what we see. [...] In such a scheme it seems reasonable to suppose that we could continue to progress indefinitely, getting closer and closer to the laws that govern the universe ». (Hawking, 1987).

Regarding what mathematical reality means, it is stated that, «By physical reality I understand the material world, the world of day and night, earthquakes and eclipses, and the world that physical sciences try to describe [...].

For me, and I suppose that for other mathematicians, there is another reality, which I will call "mathematical reality." [...] I believe that mathematical reality is outside of us and that our mission is to discover or "observe" it, and that the theorems that we prove and grandly describe as our "creations" are simply the notes of our observations. Many high-ranking philosophers have held this view in one form or another, from Plato to the present day, and I will use the language that comes naturally to a person holding that position. ». (Hardy, 1999).

Others affirm that in «mathematics is the origin and foundation of the Platonic theory of forms or ideas. [...] The idealization of mathematical entities is transformed into the idealization of physical and psychic entities. [...] Mathematical truth, due to its invariance over time, was the model to follow in all intellectual knowledge. The deductive method, which, starting from axioms and definitions arrived at the proof of theorems, was the prestigious model of reasoning for all knowledge. In the dialogue Meno Socrates, through questions and answers, makes a slave reach a mathematical truth through his own reasoning; thus, in a popular way, Plato exposes that mathematics is in the human soul, since the logos that governs the material world through arithmetic and geometric proportions is present in it. Only introspection is required for us to become aware of this internal knowledge » (Gobierno de Canarias, 2021).

A woman mathematician and scientist in residence at the School of the Art Institute of Chicago. [...], says: "I genuinely feel that I discover concepts and invent ways of thinking about them. When I do abstract research, I feel that I am wandering through an abstract jungle in search of things. I then invent a way to talk and theorize about them so that I can organize my thoughts and communicate them. "

"The part that I like the most is wandering through that abstract world to see what I find. And of course, that world is in my imagination, but it feels very inherent as if I had not done it, but was already there," he admits. [...] "People say that numbers are not real, because you can't touch them. But there are many things that are real but I can't touch, like hunger," she exemplifies. (BBC, 2019).

Having read the previous quotes, it can be deduced, and I am of the opinion that mathematical reality is there, in nature, in everything that surrounds us, linked to the material and immaterial, each of these with its own geometric patterns, different languages, and with its own mathematical structures, pre-established by the Creator of the Cosmos.

These languages and geometric patterns, in the case of the material world, are expressed by, axioms, theorems, propositions, and equalities or mathematical equations initiated by men who, through their studies and observations were able to discover, formulate and make them known to humanity. To understand mathematics, you have to make your brain work, and this always takes some effort. It is not possible to learn mathematics without exercising, without practicing, without making mistakes. It is not like the other sciences; you have to dedicate yourself with the desire to improve. In the end, your life will be pleasant and full of satisfaction. If you do not understand something, it does not matter, little by little and with the help of another person who understands you will be able to understand and you will have your reward, your destiny will be different. Transforming your life story is up to you.

In what follows, I present some topics related to mathematics such as: the meaning and praiseworthy mathematics, definition, characteristics, place of emergence, development and its bibliographic sources, etc., trying to relate them to Pythagoras, which is the object of this book.

1.3 The meaning of mathematics and praise of mathematics.

When you hear the word math, the first thing most people do is wrinkle their face, numbers and expressions that have letters and numbers combined come to their mind, etc. The aforementioned is due to an aversion towards mathematics. This is because many people have very little knowledge of the meaning and origin of the word “mathematics.” Therefore, it is necessary that we clarify this. Next, I share with you the following information:

«The word” mathematics ”(from the Greek μαθηματικά, « what is learned ») comes from the ancient Greek μάθημα (máthēma), which means « field of study or instruction ». The meaning is contrasted with μουσική (musiké) "that can be understood without having been instructed", which refers to poetry, rhetoric and similar fields, while μαθηματική (mathēmatikḗ) refers to areas of knowledge that can only be understood after they have been instructed (astronomy, arithmetic).

Although the term was already used by the Pythagoreans in the 6th century BC, it reached its more technical and reduced meaning of "mathematical study" in the time of Aristotle (4th century BC). Its adjective is μαθηματικός (mathēmatikós), "related to learning", which, similarly, came to mean "mathematician." In particular, μαθηματική τέχνη (mathēmatikḗ tékhnē; in Latin *ars mathematica*), it means "the mathematical art".

The plural form of mathematics comes from the Latin mathematical form (Cicero), based on the Greek plural τα μαθηματικά (ta mathēmatiká), used by Aristotle and which roughly means "all mathematical things". (Wikipedia, 2021).

Therefore, mathematics refers to, "what should be learned", but "learn", what is worthwhile, what is worth learning.

«Etymologically speaking, the mathematical word comes from *mathema*, which means erudition, *manthánein* which is the infinitive of learning, the radical *mendh*, means, in passive, science, to know, that is, it is relative to learning. In an implicit mathematical sense, it means, "What is worth learning." Paraphrasing it, one can say, "What must be learned, due to its great value". » (Luis-Puebla, 2005).

Speaking of the time of the Pythagoreans, it is stated that, "This exaggerated veneration enveloped both the school and the teacher in the legend, [...]. These were divided into two classes: the *mathematicians or connoisseurs* (*μαθηματικός matematikoi*) and the *accusmatic or hearers* (*akousmatikoi*); the first were initiates who enjoyed the privilege of being in direct contact with the teacher and his teachings; the latter received the teachings but not the fundamentals".

We are also told that: «To Pythagoras we owe the *mathematical* word (*μαθηματικά*): [...] the *mathematicians or* (*matematikoi, μαθηματικός*) were the initiates of the school, those who had personal access to the teacher; that is to say to Pythagoras; that is to say to *the man*. Moreover, the *mathematicians* were the ones who generated a conceptual world, rational, linked at its fundamental root to aspects that today still constitute a study problem. "(Jiménez, 2016).

Pythagoras was the first mathematician, the first philosopher and the first metempsychotic. He was the one who first invented (or used) the words 'Mathematician', 'philosopher' and 'metempsychotic' (Strathern, 2021).

With regard to mathematics, it is said that: “Plato (400 BC), considerably stimulated the development of mathematics in his academy of Athens, since he considered it *a science and art*, the study of which was essential for the formation of individuals in their dialectics and philosophical teaching.

Plato's conception of mathematics placed them as intermediaries between the world of ideas and that of things. Thus, for him, and those who subscribe to his beliefs, this discipline not only played a fundamental role in knowledge, but was considered the only bridge between pure and immutable ideas and the physical, imperfect and changing world "(Zenil, 2004).

It is stated, "There is no definition of what mathematics is. However, it is said to be a" collection of ideas and techniques for solving problems that come from any discipline including mathematics itself. “» (Luis Puebla, 2006).

«The enormous utility of mathematics in the natural sciences is something that borders on the mysterious, and there is no explanation for it. It is not at all natural that there are "laws of nature", much less that man is capable of discovering them. The miracle of how appropriate the language of mathematics is for formulating the laws of physics is a wonderful gift that we neither understand nor deserve. » « From the earliest times, the study of mathematics was of great importance in the development of human progress. This characteristic persists even today.

“Mathematical knowledge, at all its levels, gave man the dimension of a thinking being, an enlightened being and a creator of abstract ideas.

Mathematics is one of the oldest sciences; ideas of form and number arose possibly in the most ancient cultures of which we are aware. In all ages, mathematics was the basis of the knowledge that emerged from the human mind. Due to her accuracy, she is at the peak of man's knowledge. »

«The ideal of the perfection of mathematics led the greatest philosophers of ancient times to study it; thus, Thales of Miletus, Pythagoras, Democritus, Anaxagoras, Plato, Aristotle, among others, dedicated their lives to its study, making valuable contributions to Greek culture. Socrates (470 BC), the Great Teacher, exclaimed: »« “Have you ever noticed that those who know how to calculate naturally, and without difficulties, are endowed with an intelligence capable of making rapid progress in all the arts, and that mentally retarded creatures with less open mind do become more ingenious and more intelligent when they are drilled in arithmetic? “

Hippocrates, illustrious doctor of ancient time, said: "By mathematical knowledge, you will not only achieve more glory and usefulness in human things, but it will also make your mind more intelligent and more fitting in regards to matters related to medicine." (Martínez & Arsuaga, 2002).

Here are some characteristics of mathematics that make it different from other science.

1.4 Some Characteristics of mathematics

The following characteristics are listed below:

These characteristics of mathematics make it different from the other disciplines of study. We will mention the following four characteristics:

«The first characteristic is that it is very difficult to describe or define its object or subject of study, this is clear in disciplines such as Astronomy or Biology. This is because the objects it studies are abstractly defined concepts that are chained to others previously defined. The objects of study of mathematics are immaterial, they are not objects of nature, and they are mental abstractions of material objects, whose description is reduced to formal definitions requiring neural connections that need time to be carried out. «It is necessary to acquire mathematical maturity, through mathematical training, so that the human being can assimilate a good amount of abstract ideas. »

“The second characteristic is that it has perfectible logic. Euclid's mathematics is as valid today as it was then. ”

The third characteristic is the conclusiveness of mathematics; that is to say, the different disciplines reach conclusions based on mathematical results." "The fourth characteristic is its independence. It does not require expensive equipment as in experimental sciences, sometimes a paper and pencil, or not even this is enough, just your mind." (Luis-Puebla, 2005).

Next, mathematics, its development and importance.

1.5 Human progress and current mathematics.

It is said: «Due to its characteristics, we can affirm that mathematics has a long historical trajectory linked to the progress of humanity. In each cultural moment, mathematics has solved crucial problems and thus, little by little it has been achieving a prestige and an interest that has justified its insertion in the process of training people » (Alsina et al, 2002).

A professor says it: «with regard to current mathematics, this is characterized by the predominance of algebra, and it is often spoken of the algebrization of all branches of traditional mathematics. The tendency to unification is another characteristic of current mathematics, whose origin is found in analytical geometry, created mainly by Descartes who also says: « the careful care that is put into its foundations is characteristic of current mathematics. Without the Cantorian theory of sets, current mathematics would not have been able to reach the level of generality and abstraction of which it prides itself. He shows how the emergence, in our century, of the new mathematics was due to the desire to deepen and give rigor to the mathematics of the seventeenth and eighteenth centuries, rich in new results and fruitful methods, but poor in logical rigor » (Babini, 1967).

In my opinion, the study of mathematics is important because: It teaches how to think... It allows us the development of reasoning, and the use of reason. It stimulates logical thinking, to use common sense when thinking. It increases our creativity, and helps us to be creative and original. It helps us in our daily activities to follow sequences.

1.6 A little history of the emergence of mathematics.

«Mathematics has grown within a frame of reference, constituted by time and space. Space is the world we live in and the time is from around 4000 BC until today, which brings us to about 6000 years... The space of its beginnings is the Middle East with the civilizations of Babylon and Egypt. In this frame of reference, the main sources of information are written records, which have been one of the most powerful forces in scientific development, supplemented by oral and visual records that have allowed the advancement of the current era. However, the media evolved very slowly: The alphabet, for example, was invented around 850 BC; the printing press just over 500 years ago and the industrial production of paper began in the second half of the last century. . The following are the oldest civilizations that provide us with sources of information, through written records:

A) Babylon. It begins around the year 4500 BC, in the region of Mesopotamia, between the Tigris and Euphrates Rivers in southwest Asia and currently corresponds to the countries of Iran, Iraq and Kuwait.

B) Egypt. It begins around 4000 BC, along the Nile River in northeastern Africa. " « [...], The most widely accepted hypothesis, establishes that they arose from the practical needs of development, the organization of agriculture, the control of crops and rivers, irrigation systems, construction and commerce. Another hypothesis attributes its origin to mystical revelations and religious rituals, but this is poorly accepted in the scientific realm. (Saénz Quiroga, 2005)

«It is the man who gives names to things and acts; who conserves fire and imagines traps to hunt animals; that builds houses and tombs; that observes the movement of the stars and highlights special directions; who computes distances with his body and steps; those records scenes of impressive realism. In that man and in those activities the basic concepts of mathematics are prefigured: number, measurement, and order. »« As we go from the Paleolithic to the Neolithic stage, the process is refined: new agricultural and pastoral techniques, ceramics and carpentry; the textile industry; mining and metallurgy, the barter of goods and objects, navigation and transportation, the norms that govern the nascent family, social and economic organization. In addition, these norms demand ever-greater precision in counting, measuring and ordering. »« The discovery of the deductive process and the cause-effect relationship including the inexhaustible resources of the human imagination will do the rest. Moreover, when the writing appears, as a by-product of urban culture, that mathematical knowledge, still vague and nebulous, begins to acquire consistency. »

"A plausible hypothesis about the origin of writing links it to arithmetic practices." «According to this hypothesis, the writing originates in the middle of the IV millennium BC, in Lower Mesopotamia, in the bosom of the Sumerians, whose cities were built around the temple and on an artificial hill like a stepped tower, which not only represented the spiritual unity of the community, but also contained its economic wealth ». « The goods of the temple, accumulated in its workshops and granaries, were administered by the priests. It is explainable [...], that as these goods increased with the growth of the population, it became more difficult to retain by memory the "accounts of the temple". »

Those are the data relative to the tributes that were owed to the god and the amount of seeds and cattle that were delivered to the peasants and shepherds. Hence, the need to fix conventional signs that would allow retaining these data without relying on individual memory. " «That such was the origin of the first engraved signs would be verified by the fact that the pictographic tablets of Erech from 3,500 B C., which are the oldest known, contain signs that represent a cow's head, an ear of wheat, a fish, accompanied by special signs that undoubtedly represent numerical signs. For the rest, it should be remembered that among the Sumerians, there was the custom of marking personal property objects with individual stamps, and that since the god of the city was the sole owner of the land and all its fruits, the stamps that marked the Temple goods would acquire a more conventional meaning and greater diffusion.

“«The emergence of mathematics in human history is closely related to the development of the concept of number, a very gradual process in primitive communities. They had a certain ability to estimate sizes and magnitudes, but they did not have a notion of number. Thus, numbers beyond two or three, [...] used "many" to refer to a larger set. The next step is the appearance of something close to a concept of number, [...]. The problems to be solved became more difficult and it was no longer enough just to count things and communicate to others the cardinality of the counted set. Instead, it became crucial to count larger sets, quantify time, operate with dates, and enable calculations of equivalences for bartering. It is the moment of the emergence of the names and numerical symbols ». (Pastor & Babini, 1985)

Regarding the importance of mathematics, the following is expressed:

"The extraordinary genius of man has managed with only 21" letters "(or 24, or 26, depending on the languages) to write all possible, infinite words, and with only 10" figures ", all possible, infinite numbers."

«With 31 signs, at the age of six, or earlier, we become owners of the keys that unlock the world's treasures: all the books, all the tables and all the calculations that poets, writers, physicists, astronomers and mathematicians have been able to bequeath us since man has invented those two admirable instruments: alphabetic writing and positional numbering.

Two inventions that have cost the human mind thousands of years of effort. ».

"Giving a value to the place that a figure occupies ("positional principle ") was a more difficult idea than dividing words into their component sounds, and writing them one after the other (or, in some languages, one below of others) as per the signs established for same. It was better going through the trouble of inventing and remembering a different drawing, an ideogram, for each word. "

“On the other hand, the introduction of Arabic numerals (it would be more correct, say Indian), that is, of a numeration in which the position of the figures is taken into account, is a relatively recent historical fact, of which we can even give the date. We are in 1202, in the time of Marco Polo, the Crusades. » (Radice, 1976).

1.7 Geographical location of the emergence of mathematics.

The emergence: «Concepts such as those of social classes, the construction of cities, the teaching of knowledge, imperialism, the defense of economic interests, the expansion and control of land, agricultural use, trade and many others, find in Mesopotamia (between two rivers) its first verifiable expression. »It is commented: «Mesopotamian civilization began to form itself as such, several millennia before our era. Its importance lies in being the first that leaves written testimony of its achievements, the instruments for solving its daily problems, the importance of the gods, the divination of its designs, the social structure of its people, the vicissitudes of its history. » «Among the activities of that time, the instrumental use of mathematical knowledge to solve every day economic problems stands out.

From the numerical record of goods deposited in the temples or exchanged between the different economic agents, to the geometric and algebraic calculations necessary for the construction of irrigation canals in the fields, mathematics is being constituted from an eminently practical perspective, far from all abstract approach. " "Mathematics does not exist as such, but is a mere instrument for solving problems and, from this point of view, the calculations that generate a solution are being constituted as algorithms whose steps must be followed in the proper order." "Thousands of tablets are still to be studied, many of them of an accounting nature. The geographic space of ancient Mesopotamia is almost coincident with present-day Iraq. "

«A place that has always been a territory that has been convulsed to a greater or lesser extent, particularly during the last century, in which the excavations promoted by various European countries made it possible to advance a lot in the knowledge of the ancient cultures of the Middle East.

Many of the efforts in this direction were devoted to the excavations in Egypt, undoubtedly highly appreciated in terms of their results given the durability of their remains. (Saenz Quiroga, 2005).

An opinion of the beginning of the writing and why it arises: «The interest in Mesopotamia was more recent, as well as the decipherment of its cuneiform writing, which allowed us to enter into the testimonies left by the scribes of that time," a country between two rivers ". Here the writing began 4000 years before Christ, and it is believed because of arithmetic practices. "(Maza Gomez, 2017).

The following figure shows the place where the first human civilization, the Sumerians, settled. According to scholars, writing and mathematics were born here. (Watch Tower Bible, 2010).



Fig.1: Writing and mathematics were born here.

1.8 The sources of information available so far.

Regarding the available mathematical information, it is said, "The oldest sources are the scorched tablets of the Babylonians and the papyri of Egypt." (Saenz Quiroga, 2005).

It is said about the mathematical sources: "The mathematical sources in Babylon are derived from more than 400 clay tablets unveiled since 1850. Carved in cuneiform script, they were engraved while the clay was wet and subsequently baked in an oven or dried in the sun. Some of them appear to be graded assignments. The first written mathematical evidences date back to the ancient Sumerians, in Mesopotamia, who developed a system of metrology as early as 3,000 BC. Around 2500 B.C. Henceforth, the Sumerians wrote multiplication tables on clay tablets, tried geometric exercises and division problems. The earliest signs of Babylonian numerals also date from that period.

The clay tablets recovered date from 1800 to 1600 BCE, and include fractions, algebra, quadratic and cubic equations, and the calculation of reciprocal regular twin primes, include multiplication tables and methods for solving linear and quadratic equations. The Babylonian tablet YBC 7289 gives an approximation of $\sqrt{2}$ to five decimal places. Babylonian mathematics was written using a sexagesimal (base 60) numbering system. Hence, the division of a minute into 60 seconds, the hour into 60 minutes, the circle into 360 (60×6) degrees and the sexagesimal subdivisions of this unit of angle measurement in minutes and seconds. The advances in mathematics became easier because the number 60 has many divisors.

The Babylonians had a positional numerical system whereby the digits written to the left represented higher order values, like our decimal numbering system. It lacked an equivalent to the decimal point and thus the value had to be deduced from the context. With regard to mathematical texts, it is said that:

The oldest available mathematical texts are the Plimpton clay tablet 322 (c. 1900 BC), the Moscow papyrus (c. 1850 BC), the Rhind papyrus (c.1650 BC), and the Vedic texts Shulba Sutras (c. 800 BCE). In all these texts, the Pythagorean Theorem, $a^2 + b^2 = c^2$, is applied, which seems to be the oldest and most widespread mathematical development, after basic arithmetic and geometry. (Aleksandov et al, 1980).

1.9 A final reflection on mathematics

It is said that: «to understand mathematics you have to make your brain work, and this always requires some effort. [...] He who is not frightened by the efforts of the mind, who is not discouraged if, here or there, he does not understand something, at first sight; [...], must read carefully, little by little, skipping the most difficult things or having someone who has studied more explained them. (Radice, 1976). The modern world would not exist without mathematics. It hides behind almost everything around us and subtly influences almost everything we do now. “It is also said that: As can be seen, mathematical theories and results are the work of individuals and their social environment. (Fry, 2021),

That is why mathematics is not a collection of universal, infallible truths, above history and individuals. It is a science subject to error, change, movement. It is a science, like the others, dynamic and historical. (Ruiz & Barrantes, 2011).

Chapter 2: Tales from the Life of Pythagoras

In what follows, a collection of stories about the life of Pythagoras obtained from different bibliographic sources is presented.

2.1 Story 1: Pythagoras, philosopher, mystic and aristocrat.

In (Baldor D., 1998) it is said: «Pythagoras. Mystic and Aristocrat, mixed his science with some religion and magic, the starry pentagon being the symbol of his sect. The starting concept of his geometric teachings is the point, which for him was the simplest thing that existed; and which he said, “the unit that has a position”. » «All other geometric bodies are" plurality ", because they are made up of an infinite number of points. On triangles, he established his theorem»

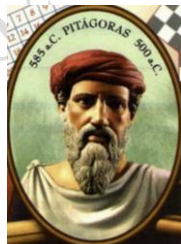


Fig. 2: Image of Pythagoras.

(Baldor D. , 2008) mentions that: «Pythagoras (585-500) A.C. Famous Greek philosopher born in Samos and died in Metaponto. Following studies in his hometown, he traveled to Egypt and other Eastern countries. Upon his return, he founded the School of Crotona, which was a political-religious secret society, which achieved great preponderance. He was the first to place fundamental concepts of mathematics at the base of philosophical speculations. The number is the universal principle per excellence.

2.2 Story 2: Pythagoras, founder of a sect, traveler and philosopher.

As the founder of the Pythagoreans, it is said that: «Pythagoras (Island of Samos, present-day Greece, c. 572 BC - Metaponto, non-existent today, present-day Italy, c. 497 BC) Greek philosopher and mathematician. [...] his name is linked to the Pythagorean theorem [...] to the development of mathematics in ancient Greece, the relevance of Pythagoras [...] reaches the history of ideas [...] that, through Plato, would leave a deep imprint on the western tradition, [...]. It seems certain that he was the son of the merchant Mnesarco and that the first part of his life was spent on the island of Samos, which he probably left a few years before the execution of the tyrant Polycrates, in 522 BC. It is possible that he then traveled to Miletus, and then later visit Phoenicia and Egypt; [...], Pythagoras could have studied geometry and astronomy.

Some sources say that Pythagoras later went to Babylon with Cambyses II, to learn the arithmetic and musical knowledge of the priests there. There is also talk of trips to Delos, Crete and Greece before finally establishing his famous school in the city of Crotona, one of the colonies that the Greeks had founded two centuries earlier in Magna Graecia (present-day southern Italy). , where he enjoyed considerable popularity and power. The community led by Pythagoras ended, [...] in a revolt that forced Pythagoras to spend the last years of his life in the also Greek colony of Metaponto, north of Crotona. [...] It seems that his disciples had to [...] always keep strict secrets about the teachings they received. Women could be part of fellowship; the most famous of his adherents was Teano, wife perhaps of Pythagoras himself and mother of a daughter and two sons of the philosopher. (Ruiza et al, 2004)

2.3 Story 3: Pythagoras and his philosophy.

About the biography of Pythagoras and his philosophy, it is said:

«Pythagoras left no written work, and to such an extent it is impossible to distinguish the ideas of the master from those of his disciples who only can expose the thought of the school of Pythagoras. In fact, externally Pythagoreanism seems more like a mystery religion (like Orphism) than a philosophical school; In this sense, it was a lifestyle inspired by an ascetic ideal and based on the community of goods, whose main objective was the ritual purification (catharsis) of its members. However, such purification (and this is its main singularity with respect to the mystery cults) was carried out through the cultivation of a knowledge in which music and mathematics played an important role. The way to this knowledge was philosophy, a term that, according to tradition, Pythagoras was the first to use in its literal sense of "love of wisdom"; when the tyrant Leontes asked him if he was a sage, Pythagoras politely replied that he was "a philosopher," that is, a lover of knowledge.

Pythagoras is also credited with transforming mathematics into a liberal teaching (without the utility for example of a surveyor that they had in Egypt) by abstractly formulating its results, regardless of the material context in which some of them were already known. "This is especially the case with the famous Pythagorean theorem, which establishes the relationship between the sides of a rectangle triangle: the square of the hypotenuse (the longest side) is equal to the sum of the squares of the legs (the short sides that form the rectangle angle). »

«Of the practical use of this relation there are testimonies coming from other civilizations before the Greek (such as the Egyptian and the Babylonian), but the first proof of the theorem is attributed to Pythagoras, as well as numerous other advances to the school of him. The effort to rise to the generality of a mathematical theorem from its fulfillment in particular cases exemplifies the Pythagorean method for the purification and perfection of the soul, which taught to know the world as harmony. » “By virtue of this, the universe was a cosmos, that is, an ordered set in which the celestial bodies kept a harmonic arrangement that caused their distances to be in proportions similar to those corresponding to the intervals of the musical octave. The celestial spheres, when turning, produced the so-called music of the spheres, inaudible to the human ear because it was permanent and perpetual. In a sensible sense, the harmony was musical; but its intelligible nature was numerical, and if everything was harmony, number turned out to be the key to all things. [...]. »

«The Pythagoreans saw such a principle in numbers: numerical laws and proportions govern natural phenomena, revealing the order and harmony that prevails in the cosmos. »« Only with the discovery of such laws and proportions do we arrive at an exact and true knowledge of things. » [...] «The sanctity preached by Pythagoras implied a whole series of hygienic norms based on taboos such as the prohibition of consuming animals, which seems to have been directly related to the belief in the transmigration of souls; It is said that Pythagoras himself declared himself the son of Hermes, and that his disciples considered him an incarnation of Apollo.

The belief in metempsychosis, an Orientalizing idea and foreign to the Greek tradition, implied the conception of the soul [...] that it would reincarnate after the death of the body.

“«More than a century after the death of Pythagoras, during a trip to southern Italy made before the founding of the Academy, Plato learned of Pythagorean philosophy through his disciples. [...] The influence of Pythagoras is clear at least in the Platonic doctrine of the soul (immortal and prisoner of the body), which also in Plato achieves its liberation through knowledge. [...] Still in the seventeenth century an astronomer as distinguished as Kepler, who was responsible for the discovery of the elliptical orbits of the planets, continued to believe in the music of the spheres. [...] Pythagoras is seen as the forerunner [...] from Galileo's scientific revolution: of the mathematical formalization of knowledge. » (Ruiza et al, 2004)

2.4 Story 4: Pythagoras, the first pure mathematician.

In this account of Pythagoras' life, we are told the following:

«Pythagoras of Samos (ancient Greek Πυθαγόρας) (ca. 569 BC - ca. 475 BC) was a Greek philosopher and mathematician considered the first pure mathematician. He contributed significantly to the advancement of Hellenic mathematics, geometry and arithmetic, derived particularly from numerical relations, and applied for example to the theory of weights and measures, to the theory of music or to astronomy. He is the founder of the Pythagorean Brotherhood, [...] he was also interested in medicine, cosmology, philosophy, ethics and politics, among other disciplines.

Pythagoreanism formulated principles that influenced both Plato and Aristotle and, more generally, the later development of mathematics and rational philosophy in the West. No original writing of Pythagoras has been preserved. His disciples - the Pythagoreans - invariably justified his doctrines by citing the authority of the teacher indiscriminately, making it difficult to distinguish between the findings of Pythagoras and those of his followers. «Pythagoras is credited with the theory of the functional significance of numbers in the objective world and in music; other discoveries, such as the incommensurability of the side and the diagonal of the square or the Pythagorean Theorem for rectangle triangles, were probably developed by the Pythagorean School.

Pythagoras's father was Mnesarco, a merchant from Tíre; and his mother, Pythais, originally from Samos in Ionia. Most historians agree that he flourished around 532 BC, under Polycrates and Tarquinius the Proud. Pythagoras lived the first years of his life on Samos and accompanied his father on many of his travels; he was certainly educated: he learned to play the lyre, to write poetry, and to recite Homer. It is possible that his father took him to Tíre and there he received instruction from Chaldeans and learned men from Syria. Among his teachers, three philosophers are mentioned: Ferécides de Siros, who is often described as the teacher of Pythagoras; Thales and his pupil, Anaximander.

According to Iamblichus, in his *Life of Pythagoras*, at the age of 18 or 20, Pythagoras visited Thales at Miletus. Although Thales must have been an old man at the time, he would have made a strong impression on the young Pythagoras, got him interested in mathematics and astronomy, and advised him to visit Egypt to learn more about these issues.

Anaximander taught Thales' teachings, readings attended by Pythagoras, and many of his ideas on geometry and cosmology influenced his own vision. It was a common belief in antiquity that Pythagoras undertook extensive travels in order to collect the available scientific information of his time directly from sources. To this end, he would have visited not only Egypt, but also Arabia, Phoenicia, Babylon, and even India. “

«The passage of Pythagoras through Egypt can be seen as more than probable; Polycrates had established an alliance and strong ties existed between the island of Samos and Egypt at that time. In 525 BC Cambyses II, king of Persia, invaded Egypt. The alliance with Polycrates was broken and, after the Battle of Pelusium, Cambyses captured Heliopolis and Memphis. According to Iamblichus, Pythagoras was led to Babylon as a prisoner of war by the followers of Cambyses. There, he would associate with the "magies", instructing himself in their sacred rites and the "mystery cults of the gods", as well as the mathematical sciences cultivated by the Babylonians. It is not clear how Pythagoras obtained his freedom, although the deaths of Polycrates and Cambyses - both in 522 BC - may have been determining factors in Pythagoras' emigration to southern Italy and settling in the city of Crotona. There is no certainty about the time that Pythagoras spent in Egypt or the East, nor about the vicissitudes of him in Samos or other Greek cities before his arrival in Italy. There is also no direct evidence of the type and amount of knowledge he may have acquired, nor of how he arrived at the definitive philosophical views of it. Some accounts suggest that he visited the temples and participated in discussions with the priests, initiating the rites and beliefs that he would later impose on the society that he founded in Italy. ”

Among the new customs that Pythagoras adopted, these can be mentioned: secrecy, vegetarianism, refusing to wear clothes made of animal skins and his stubbornness in purity. [...], To what extent Pythagoras is a tribute to the Egyptian priests, or even if he owes them something at all; [...] Pythagoras learned geometry from the Egyptians, but given the purely practical nature of the cultivation of this science by Egyptian mathematicians, direct influence from the teachings of Thales and Anaximander seems more likely.

Of the visits that he made to various places in Greece -Delos, Sparta, Fliunte, Crete, and others with whom he is related-, Pythagoras is commonly portrayed in his character as a religious or legislator. [...] According to Diogenes, [...] he founded a philosophical and religious school.

Pythagoras was the head of this society within a restricted circle of adepts known as matematikoi. According to some accounts, he married Teano, from Crotona, and they had a daughter -Damo- and a son Telauges. Others say they were two daughters -Damo and Myia-; others report that he already had a wife and daughter when he arrived in Italy. The evidence for the place and year of Pythagoras' death is uncertain. In 508 BC the Pythagorean Society of Crotona was violently attacked and Pythagoras escaped to Metaponto, where he would end his days (some authors claim that he starved himself to death). "" Iamblichus relates the following version of events: Cylon, a noble citizen of Crotona, leader by birth, rich and powerful, but also violent and tyrannical, eagerly wanted to participate in the Pythagorean way of life. He approached Pythagoras, by then an older man. Cylon, however, was rejected by virtue of the aforementioned character flaws.

Cylon decided to take revenge and vowed to pursue the Pythagoreans to their last man. This is the version mostly accepted by historians, but Iamblichus disputes it, arguing that Cylon's attack was a minor affair and that Pythagoras returned to Croton. [...]. The Pythagorean Society prospered [...]. After this event, it spread to other Italian cities. His tomb was exhibited at Metaponto in Cicero's time. "(Academia.edu, 2021).

In regards to the mathematical discoveries of the Pythagoreans among others, the following are mentioned:

«The Pythagorean theorem. In a rectangle triangle: "the sum of the squares of the legs (cathetus) is equal to the square of the hypotenuse." The Pythagoreans were the first to enunciate a formal proof of the theorem; [...]. "The square of a number" was not interpreted as "a number multiplied by itself", [...] but as a geometric square.

- Perfect solids. The Pythagoreans proved that there are only 5 regular polyhedrons. [...]
- Interior angles of a triangle. The sum of the interior angles of a triangle is equal to two straight lines, [...]
- A triangle inscribed in a semicircle is a right- angle triangle.
- Construction of figures given a determinate area.
- The irrationality of the square root of 2. This event marks the discovery of the irrational numbers, [...].

- The discovery of the perfect numbers and the friendly numbers.
- Halves. The arithmetic half, the geometric half, the harmonic half, and the relationships between them.
- The discovery of polygonal numbers, triangular, quadrangular, pentagonal, hexagonal, etc.
- Tetraktys. The triangular figure composed of ten points arranged in four rows. It was a symbol of special importance to the Pythagoreans, who used to swear in its name. “(Academia.edu, 2021).

2.5 Story 5: Pythagoras the philosopher, astronomer and musician..

Regarding these three facets of the life of Pythagoras, it is said:

«Pythagoras (570-480 BC; 586-500 BC). He is one of the most famous and enigmatic men of Ancient time. Several biographies were written, but they have been lost and it is difficult to separate in his biography what is historical from what is legendary. He was born on the Island of Samos, next to Miletus, the son of a rich merchant Menesarco. He was a Greek philosopher responsible for important advances in Mathematics, Astronomy, and Music. Three stages in his life can be distinguished according to the trips made: to India where he met mathematical and religious knowledge, to Egypt where he studied Geometry and Astronomy and to Babylon where he learned the arithmetic half the geometric half and the harmonica.

»

«Upon returning to Samos and finding it dominated by the Persians, he decided to immigrate to southern Italy, establishing one of the

most flourishing Greek colonies in Croton, where he founded the Pythagorean School.

The world's first boarding school, which became a partially religious, scientific and philosophical association, supported by the belief in the immortality of the soul, the doctrine of reincarnation, the practice of vegetarian eating and an educational system based on gymnastics, mathematics and music. He began to teach philosophy and mathematics with such success that among the audience, he had people from different social classes and even women who attended, and which inflicted a regulation that prohibited them from attending public meetings. Among them was Theano, daughter of Milo, innkeeper of Pythagoras, beautiful young man whom he married and who wrote a biography of him and got lost. His religious, social and political beliefs and his sense of superiority offend the natives of the port of Crotona and generate rebellions during which they set fire to the house where he resides and consequently flees to Metaponto, [...]. He addressed speeches with content of harmony and justice [...]. Towards the end of his life, his influence and that of the created school grows. He died at the age of 80. “(Sauce.pntic.mec.es, 2021).

As contributions of the Pythagorean School, these are mentioned:

- «Invention of the multiplication table.
- Proof of the theorem that bears his name.
- Construction of the regular pentagon and the five regular polyhedrons.
- Discovery of the existence of irrational numbers.
- Discovery in geometry of such perfect proportions that he thought them divine without suspecting that they were closely linked to a number belonging to the same group.

- The Pythagoreans were the first to establish mathematical proofs by deductive reasoning.
- Formation of square numbers starting from the unit and adding the ascending series of odd numbers.
- Use of the word number only for the sum of equal whole numbers.
- He showed that the intervals between musical notes could be represented by whole number ratios using a kind of guitar with a single string, called a monochord.
- He discovered the relationship between the harmony of a pitch interval and the proportions that produce it... “(Sauce.pntic.mec.es, 2021).

2.6 Story 6: Pythagoras: Genius, Mathematician, and Philosopher.

To highlight these characteristics of Pythagoras' personality, this story makes it known to us the following:

“Pythagoras was possibly the first genius that Western culture produced and seemingly set a precedent by representing that mixture of privileged intellect and outstanding insanity that would later become such a recurring trait within this subspecies. It is also possible that Pythagoras was the first mathematician, the first philosopher, and the first metempsychotic. This is not due to the circumstance that he was the first person to use numbers, the first to seek a rational explanation for the world; nor is it because he was the first person to have the conviction that, in a previous life, his soul had occupied a prominence, a pharaoh, or who knows what, etc.” «He was the one who invented (or used) for the first time the words "Mathematician", "philosopher" and "metempsychotic".

With the sense that they currently have, a fact for which he quickly attributed them to himself; he also invented the word "cosmos", which he attributed to the world. In Greek, "cosmos" means "order", and Pythagoras assigned it to the world for its "perfect harmony and concert." We have few reliable data on the life of Pythagoras, to which it must be added that all the discoveries attributed to him could very well be the result of the work of any of his disciples. Therefore, it is possible that even the famous theorem that bears his name is not his work. Moreover, in this sense, too, Pythagoras sets a precedent in the prevailing conception of "genius" today, in which outstanding history-making discoveries are attributed to luminaries that are often nothing more than the result of the teamwork of an entire laboratory, and in which paintings attributed to geniuses are produced entirely by apprentices. »

“Bertrand Russell described Pythagoras as” one of the most important intellectuals who ever lived, both because of his wisdom and because of his foolishness.” The fundamental principle of Pythagoras was, "everything is number." This seems to coincide with the two categories pointed out by Russell. It states: It's quite evident that the world is made up of more than numbers, even though almost two and a half millennia later Einstein would base his work on a remarkably similar conception. On the other hand, Pythagoras has been credited with many displays of pure genius, and the most memorable of these is, of course, his theorem. This establishes that given a right-angle triangle whose sides measure a , b and c , where c is the side opposite the right angle: $a^2 + b^2 = c^2$ »,

«However, the most important example of Pythagoras' genius is, in all probability, the fact that he proved the theorem that bears his name.

Such an action introduced the concept of proof in mathematics, which in turn signifies the beginning of deductive reasoning. As a result, mathematics ceased to be a series of formulas to become an elaborate and powerful logical structure of great beauty. Logic was used in mathematics almost two centuries before Aristotle "invented" it ».

«The best example of Pythagoras' folly was, without a doubt, the religion he founded, which contained a long list of nonsensical precepts that his practitioners should follow. They could not pick up anything that had fallen, or jump over a stick; they also had to refrain from picking flowers or laying hands on white chickens; and the consumption of beans was completely prohibited. Pythagoras justified this last prohibition by claiming that if a Jew is placed in a new grave and then covered for forty days, the Jew takes human form. How a mind capable of such a consummate mathematical genius can believe in such a string of absurd things is hard to imagine; yet Pythagoras managed to do so, which should possibly only increase our admiration for the power of his mind.

«Pythagoras was born in approximately 565 BC, on the Greek island of Samos, located in the eastern Aegean. It is said that he was the son of a wealthy engraver and merchant named Mnesarco, although other sources claim that he was the son of Apollo, the Greek god of music, poetry and dance. In Russell's words, "he left it to the reader to choose between these alternatives."

“In the century before the birth of Pythagoras, Samos had become the richest island in the Aegean.

This wealth is said to have originated from a legendary journey between the Pillars of Hercules (that is, beyond Gibraltar, towards the Atlantic Ocean). The ships of Samos returned laden "with riches that were proverbial."

“As a result of this mysteriously acquired wealth, Samos was able to become a major trading power possessing colonies as far away as Egypt or Spain. It established a colony in southern Spain, in Tartesos (a region referred to as "Tarshish" in the Old Testament, and which appears in prehistoric Greek mythology). It had silver mines and was located on the southwest coast, beyond the Pillars of Hercules, which may well explain the object of the legendary journey. "

«The first stage of the life of Pythagoras developed at the beginning of the golden age of ancient Greek culture. The Greeks had extended their dominions to the Black Sea and to the south of the Italian peninsula (which they knew as Megale Helias and the Romans as Magna Graecia).

«In Athens, the first marble temples began to be built on the Acropolis, and the first philosophers made their appearance in Miletus and on the Ionian continent (present-day Turkey). One of these philosophers, Anaximander, would eventually become Pythagoras' teacher. Western philosophy begins with Thales of Miletus, some 20 years before the birth of Pythagoras, which means that it was still a novelty with full possibilities yet to be discovered.

»

«Considered as the Internet of its time, it attracted a similar volume of child prodigies, wizards, and baddies.»

«It was Thales who first had the idea that the world had originated from a single substance. As he walked through the hills overlooking Miletus, he found some fossilized shells from which he deduced that, ultimately, the entire world was made up of water. Anaximander was the second philosopher of the so-called School of Miletus, although he was a much more interesting thinker than his teacher, Thales was. He also was the first philosopher who tried to provide a completely rational explanation of the origin of the world. As we shall see later, he was quite off the mark, but at least he was rational. »

«On the contrary, Thales' explanation could be included rather in the field of inspired supposition. Although Anaximander undoubtedly learned the trade, thanks to Thales, his imagination reached far wider horizons, and his point of view was far more scientific. He was the first of the ancients who drew a column supported on its side. We occupied the upper surface of said cylinder. »

«The observation of the sun led Anaximander to the invention of the sundial, which is a decisive discovery for Western culture, since, until then; there had been no precise method of measuring time. Cronos (time) had been a god up to that point: now it was an instrument, and the notion of duration made its appearance on the scientific scene. In symbolic terms, men had taken control of time.

Observing the sun also led him to the conclusion that this star was much larger than Earth. This was a sensational theory, contradicting the evidence that everyone could see with his or her own eyes.

It was a slap on the cheek of all contemporary 'common sense', as unthinkable as Einstein's theory of relativity was which instituted that space was curved and time relative. »

«Common sense is a collection of prejudices acquired at eighteen," as Einstein put it. On page 6, he tells us: Anaximander calculated that the sun was 28 times larger than the Earth. Considering that the only technology he had was the human eye, and the only computing power lay in mental arithmetic, such a conclusion is still surprisingly correct. »

«(Actually, the sun is slightly more than a hundred times bigger than the Earth.). Anaximander also made a celestial map, and conjectured that ray falling from the sun on a swampy area generated the first living things. All ideas he compiled in a work entitled On Nature, which was widely accepted in the intellectual circles of the eastern Mediterranean. "Unfortunately, only a fragment of this work is preserved, which is part of a document dated more than a thousand years later. It describes the way in which things "compensate and repair one another for their injustices, according to the disposition of time." These are the first known written words on philosophy, and as so often occurs with philosophy, thereafter, no one except the author knows for sure what they mean. »

«But we know from other sources that Anaximander was convinced that the world was made up of a single fundamental substance, and this was not any known element, such as water.

Anaximander referred to it as "indeterminate" (apeirus), and classified it as infinite, timeless, and indestructible. As we will see later, this definition bears a more than casual resemblance to the Pythagorean conception of number. It is already known that man is a product of his time, and Anaximander was, apparently, the best teacher that the founding genius of Western culture could have. So where did Pythagoras' outlandish ideas come from?

The seed of such ideas seems to have been sown by his other teacher known to him: Pherecides. If Anaximander was pigeonholed into the role of prodigious child, Pherecides was undoubtedly a wizard of the ancient philosophical internet. »

«He could be defined as a curious combination of philosopher and storyteller. Some hold him responsible for the invention of the metempsychotic doctrine (the transmigration of souls). According to this theory, after death the soul passes to another body, better or worse than the previous one, depending on recent behavior. This other body can be of human nature, animal, and in the worst case, even vegetable. The aspiration of the soul should be to behave in the best possible way; in this way, you will avoid the quiet and sunny life of an olive tree, and will even rise above the exemplary and tormented existence of a saint. Thus, freeing one self, at last, from the circle of birth, life, and death.

This idea appears under different forms in majority of cultures, where often, it originates spontaneously. »

Like human sacrifice, it can be seen as a stage in the evolution of our psyche, and if so, there is no reason why Pherecides could not have come up with the idea. "

“Other sources suggest that he stole the idea from the Egyptians and put it forward as his own. Those who remain skeptical of the possibility of an original idea (be it Pherecides or the Egyptians) suggest that it came from India, where it is still a widespread religious belief today. Nevertheless, one thing is for sure: Pythagoras took the idea of metempsychosis, along with a host of other nonsense, from the fabulist philosopher Ferécides. In addition, it seems, he did so with the skepticism of youth; but without a doubt, he assimilated them, and they remained lethargic inside. For according to an authority of the rank of Aristotle, Pythagoras "worked first with mathematics and arithmetic, and later, for a time, deigned to work wonders in the manner of Pherecides.

“But such an aberration would not resurface until many years later. It is more important to determine when Pythagoras "first worked with mathematics and arithmetic." Anaximander was a philosopher-scientist, Pherecides was a philosopher-magician: neither of them was a mathematician. It seems that Pythagoras acquired mathematical knowledge from him during his travels to Egypt.

“In those days, trips to the East were not considered a way not to hallucinate, but to open the mind. Egypt was thought to be more cultured than Greece, and in fact, it probably still was (although that would not last long).

According to Aristotle: "Mathematical sciences had their origin in Egypt, since all the priests had free time there."

«Previously, the Greeks had been too busy fighting each other to care about the finer points of abstract calculus.

During the golden age of Greek culture, the struggles would continue, but by then mathematicians were already hooked on the oriental addiction, and it was impossible for them to overcome it.) [...] However, going back to Egypt and what Pythagoras could have learned during his travel, the Egyptians had also discovered geometry in addition to arithmetic. This word literally means "measurement of the land," and their techniques were originally used to measure property boundaries, a process they had to repeat each time the Nile overflowed. »

«It is mentioned: [...] «in relation to Pythagoras and his theorem, the fact is that the Egyptians knew that a triangle whose sides are 3, 4 and 5 units long must necessarily be a right-angle triangle. There is historical evidence that they knew more properties of such triangles, including a rudimentary trigonometry. Tradition says that Tal was able to measure the height of the pyramids using the shadow they cast, for which it is almost certain that he used a trigonometric technique developed centuries ago by the Egyptians. »

«From Samos, Pythagoras headed west, finally reaching Magna Graecia around 529 BC. Once here, he settled in the Greek colony of Croton (today's Crotona), on the sole of the boot that forms Italy. By then Pythagoras describes himself as a philosopher, and settles as a professor of this subject, quickly attracting a group of followers who apparently knew how to see his exceptional qualities from the beginning. »

« "In Greek, philosopher means" lover of wisdom, "and Pythagoras is the first man to describe himself as such. Earlier philosophers were known as sophists, a term that means "wise men."

The meaning of the latter term has been much questioned. Some suggest that Pythagoras, in his modesty, did not consider himself a sage, but only as a man hopelessly attracted to wisdom, after which he was constantly running, though never reaching it. This does not seem very likely (such shyness would have been that after half a century of life, the sophists had already managed to cultivate a not very good reputation. As with any type of "wisdom" since the beginning of time, philosophy, as it turned out to be, was irresistibly appealing to all manner of intellectual charlatans and swindlers. Calling himself a philosopher, Pythagoras probably wanted to distance himself from all this falsehood, despite the fact that, as we can see, his conception of philosophy would far exceed the antics previously practiced by any imposter. » «“Until the 19th century, there would not be a clear distinction between philosopher and sophist. Hegel compared this ancient dichotomy with the difference between a wine-loving gourmet and an old drunkard. The Greeks would bear a striking resemblance to the latter image for several centuries to come. It is almost certain that it was during those years in Croton that Pythagoras carried out the bulk of his mathematical work, including the discovery of his celebrated theorem (if indeed he discovered it and not one of his followers: a controversial matter to which I will return later). One last thing: the Latin writer, Aulus Gellius, provided an ingenious explanation of Pythagoras' ban on eating beans. According to his version, the exact words of Pythagoras were “Wretched, final wretches, don't ever put your hands on the beans! “Which was not exactly what it sounded like. In those days, "Jewish" was used to euphemistically designate the testicles, so Pythagoras' prohibition pertained to sexual activity. » (Strathern, 2021).

2.7 Story 7: The Island of Samos and Polycrat in the time of Pythagoras

About Samos in the time of Pythagoras, it is commented:

“Samos was at the beginning of the 6th century BC, one of the most flourishing islands in Ionia. The bay of its port opened in front of the mountains, from where all the luxuries and all the seductions came. In a wide bay, the city stretched out on the shore and appeared in an amphitheater on the mountain, crowned by the temple of Neptune. ““The colonnades of a magnificent temple dominated it. The tyrant Polycrat reigned there. Having deprived Samos of its freedoms, he had given it the luster of the artisan Asian splendor. The hetairas of Lesbos, called by him, had settled in a neighboring palace to his and invited the young people to parties, with music, dances and feasts. The fate of Polycrat was proverbial throughout Greece. He had as a friend, the Pharaoh Amasis, who warned him several times that he, distrusted such continuous happiness and above all that he did not praise it. » «Polycrat responded to the Egyptian monarch's advice by throwing his ring into the sea:" I make this sacrifice to the Gods, "he said. The next day, a fisherman brought the tyrant the precious ring that he had found in the belly of a fish. When the pharaoh found out, he declared that he was breaking his friendship with Polycrat, because such insolent luck would bring him the vengeance of the Gods. Whatever the anecdote, the end of Polycrat was tragic. One of his satraps lured him to a neighboring province, made him expire in torment, and ordered his body tied to a cross on Mount Mycale. In this way, the inhabitants of Samos could see in a bloody sunset the corpse of their crucified tyrant on a promontory, in front of the island where he had reigned in glory and pleasures. » “(Schure, 1992)

2.8 Story 8: Greece in the time of Pythagoras.

Regarding Greece in the time of Pythagoras, we are told that:

«At the end of the seventh century, Greece was beginning to decline. Before, the Greeks had a pure doctrine kept in the temples, a plastic religion and a high court of justice centralized in the temples. Now the orders were no longer respected; the sacred territories were violated, the intellectual and moral level of the temples had been corrupted, the priests were sold to the political powers. The general appearance of Greece had changed. The ancient priestly and agricultural majesty was replaced by an outright tyranny, or an anarchic democracy. The temples were already powerless to prevent the threatening dissolution.

There was a need for new help. The science of the temples was needed to pass to the lay orders. In this context, under various disguises, science penetrated the corporation of civil legislators, the schools of the poets, the porticoes of the philosophers. They felt, in their teachings, the need for two doctrines, one public, and the other secret that expounded the same Truth, in a different measure and in different ways that were appropriate to the development of their disciples. This evolution gave Greece its three great centuries of artistic creation and intellectual splendor. It allowed the Greek thought to concentrate all its light and to radiate it over the whole world, before collapsing under the blows of Macedonia and completely sinking under the iron hand of Rome. «The evolution we are talking about had many architects. It raised physicists like Thales, legislators like Solon, poets like Pindar, heroes like Epaminondas; but it had a recognized leader, an initiate of first order, a sovereign, creative and ordering intelligence person.

Pythagoras is the teacher of secular Greece, as Orpheus is of priestly Greece. He translated, continued the religious thought of his predecessor and applied it to the new times. However, the translation of him is a creation. He coordinates the orphic inspirations into a complete system; he gives the scientific proof in his teaching and the moral proof in his institute of education, in the Pythagorean order that survives him ». «Although Pythagoras outshines in the pages of history, he is an almost legendary character. The main reason for this is the fierce persecution of which he was a victim in Sicily and which cost the lives of so many Pythagoreans. Some were crushed under the remains of his burnt school, others starved to death in a temple. The survivors who were able to flee to Greece only perpetuated the memory and the doctrine of the master. Plato, with great labor and at great price, procured, by means of Archytas, a manuscript of the master, who, moreover, never wrote his doctrine other than by means of secret signs and in symbolic form. » «His true action, like that of all the reformers, was exercised by oral teaching. However, the essence of the system subsists in the *Golden Verses of Lysis*, in the commentary of Hierocles, in the fragments of Filolaus and Archytas, as well as in *Plato's Timaeus* that contains the cosmogony of Pythagoras. The writers of antiquity, in short, are full of the philosopher of Crotona. “«They have no shortage of anecdotes that depict their wisdom, their beauty and their marvelous power over men. The Alexandrian Neo-Platonists, the Gnostics, and even the early Church Fathers cite him as an authority. Seen as a whole, open with the keys of esotericism, his doctrine presents a magnificent whole, a solidarity synthesis whose parts are linked by a fundamental conception. In it, we find a reasoned reproduction of the esoteric doctrine of India and Egypt, to which he gave Hellenic clarity and simplicity, uniting with them a more energetic feeling, a clearer idea of human freedom. »

«At the same time and in various parts of the globe, great reformers were popularizing similar doctrines. Lao-tzu emerged in China from the esotericism of Fo-Hi; the last Buddha Shakia-Muní preached on the banks of the Ganges; in Italy, the Etruscan priesthood sent to Rome an initiate provided with sibylline books, King Numa, who tried to curb the threatening ambition of the Roman Senate. Moreover, it is not by pure chance that these reformers appear at the same time in such diverse peoples "" Their different missions concur to a common goal. They prove that at certain times the same spiritual current mysteriously crosses all of Humanity. Where does it come from? Of that divine world that is out of our sight, but of which the geniuses, the prophets and witnesses are sent... Pythagoras traversed the ancient world before preaching to Greece. He saw Africa and Asia, Memphis and Babylon, their politics and initiation. His stormy life is as a ship launched in a storm; sails unfolded, it pursues its end without deviating from the path, an image of calm and strength. His doctrine gives the sensation of a cool night succeeding the sharp fires of a bloody day. » (Schure, 1992).

2.9 Story 9: Pythagoras' possible place of birth, his parents.

Pythagoras is said to have been born on the Island of Samos (Google Earth, 2017)



Fig.3: Island of Samos, birthplace of Pythagoras?

It is also stated that, Pythagoras was born in Sidon Phoenicia, here an ancient map of the place of the possible birth of Pythagoras, obtained from (Watch Tower Bible, 2010)



Fig.4: Sidon, Phoenicia, present day Lebanon

Regarding the progenitors of Pythagoras, the following is said:

«Pythagoras was the son of a rich Samos ring merchant named Mnesarco and of a woman named Parthenis. When Pythagoras' parents consulted the Pythoness of Delphi, she told them that they would have: "A son who would be useful to all men, at all times," In addition, she sent them to Sidon, in Phoenicia, so that their son could be born there. When he was one year old, his mother, following advice given by the priests of Delphi, took him to the temple of Adonai, in a valley in Lebanon. There the high priest had blessed him. Then his family took him to Samos. "«The son of Parthenis was very beautiful, sweet, moderate, and full of justice. Only intellectual passion shone in his eyes and gave his acts a secret energy. Far from antagonizing him, his parents had encouraged his early penchant for the study of wisdom.

He had been able to confer with the priests of Samos and with the intellectuals who were beginning to form a school in Ionia where they taught the principles of Physics. »

«At eighteen, he had followed the lessons of Hermodamas of Samos; at twenty, those of Pherecides, in Syros; he had also conferred with Thales and Anaximander at Miletus. Those teachers had opened new horizons for him, but none had satisfied him. »

«Among his contradictory teachings, he was searching internally for the bond, the` synthesis, the unity of the great Whole.

Now, the son of Parthenis had reached one of those crises in which the spirit, overexcited by the contradiction of things, concentrates in a supreme effort all the faculties to glimpse the end, to find the path that leads to the Sun of Truth, to the center of Life. (Schure, 1992). »

2.10 Story 10: The influence of Pythagoras to this day.

Referring to the influence of Pythagoras, it is said that:

«He lived thirty years in Crotona. In twenty years, this extraordinary man had acquired such power that those who called him a demigod did not seem to exaggerate. That power was a marvel: never had a philosopher wielded it like this. His influence extended not only to the Crotona school and its ramifications in the other cities of the Italian coast, but also to the politics of all these small states. Pythagoras was a reformer in all sense of the word. Crotona, an Achaean colony, had an aristocratic constitution.

The council of a thousand, composed of the great families, exercised legislative power there and oversaw the executive power. Popular assemblies existed, but with restricted powers. Pythagoras, who wanted the State to be an order and harmony, was not satisfied with either oligarchic pressure or the chaos of demagoguery. Accepting the Doric constitution as it was, he simply tried to introduce a new gear into it. The idea was daring: to create a scientific power over political power, with a deliberative and consultative voice on vital issues, and that would be the key to the vault, the supreme regulator of the State. On the council of the thousand, he organized the council of the three hundred, elected by the first, but recruited from among the initiates only. Their number was enough for such work. Porfirio tells that two thousand citizens of Cretona gave up their usual life and came together to live together with their wives and children, after having put their property in common. Pythagoras wanted, at the head of the state, a scientific government, less mysterious, but placed as high as the Egyptian priesthood.»

«What he accomplished for a moment was the dream of all initiates who dealt with politics: to introduce the principle of initiation and examination in the government of the State, and to reconcile, in this higher synthesis, the elective or democratic principle with a government constituted by the selection of intelligence and virtue. The Council of three hundred formed a kind of political, scientific and religious order, of which Pythagoras was the visible head. They committed themselves to it by a solemn and terrible oath to an absolute secret as in the Mysteries.»

«These societies or hetarios spread from Cretona, where the mother society was, to almost all the cities of great Greece, where they exercised a powerful political action.

The Pythagorean order also tended to conquer the head of state throughout southern Italy. It had ramifications in Tarentum, Heraclea, Maetaponte, Regium, Himere, Catania, Agrigente Sybaris, according to Aristoxene; even among the Etruscans. As for the influence of Pythagoras on the government of the great and rich cities, nothing could be imagined higher more liberal, more pacifying. Everywhere he appeared, he restored order, justice, harmony. Called by a Sicilian tyrant, he decided by his eloquence alone to resign the ill-gotten riches and abdicate a usurped power.

As for the cities, he made them independent and free, subject to each other. His action was so well done that when he went to the cities, they said: "It is not to teach, but to heal". The sovereign influence of a great spirit and a great character, that magic of the soul and intelligence, excites jealousy all the more fearsome, hatred all the more violent, since it is unassailable. The empire of Pythagoras lasted for a quarter of a century, the indefatigable adept reached the age of ninety, when the reaction arrived. The spark came from Sybaris, Crotona's rival. There was a popular uprising there and the aristocratic party was defeated.» «Five hundred exiles requested asylum to the Crotoniatas, and the Sybarites demanded their extradition. Fearing the wrath of an enemy city, the magistrates of Crotoniatas were going to agree to their demand, when Pythagoras intervened.» «At his behest, he refused to hand over those unfortunate supplicants to relentless adversaries.

Then, Sybaris declared war on Croton. But the army of the Crotoniates, commanded by a disciple of Pythagoras, the famous athlete Milo, completely defeated the Sybarites.»

«The Sybaris disaster was total and the city was taken, plundered, destroyed and transformed into a desert. It is impossible to admit that Pythagoras approved such reprisals. They were contrary to his principles and those of all initiates. But neither he nor Milo could restrain the unleashed passion of a victorious army, fueled by ancient jealousy and flared by an unjust attack. All revenge, whether of individuals, or of peoples, brings a backward shock of the passions. The nemesis of this one was terrible; the consequences fell on Pythagoras and his order. After the taking of Sybaris, the people asked for the distribution of the lands.

Not content with having obtained it, the democratic party proposed a change in the constitution that took away its privileges from the Council of a thousand and abolished the Council of three hundred, admitting no more than a single authority: universal suffrage. Naturally, the Pythagoreans who were part of the Council of a thousand opposed a reform contrary to their principles and which undermined the master's patient work at its base. Already the Pythagoreans were the object of that deaf hatred which mystery and superiority always excite among the masses. Their political attitude raised fury of demagoguery against them, and a personal hatred against the teacher brought about the explosion. A certain Cylon had once come to the School. Pythagoras, very severe in admitting disciples, rejected him because of his violent and imperious character. That dismissed candidate was a poisonous adversary. »

«When public opinion began to stir against Pythagoras, he organized a club of great popular society against the Pythagoreans. He succeeded in attracting to him the main leaders of the town and prepared in their assemblies a revolution that was to begin with the expulsion of the Pythagoreans. Before a stormy crowd, Cylon climbs to the popular platform and reads stolen excerpts from Pythagoras' secret book, entitled: The Sacred Word (hieros logos). He disfigures them, disguises them. »

«Some speakers try to defend the brothers from silence, who respect even the animals. They are answered with laughter. Cylon goes up and continues to the rostrum, to show that the religious catechism of the Pythagoreans is an attack on freedom. "And it is little to say, adds the tribune." » «What is this teacher, this so-called demigod, who is blindly obeyed, and who has only to give an order for all his brothers to exclaim: The teacher has said it! What is he, I repeat, but the tyrant of Crotona and the worst of tyrants, a hidden tyrant? » «Of what is formed that indissoluble friendship, which unites all the members of the Pythagorean hetaires, if not of disdain and contempt for the people? They always have in their mouth that word of Homer, that the prince must be the shepherd of his people. «That means that, for them, the town is just a vile herd. Yes, the very existence of the order is a permanent conspiracy against popular rights. As long as it is not destroyed, there will be no freedom in Crotona". One of the members of the popular assembly, animated by a feeling of loyalty, exclaimed "That at least Pythagoras and the Pythagoreans be allowed to come and justify themselves to our rostrum, before condemning them." »«However, Cylon replied haughtily, "Have not the Pythagoreans taken from you the right to judge and decide on public affairs? With what right could they ask today to be heard?

«They have not listened to you when they deprive you of the right to exercise justice; Now then, punish without listening!». Thunder of applause responded to those vehement words and the spirits became more and more exalted. «One afternoon when the forty leading members of the order were gathered at Milo's house, the tribune mutinied its partisans. «They surrounded the house. The Pythagoreans, with the master among them, bolted the doors. The angry crowd set the house on fire. Thirty-eight Pythagoreans, the best disciples of the master, and Pythagoras himself perished, the flower of the order, some in the flames of the fire and the others murdered by the people. Only Archippo and Lysis escaped the slaughter. “Thus, died that great sage, that divine man, who had tried to bring his wisdom into the government of men.” «The massacre of the Pythagoreans was a sign of a democratic revolution in Crotone and in the Gulf of Taranto.

The cities of Italy drove away the unfortunate disciples of the master. The order was dispersed, but its remains spread throughout Sicily and Greece, spreading the word of the master everywhere. Lysis became Epaminondas's teacher. After new revolutions, the Pythagoreans were able to return to Italy on the condition that they no longer form a political body. A moving brotherhood always united them; they considered themselves as one big family. One of them fell into misery and fell ill. An innkeeper picked him up. » «Before dying he drew some mysterious signs on the door of the house, and said to his guest: 'Do not worry; one of my brothers will pay my debt. » «A year later, a foreigner passed by the same inn, saw those signs and said to the innkeeper, “I am a Pythagorean; one of my brothers has died here: tell me what I owe you for him”. The order subsisted for 250 years in terms of ideas. The traditions of the master live to this day. » (Schure, 1992)

2.11 Story 11: Family and version of Pythagoras's death. .

Regarding the family of Pythagoras, in (Schure, 1992) the following is related:

«Among the women who followed the teacher's teaching, there was a young woman of great beauty. Her father was from Crotona and his name was Brontinos. Her name was Theano. Pythagoras was then in his sixties. “But the great mastery of his passions, and a pure life devoted entirely to his mission, had kept his virile strength intact. The Greek magician was not in decline, but at the height of his power. Theano was drawn to Pythagoras by the almost supernatural radiance emanating from her person. Serious, reserved, she had sought next to the teacher the explanation of the mysteries that she loved without understanding, but she silently fell in love with the teacher with boundless enthusiasm and passionate love. Pythagoras had not tried to attract her. Her affection belonged to all of his disciples. He only thought of his school, Greece, and the future of the world. Like many great adepts, he had renounced women to give himself to his work. The magic of his will, the spiritual passion of so many souls he formed, were bound to him like an adored father. » « But one day when he was meditating alone on the future of his School, in the crypt of Proserpina, he saw coming toward him, serious and resolute, that beautiful virgin to whom he had never spoken alone.

Theano knelt before him and without raising her head lowered to the ground, she pleaded with the master, - To him, who could do everything! - To free her from an impossible and unfortunate love that consumed her body and devoured her soul. “Pythagoras wanted to know the name of the one whom she loved.

After long hesitations, Theano confessed that it was he, but that imminently, she would submit to his will. » «Pythagoras answered nothing.

Encouraged by that silence, she raised her head and gave him a pleading look, from which the sap of a life escaped and the perfume of a soul offered as a holocaust to the master. The philosopher was moved; he knew how to conquer his senses; he had mastered his imagination; but the lightning of that soul had penetrated his. In that virgin matured by passion, transfigured by a thought of absolute self-denial, he had found his companion and glimpsed a more complete realization of his work. »

«An emotional Pythagoras lifted the girl, and Theano could read in the eyes of the teacher that her destinies were united forever. ». By his marriage to Theano, Pythagoras stamped the seal of accomplishment on his work. The association, the fusion of the two lives was whole. One day when they asked the teacher's wife how long a woman needed to be pure again, after having had relation with a man, she replied: "If with her husband she purifies herself at the same moment, if with another, never". Many women will answer smilingly that to say those words it is necessary to be the woman of Pythagoras and to love him as Theano loved him.

She is right. It is not marriage that justifies love; it is love that justifies marriage. »

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Theano entered so completely into the thought of her husband that after his death, she served as the center of the Pythagorean order, and a Greek author cites her opinion on the doctrine of numbers as her authority. » She gave Pythagoras two sons: Arimmestes and Telauges, and a daughter: Damo.

Telauges was later the teacher of Empedocles and transmitted the secrets of the doctrine to him.

The Pythagoras family offered the order a true model. His house was called the Temple of Ceres and his courtyard the Temple of the Muses. At domestic and religious festivals, the mother led the women's choir and Damo the youth choir. Damo was in every way worthy of her father and her mother. Pythagoras had entrusted her with certain writings, with the express prohibition of communicating them to anyone outside the family. After the dispersion of the Pythagoreans, Damo fell into extreme poverty. She was offered a large sum for the precious manuscript. Nevertheless, true to the will of her father, she always refused to hand it over. »

2.12 Story 12: Another version about the death of Pythagoras.

“This is Diogenes of Laertius's version of the death of Pythagoras. - According to Dicearco, quoted by Porfirio, the master escaped disaster with Archippo and Lysis and walked from city to city until he reached Metaponte, where he allowed himself to starve in the temple of the Muses. The inhabitants of Metaponte claimed that the philosopher, taken in by them, had died peacefully in their city. They showed Cicero, his house, his chair, and his family. [...] They claimed the honor of welcoming and saving him. [...] The ashes of the philosopher were disputed with the same fierceness with which the cities of Ionia disputed having seen Homer born. (Schure, 1992).

Chapter 3: Emergence of The Pythagorean Theorem.

3.1 The Pythagoras Theorem in Euclid's Book of the Elements

In this section, it is mentioned that, «I must say that you cannot write about Pythagoras and his theorem without mentioning Euclid. » «Even today, after 2300 years since Euclid composed them, the reading of The Elements continues to provoke a deep admiration. [...] for the [...] ingenuity, logic, rigor, didactics, accuracy, certainty, purity, beauty, coherence and elegance. [...] The Elements of Euclid is the Mathematical treatise that has had the greatest influence throughout the entire History of Culture - even beyond Mathematical science and related sciences - and that has been edited the most times after the Bible. » Of the content of Book I, it says, «Book I. Elementary constructions with ruler and compass, congruence of triangles and quadrilaterals, inequalities relative to angles and sides of a triangle. Parallelism. Pythagoras theorem. » It speaks of a Proposition: « Proposition I.47. In rectangle triangles, the square of the side that subtend the right angle is equivalent to the squares of the sides that comprise the rectangle (Pythagorean Theorem). It is illustrated in the following figure: (González Urbaneja, 2017).



Fig.5.: Illustrates Proposition I.47 of Euclid.

About the place where the bulk of Pythagoras' mathematical work was carried out, the following is related:

«From Samos, Pythagoras headed west, finally reaching Magna Graecia around 529 BC. Once here, he settled in the Greek colony of Croton (today's Crotona), on the sole of the boot that forms Italy. It is almost certain that it was during those years in Croton that Pythagoras carried out the bulk of his mathematical work, including the discovery of his celebrated theorem (if indeed he discovered it and not one of his followers: [...]). As we have seen, the Babylonians needed a real wise person to discover the formula that today we know as the Pythagorean Theorem. They knew that if the legs of a right triangle were 3 and 4 units respectively, its hypotenuse would be 5 units long. Moreover, not only that: a cuneiform tablet even includes a list of 15 different triads of numbers that constitute different combinations of right-angle triangles. However, it was in all probability, Pythagoras (or one of his followers) who came up with the definitive formula: $a^2 + b^2 = c^2$, for a rectangle. »

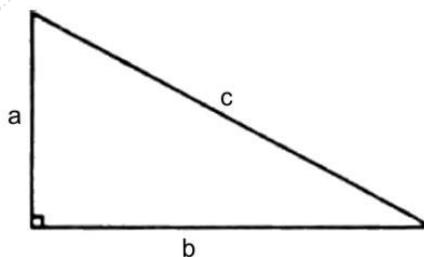


Fig.6: Rectangle triangle. (Pythagorean)

« This formula is revolutionary for several reasons: on the one hand, it characterizes the distinctive Greek contribution to mathematics, which is why the Greeks are still considered the founders of this discipline in many respects. »

« On the other hand, the Greeks were the first to turn mathematics into a purely theoretical object of study whose methods could be applied in a general way; they went even further, confirming these widespread procedures with tests. Because of their unfamiliarity with algebra, they were unable to state them in general terms. Only when this had been done could these propositions be proved or discarded with the help of deductive reasoning. Interestingly, the Greeks lacked a word for this procedure. Like many other words of Arabic origin, such as alcohol, alchemy or almanac, the word algebra began to be used in the Middle Ages, and has its origin in the Arabic term al-jabr, which means "meeting", "equation" by inference. Abstraction, proof, deductive reasoning are three primary mathematics characteristics elaborated probably by Pythagoras himself and introduced by the ancient Greeks. »

«As Pythagoras did not leave anything written, we have no record of the tests that he carried out to prove his theorem. About two centuries later, the geometer, Euclides, wrote several proofs of this theorem in his work Elements, the bible of geometry for more than two millennia and, in all probability, at least one of the proofs is of Pythagorean origin. The first authority to attribute the discovery of the Pythagorean Theorem to Pythagoras himself was the 1st century BC Roman architect. C. Vitruvius (Marcus Vitruvius Pollio), who is best known today for his theory of human proportions, according to which a human figure can be included in a square circumscribed by a circle, as Leonardo's famous illustration shows. In fact, we know so little about the life of Pythagoras that it is virtually impossible for us to distinguish his ideas from those of his followers. »

«Due to the lack of works by Pythagoras himself, we cannot help but trust those of the Pythagoreans and later critics; however, since the Pythagoreans were in the habit of attributing all their works to their teacher, the help they can provide is questionable at best. For the moment, I will continue to attribute the main Pythagorean ideas to Pythagoras himself, suggesting below the discoveries that may have been the work of his disciples. The discovery of the Pythagorean Theorem leads to a series of fascinating discoveries about right-angled triangles with sides whose length is an integer (now known as Pythagorean triangles). For example, the triangle whose sides are 3, 4, and 5 has some properties that other Pythagorean triangles do not have: it is the only one whose sides are in geometric progression and the only triangle of the type with full-length sides whose area is half its perimeter. There are only two Pythagoras triangles whose areas equal their perimeters (5, 12, 13 and 6, 8, 10). »

«The transformation of mathematics into a purely abstract science, which we owe to the Greeks, allowed such possibilities to be explored. There was room for speculation, gambling, and discovery. One could simply follow a line of reasoning and calculate what came of it: the door to the wide field of mathematical exploration had just opened. Another of the great discoveries to which the Pythagorean Theorem leads were irrational numbers. According to the Pythagorean Theorem, if the legs of an isosceles triangle (that is, one that has two equal sides) measure 1 and 1, the length of its hypotenuse must be $\sqrt{2}$; $\sqrt{1^2 + 1^2} = \sqrt{2}$. The Pythagoreans realized that the value of $\sqrt{2}$ could not be calculated. "There simply was no rational number equal to $\sqrt{2}$." (Strathern, 2021)

3.2 The classical proof of the Pythagorean Theorem.

$$a^2 + b^2 = c^2 \quad (1)$$

It's about proving equality (1)

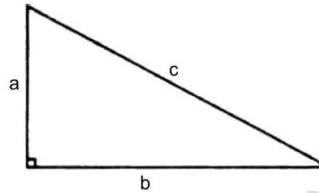


Fig.7: Illustration of the Pythagorean Theorem

«In Chou Pei suan ching, a Chinese test is included that dates between 500 a. C. [...] the simplified version of this Chinese test is the one that contains the most beauty: A square with sides $(a + b)$ contains a square with sides c inside it». (Strathern, 2021).

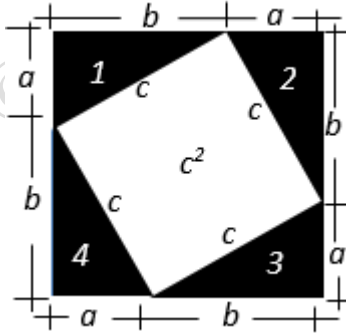


Fig.8: Square of side c , and four triangles of sides a, b

This test involves equating the total area $(a + b)(a + b)$, to the areas of the interior square $(c \cdot c)$ and the four triangles, resulting in the equation:

$$(a + b)(a + b) = (a + b)^2 = 4\left(\frac{1}{2}ab\right) + c^2;$$

$$a^2 + b^2 + 2ab = 2ab + c^2 \text{ Simplifying } 2ab, \text{ we get:}$$

$$a^2 + b^2 = c^2 \text{ with which it is proved.}$$

3.3 Brief explanation of the proof of the Pythagorean Theorem.

Be it the triangle a , b and c , we copy this four times and form a square with side $(a + b)$, as shown in the following graph:

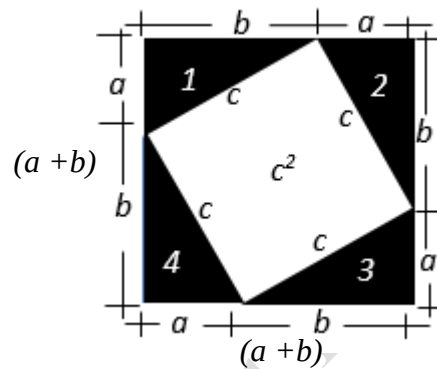
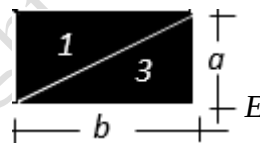
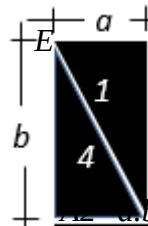


Fig.9: Square of side $(a + b)$

The four right triangles in the previous figure, when joined by the hypotenuse, 1 with 3 and 2 with 4, form two rectangles: the area of each one is, $A = ab$, since there are two, the total area $A_t = 2ab$.



$$A_1 = a.b$$



$$A_t = 2a.b$$

Fig. 10: Two rectangles of four triangles

Now, let us join the two rectangles at the corner E, we are left with the graph of figure 12, which has the same area as that of figure 11.

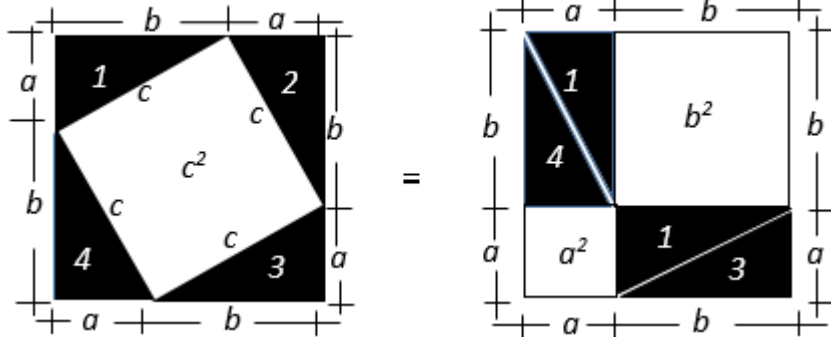


Fig. 11.: Area $A = c^2 + 2ab$ (four triangles)

Fig.12: Area $A = a^2 + b^2 + 2ab$ (two rectangles)

The area of Figure 11, $A = c^2 + 2ab$, is the same as that of Figure 12, $A = a^2 + b^2 + 2ab$; as you can see in the previous graphs.

Now let's remove the shaded areas from figures 11 and 12, that is, $2ab$, and we are left with the following:

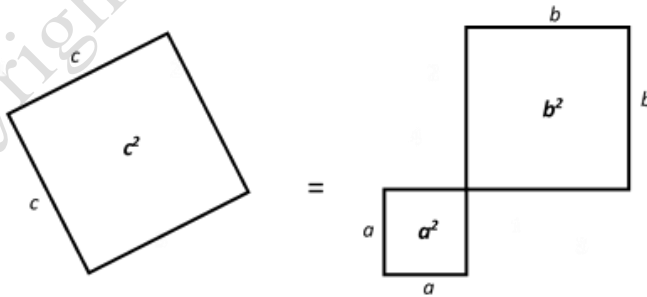


Fig.13: A square equal to Sum of two squares

Therefore, it is shown that: $c^2 = a^2 + b^2$

3.4 A Numerical Proof of the Pythagorean Theorem.

Next, the proof of the Pythagorean Theorem, using the natural numbers 3, 4, 5, which correspond to the sides of the Egyptian triangle, the oldest, best known, and most famous; The three numbers are known as *Pythagorean triad or triple*.

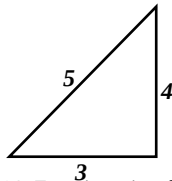


Fig.14: Egyptian triangle with sides 3, 4, 5

The following image is a stamp issued in Greece on August 20, 1955, commemorating the 2500th anniversary of the founding of the first School of Philosophy of History. (Gonzalez Urbaneja, 2017).



Fig.15: Image of Egyptian triangle.

The Egyptian triangle $a = 3$, $b = 4$, and $c = 5$; creates three squares, the one with side 3 has area $= 3^2$, 9 resulting in 9 quadricles; the one with side 4 with area $= 4^2$ gives 16 quadricles; added the two gives 25 grids. These are equal to the quadricles of the square with side 5 whose area is $= 5^2 = 25$ quadricles, this shows that: $3^2 + 4^2 = 5^2$. We clarify that the unit of area is a quadricle.

From the Egyptian triangle, $a = 3$, $b = 4$, $c = 5$, we draw two squares with side $a + b$, or $3 + 4 = 7$ squares or quadricles, the area of each of them is 49 squares, as shown see in the following figure:

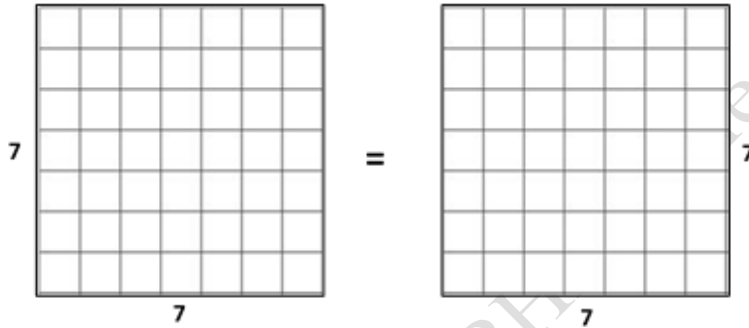


Fig.16: Two squares of 7×7 , with areas of 49 squares each one

Now let's draw inside the square on the left, the square of side 3 and the square of side 4, and in the one on the right the square with of 5, the graph looks like this:

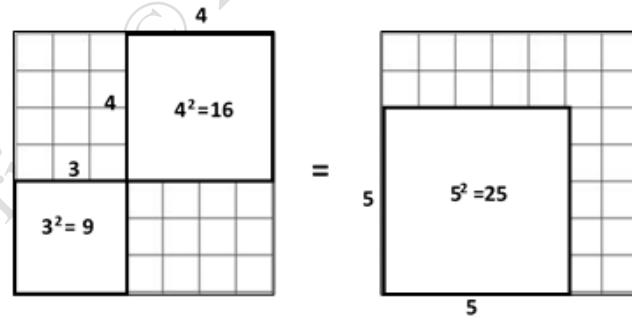


Fig.17: Two squares of 9 and 16 squares equal to one of 25 quadricles

$$\text{Net area} = 49 - 24 = 25$$

$$\text{Net area} = 49 - 24 = 25$$

$$\text{It is shown: } 9 + 16 = 25 \rightarrow 3^2 + 4^2 = 5^2$$

$$\text{Therefore: } a^2 + b^2 = c^2$$

3.5 Another graphical and numerical proof of: $a^2 + b^2 = c^2$

For this section, it is necessary to define what a "gnomon" is. For which it is said, «The second book begins with the definitions of rectangle and "gnomon", which is a kind of L obtained by removing a similar smaller rectangle at a vertex. (Navarro Loidi, 2002).

Another source says, «The name of gnomon was given to the figure resulting from extracting another smaller square from a square, (Jiménez, 2016). For example, from square 1 of 49 quadricles, we subtract square 2 of 25 quadricles; by doing so, we have a "gnomon" of 24. (Jiménez, 2016).

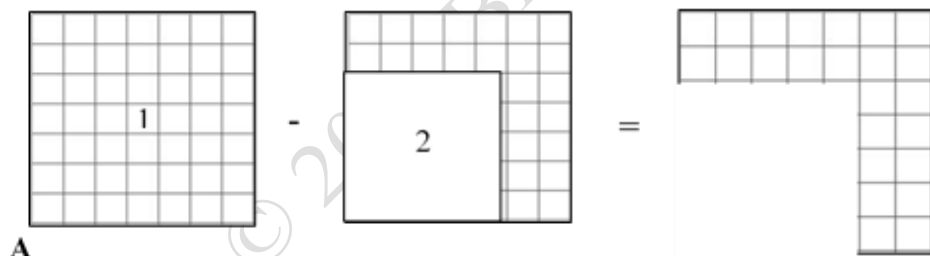


Fig.18: Obtaining a "gnomon" or square of 24 squares

This means that:

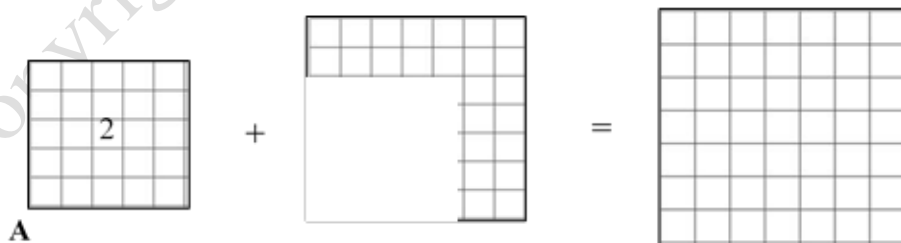


Fig.19: A square of 16 grids plus a "gnomon" of 9 grids results in a larger square of 25 grids

3.6 Example of application of the “gnomon” in $a^2 + b^2 = c^2$

In the following example, we make use of the concept of "gnomon" and the first Pythagorean triple 3, 4, 5, the sides of the Egyptian triangle.

The procedure is described below

1. With the quadricle of $3 \times 3 = 9$ squares, we convert it into a "gnomon" or square of 5×5 , also with 9 squares like this:

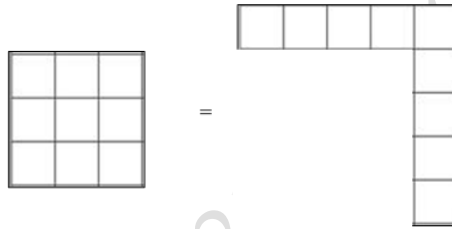


Fig.20: Square of 9 squares equal "gnomon" of $(5 + 4)$ squares

2. We add the square $4 \times 4 = 16$ to the 5×5 square and the result is a 5×5 square, as seen in the figure.

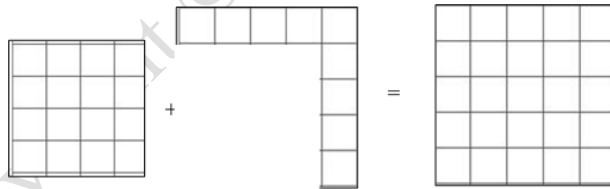


Fig.21: Square of $16 +$ "gnomon" of 9, equal to the larger square (25 grids)

Therefore the "gnomon" obtained from the square of the minor side a , added to the median square, of side b , fulfill the Pythagorean Theorem:

$$a^2 + b^2 = c^2.$$

In this case: $3(3) + 4(4) = 5(5)$; $3^2 + 4^2 = 5^2$; $9 + 16 = 25$

3.7 Comments on the proofs of $a^2 + b^2 = c^2$

Finally, I present this graph and some comments on the proofs of the Pythagorean Theorem.

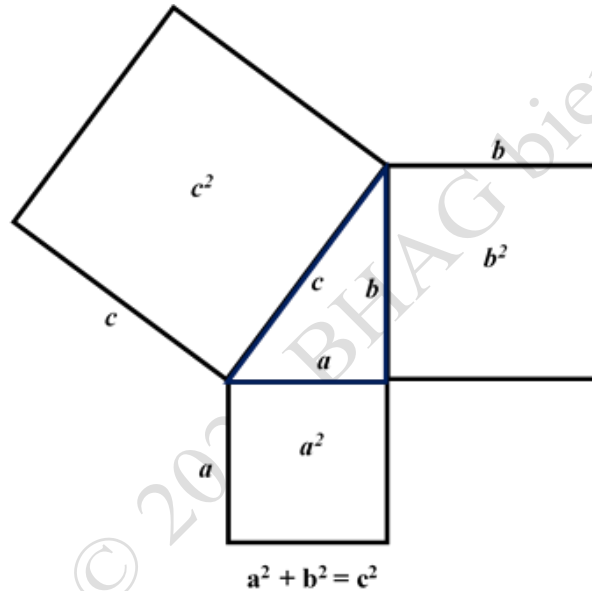


Fig. 22: Final graph of the Pythagorean Theorem

«There are about 400 known proofs of the Pythagorean Theorem, more than for any other mathematical theorem. People of very different backgrounds - including a Babylonian magician, a 14-year-old Ohio student of normal intellectual level, and the 21-year-old mathematical genius Galois, who was killed in a duel, have devised these tests.

A similar fate fell to another famous author of a proof of the Pythagorean Theorem, James A. Garfield, who became President of the United States in 1881, and was shot only three months after taking office. (Strathern, 2021). »

3.8 Pythagorean triads. The way to get those.

Regarding the Pythagorean triples, it is said that, «If (x, y, z, k) are integers and (x, y, z, k) satisfy $x^2 + y^2 = z^2$, $z = (y + k)$, then there are infinite Pythagorean triples with different configuration, as shown below:

$$[x, y, (y + 1)], [x, y, (y + 2)], [x, y, (y + 8)], [x, y, (y + k)] \text{ »,}$$

(Monografias.com, 2021)

Another source says, "Pythagoras himself is credited with the discovery of Pythagorean triplets, that is, triads or triples of numbers that fulfill the equality: $a^2 + b^2 = c^2$ " (Strathern, 2021).

Another Author also says, «The formula to find Pythagorean triplets is the following:

$$\begin{aligned} a &= n \\ b &= \frac{1}{2}(n^2 - 1) \\ c &= \frac{1}{2}(n^2 + 1). \end{aligned} \text{ As long as } n \text{ is odd}$$

Three numbers that correspond to the measures of a right triangle, is called a Pythagorean triad or triple [...], (3, 4, 5). It is a Pythagorean triple, since, $3^2 + 4^2 = 5^2$. The Pythagoreans found a method to find an infinity of triples, like this; If $n \in \mathbf{N}$, $n > 1$ is odd, then the numbers

(The symbol $\in$ reads, "belongs to"); $\mathbf{N}$: Natural numbers

$$\begin{aligned} x &= n; \\ y &= \frac{1}{2}(n^2 - 1), \\ z &= \frac{1}{2}(n^2 + 1) \end{aligned}$$

They satisfy the equation: $x^2 + y^2 = z^2$. Therefore, (x, y, z) is a Pythagorean triple, with $z = y + 1$ » (Jimenez Santamaria, 2012).

Application examples to find Pythagorean triples

n, impar	x=n	$y = \frac{1}{2}(n^2 - 1)$	$z = \frac{1}{2}(n^2 + 1)$	Terna (x, y, z)
3	3	$y = (3^2 - 1)/2 = 4$	$z = (3^2 + 1)/2 = 5$	(3, 4, 5)
5	5	$y = (5^2 - 1)/2 = 12$	$z = (5^2 + 1)/2 = 13$	(5, 12, 13)
7	7	$y = (7^2 - 1)/2 = 24$	$z = (7^2 + 1)/2 = 25$	(7, 24, 25)
9	9	$y = (9^2 - 1)/2 = 40$	$z = (9^2 + 1)/2 = 41$	(9, 40, 41)
11	11	$y = (11^2 - 1)/2 =$	$z = (11^2 + 1)/2 = 61$	(11, 60, 61)
13	13	$y = (13^2 - 1)/2 = 84$	$z = (13^2 + 1)/2 = 85$	(13, 84, 85)
15	15	$y = (15^2 - 1)/2 =$	$z = (15^2 + 1)/2 = 113$	(15, 112, 113)

Table of Pythagorean Triples with n odd

The philosopher, Plato, found the general formula that models triples of the form (8, 15, 17), (12, 35, 37), (16, 63, 65) ..., $n \in \mathbb{N}$: "ene" is a Natural number (those that serve to count). If $n \in \mathbb{N}$ then the numbers defined by:

$$\begin{aligned}x &= 4n \\y &= 4n^2 - 1 \\z &= 4n^2 + 1,\end{aligned}$$

They satisfy $x^2 + y^2 = z^2$. Therefore, we conclude that the triple (x, y, z) is Pythagorean with $z = y + 2$ »

n	x=4n	$y = 4n^2 - 1$	$z = 4n^2 + 1$	Terna (x, y, z)
1	4	$y = 4(1)^2 - 1 = 3$	$z = 4(1)^2 + 1 = 5$	(4, 3, 5)
2	8	$y = 4(2)^2 - 1 = 15$	$z = 4(2)^2 + 1 = 17$	(8, 15, 17)
3	12	$y = 4(3)^2 - 1 = 35$	$z = 4(3)^2 + 1 = 37$	(12, 35, 37)
4	16	$y = 4(4)^2 - 1 = 63$	$z = 4(4)^2 + 1 = 65$	(16, 63, 65)
5	20	$y = 4(5)^2 - 1 = 99$	$z = 4(5)^2 + 1 = 101$	(20, 99, 101)
6	24	$y = 4(6)^2 - 1 = 143$	$z = 4(6)^2 + 1 = 145$	(24, 143, 145)
7	28	$y = 4(7)^2 - 1 = 195$	$z = 4(7)^2 + 1 = 197$	(28, 195, 197)

Table of Pythagorean Triples with n a natural number.

« Euclid 300 BC, discovered the method to find all Pythagorean triples. However, he did not make the general demonstration.

Euclid's Theorem

Let m, n, r , be arbitrary positive integers, $m > n$ where m and n are relative primes (without common factors), one of them is even and the other is odd, then the integers:

$$x = r (m^2 - n^2)$$

$$y = 2r mn$$

$$z = r (m^2 + n^2)$$

They satisfy the equality $x^2 + y^2 = z^2$.»

(Jimenez Santamaria, 2012)

Illustrating the previous equalities, I present the following graph:

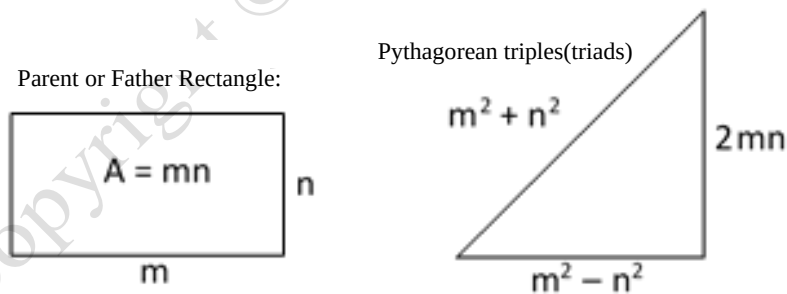


Fig. 23: Rectangle that originates Pythagorean triples.

Parent or Father Rectangle: Is the one that gives rise to the Pythagorean triples. Here, m is greater than n , if m is even; n is odd and vice versa, if one is even, the other is odd. They can be consecutive or not.

We illustrate the obtaining of a primitive Pythagorean triple with the "parent rectangle" of the first natural numbers 1, 2, which give rise to the triple 3, 4, 5 the sides of the Egyptian triangle.

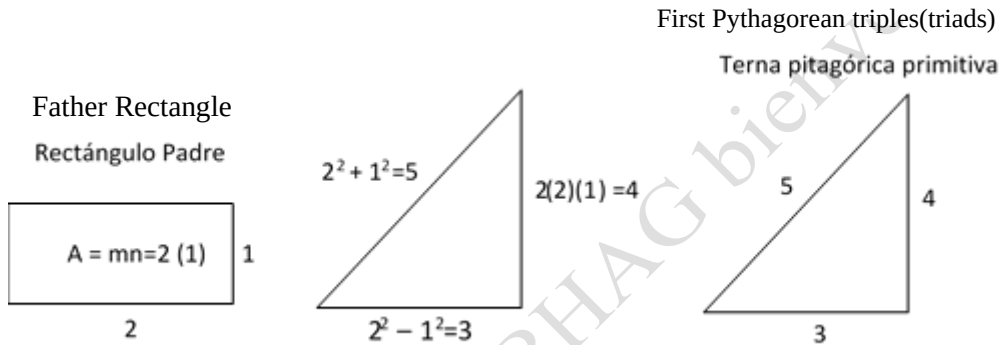


Fig. 24: Obtaining the first Pythagorean triples (triads)

3.9 Pythagorean triads in ancient times.

The Babylonian Plimpton tablet with numbers that satisfy the Pythagorean Theorem.



Babylonian tablet from Plimpton 322 (1800-1900 B.C). (Rotger, 2011)

Tablets with Pythagorean triples, from the Plimpton tablet.

a	b	c
119	120	169
3367	3456	4825
4601	4800	6649
12709	13500	18541

(Rotger, 2011)

Chapter 4: Reading, Basic Geometry and Number Sets.

4.1 What it means to know how to read.

What does it mean to know how to read? I am going to explain to you what I understand by "knowing how to read". Knowing how to read is not knowing the words and pronouncing them, whether you read aloud or softly. Knowing how to read means, you can do the following:

1) "Know the words", that is, how they are pronounced and what they mean. Therefore, when you read you should have a good dictionary at hand, in case you find an unknown word, you can look it up immediately, with this you guarantee that you understand the sentence you read, understand?

2) "Record or remember" in your mind what you have just read. Remember what you are reading, making an "image in your mind" of what the paragraph or sentence you read says. If for example, the sentence says, "the church bells rang ", you imagine the church, look at the bell tower in the church tower, imagine the person who is pulling the rope of the bell and hear in your mind the sound of the bell, did you copy me?

3) "Explain or tell" other people what you have read. The mother of learning is "repetition". When you read something, it is good to repeat mentally what you are reading. If after reading some content, of any document, you are able to explain it in your own words to someone, or to yourself, it means that you "know how to read".

In other words, "knowing how to read," means "learning"; therefore, the one who knows how to read, "learns", takes possession of the knowledge. He becomes wise and does well in life, especially if what he reads is the Bible.

4.2 Concepts and terms of basic geometry.

Having written the above, I move on to the part in which it is required to know the meaning of some words related to mathematics and geometry; the latter I say is the "science of forms", although it literally means "measurement of the earth".

As your reading progresses and we move forward, it is necessary to explain some concepts, which are related to the Pythagorean Theorem. In some cases, there are no precise definitions, but I will try to present you with a description, which allows you to make a mental image and makes it easier for you to understand what you are reading.

Following are the terms:

Mathematical object: "A mathematical object is an abstract object, studied in mathematics. Some typical examples of mathematical objects are numbers, sets, functions, and figures." They are mental images, they have no material existence; therefore, they are immaterial. They are abstractions of the human mind. (Wikipedia.org, Mathematical object - Wikipedia, the free encyclopedia, 2021).

Axiom: It is a grammatical sentence, proposition or statement so evident that it is considered not requiring proof. It comes from a Greek noun that means, "What seems fair" or, which is considered obvious, without the need for proof. "The whole is greater than the part" is an axiom. (Wikipedia.org, 2021).

Theorem: It is a formula, a demonstrable mathematical statement or proposition (which must be proved as true) from axioms or already proven propositions, within a formal system. (Ecured.cu, 2021).

Point: "Point is what has no parts". It is the D.I.1 definition obtained from Euclid's Elements (Gonzalez Urbaneja, 2017). The starting concept of the geometric teachings (of Pythagoras) is the point, which for him was the simplest thing that existed; it is, he said: "the unit that has a position". (Baldor D., 1998). What I understand and explain to you is that "a point" is the smallest imaginable part of which any geometric figure is composed. It is like the elementary particle of geometric shapes. It is the drawing or graph of the smallest mathematical object known in geometry. For example, at the end of this sentence you will see a point, indicating that this sentence ends here.

Imagine that you are going to write with a graphite pencil, but it is new, what you do is remove the tip with a sharpener or pencil sharpener. Now, if after sharpening it, you click on a sheet of paper that is empty, there is a mark, that mark represents a point. As shown below, but amplified:



Fig.25: Graph of a point.

Line: "It is the length without width". It is the definition D.I. 2, obtained from the Elements of Euclides (Gonzalez Urbaneja, 2017). "Continuous succession of points in space." "Curved line" "Long, narrow mark or mark that is made or formed on a body or surface." "Follow the line on the ground." (Oxford languages, 2021).



Fig.26: Graph of a curved line.

In geometry, a line is a continuous succession of points, closely linked, and that extend indefinitely, forming a path in the same plane. The lines can be straight or curved, and their size can be adjusted; hence, they are used for the creation or construction of figures. (Significados (Meanings), 2021).

To illustrate that it is a curved or straight line, grab a piece of thread with the index finger and thumb of the right hand and the left without tightening it, what you see of the thread represents a curved line. Now, tension or pull the thread from the ends, what you see is a tensioned thread. This gives us the idea of what a straight line is.



Fig.27: Curved line and straight-line

Straight line: in the realm of geometry, a line is an indefinite and continuous succession of points. The adjective straight, meanwhile, refers to that which has no angles or curves. (<https://www.bienvenue.be/2021/01/20/definition-dune-ligne-droite/>, 2021). In Euclidean geometry, the line or the straight line is a line that extends in the same direction; therefore, it has only one dimension and contains an infinite number of points, extended in only one direction. (Wikipedia.org, 2021).

Ray: A Ray is each of the two parts into which a Line is divided when it is cut at one of its points. (Matematicas10.net, 2021).

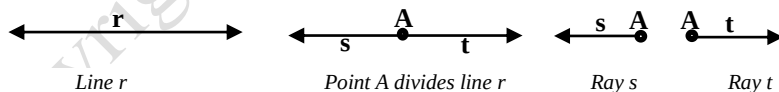


Fig.28: Ray s and t

Line segment: In geometry, the Segment is a fragment of the line that is between two points, called extreme or ends. (Wikipedia.org, 2021)

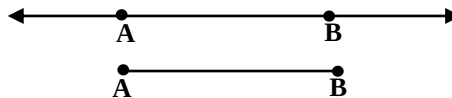


Fig.29: AB Segment

Horizontal straight line: It is that straight line that coincides with the horizon line. Simulates the surface on which we walk or lie. (Glosarios.com, 2021).



Fig.30: Horizontal line

It is the one that we draw according to the cardinal points from east to west or vice versa, from right to left or from left to right. (Arguello, 2021).

Vertical straight line: We draw the one according to the cardinal points from north to south or vice versa, from bottom to top or from top to bottom. (Arguello, 2021)



Fig.31: Vertical line.

Slanted straight line: An inclined line is any line that is neither horizontal nor vertical. (Glosarios.com, 2021).

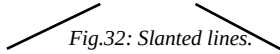


Fig.32: Slanted lines.

Angle: "Angle" is the amplitude between two lines of any type that meets at a common point called the vertex.

(Wikipedia.org, 2021). It is an opening between two lines. The width or opening is measured in degrees; the symbol is $^{\circ}$, example 3° , it reads three degrees.

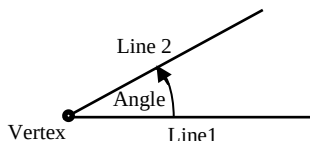


Fig.33: The Angle

The angles are identified with Greek letters such as: alpha(α), beta(β), gamma (γ), etc. According to the size of the opening, they receive the following names:

Right Angle: When the opening is 90° and its sides or lines are always perpendicular.

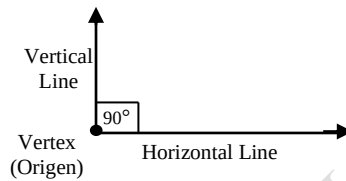


Fig.34: The Right Angle

Acute angle: Measures less than 90° and more than 0° .

Blunt angle: Greater than 90° but less than 180°

Plain angle: Measures 180° .

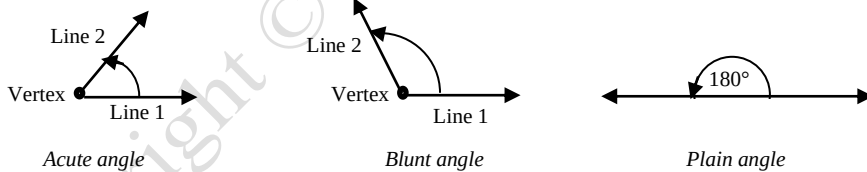


Fig.35: Types or classes of angles

4.3 The basic geometric figures and some definitions.

The basic geometric figures or shapes that we will use in the development of this chapter are the horizontal line segment, the vertical line segment, the slanted line segment, the right triangle, the rectangle and the square. Which are shown below:

1. Segments: Pieces of straight lines.

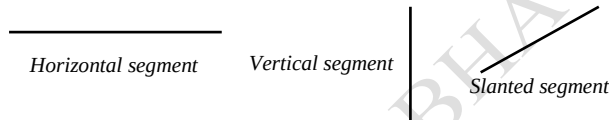


Fig.36: Types of line segments

2. Triangle: Figure (polygon) formed by three-line segments that join to form a three-sided figure. The segments can be measured, and we will say that a , b and c are the measurements of each side of the triangle. Inside the triangle, three different angles are formed that we denote as α (α), β (β) and γ (γ), it is true that: $\alpha + \beta + \gamma = 180^\circ$

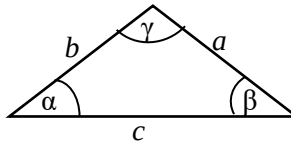
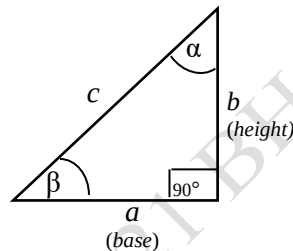


Fig.37: Triangle in general

For convenience, side a is the side opposite angle α (alpha), side b opposite angle β (beta), and side c opposite angle γ (gamma). In every triangle c is the longest side, but $a + b$ is greater than c . In this book, a will be less than b and c is greater than each of them is ($a < b < c$).

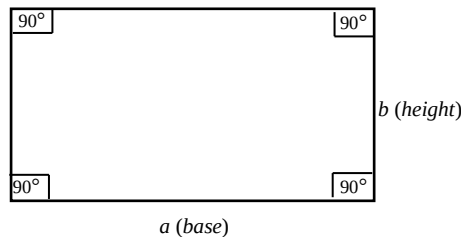
3. **Right angle Triangle:** Figure (polygon) with three sides, the first side is horizontal (a ; it is called the *base* of the triangle, the second vertical (b), is called the *height* of the triangle and the third is the inclined side (c), called the *hypotenuse*. When the three sides come together, they form what is called a *right-angle triangle*, with the following characteristic: two of its sides are perpendicular, that is, they form a right angle between them, which has 90° ; it is satisfied that: $\alpha + \beta = 90^\circ$. The hypotenuse c is the side opposite the right angle (90°).



$$A = \frac{1}{2} ab$$

Fig.38: Right angle Triangle

4. **Rectangle:** Figure (polygon) with four sides, two horizontal sides that measure the same (a), and two vertical sides that measure the same (b); when joined they form what is called a rectangle with one characteristic: four corners that are formed are perpendicular, that is, they form four right angles between them, as to say, four openings of 90° each.



$$A = ab$$

Fig.39: The Rectangle

5. **Square:** Polygon with four equal sides, measuring the same (a), with one characteristic: they are perpendicular, that is, they form four right angles between them, as to say, four openings of 90° each.

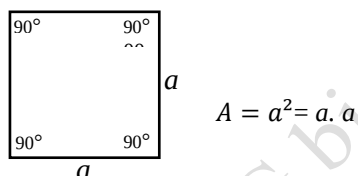


Fig.40: The square from side a

6. **Area:** «The area is the measure of a space delimited by a contour which is called the perimeter. The term surface or area is often used interchangeably in some cases, but the first refers to space, while the second refers to its measurement. That is, *area is the measurement of a surface* (Westreicher, 2020).
7. **Area of a Triangle:** [...]to calculate the area of a triangle. First, you can multiply the base (which can be either side) by the height and divide by two (we must remember that the height is the segment that joins the vertex with its opposite side, forming a 90° angle). (Westreicher, 2020)

$$A = \frac{1}{2}ab$$

8. **Area of a Rectangle:** The lengths of two contiguous sides of the figure, which are different from each other, are multiplied, Height by width.

$$A = ab$$

9. **Area of a Square:** The length of any of the sides (a) of the figure is squared (all the sides are equal).

$$A = a^2$$

4.4 A summary of the number sets.

In this section, the book will summarize the numerical sets that are studied in mathematics. The objective is to familiarize the reader with the terms and vocabulary referred to numerical sets and can be related to the Pythagorean Theorem. (Wrede & Spiegel, 2002) the following is obtained:

«The Set of Real Numbers

The following types of numbers are already known:

Natural Numbers:

1, 2, 3, 4 ... they are also called *positive integers* used to count the elements or members of a set. The symbols have varied over time, for example, the Romans used I, II, III, IV.... The sum $a + b$ and the product $a.b$ or ab of any two natural numbers a and b Is a natural number. It is said that the set of natural numbers is *closed* under the operations of *addition* and *multiplication*. (Wrede & Spiegel, 2002).

Negative Integers and Zero:

Denoted by -1, -2, -3, and 0, respectively, they allow solving equations of the type $x + b = a$, where a and b are natural numbers. This brings us to the operation of subtraction or inverse of addition. We write this as $x = a - b$. Positive and negative integers and zero form the set of integers. (Wrede & Spiegel, 2002).

Rational numbers or fractions:

Since $2/3$, $-5/4$... allow solving equations of the form $bx = a$ for all integers a and b , where $b \neq 0$ (the symbol $\neq$ is read, '*it is different from*'). This leads us to the division operation, or the inverse of multiplication, which we write as $x = \frac{a}{b}$ or $a \div b$ where a is the *numerator* and b the *denominator*. The set of *integers* is a subset of rational numbers, because integers are rational numbers in which $b = 1$. (Wrede & Spiegel, 2002).

Irrational numbers:

Such as $\sqrt{2}$, and π are numbers that are not rational, because it cannot be expressed as $\frac{a}{b}$ (called the *quotient* of a and b), where a and b are integers, where $b \neq 0$.

The set of rational and irrational numbers is called the set of real numbers. » (Wrede & Spiegel, 2002).

The Complex Number System:

It is said: «Equations like $x^2 + 1 = 0$, has no solution in the real number system. [...], Mathematicians of the late nineteenth and early twentieth centuries developed an extension to the real number system [...]

The new number system came to be known as the complex number system, which includes the real numbers as a subset of this system. Complex numbers have the form $a + bi$, where a and b are real numbers; $i = \sqrt{-1}$, is the *imaginary unit*. » (Wrede & Spiegel, 2002). »

Here are some notable products and factoring.

Chapter 5: Notable Products and Factoring.

Notable products are the result of "algebraic multiplication." Algebraic multiplication "is the operation that aims to determine an algebraic expression called product, given other algebraic expressions called multiplying and multiplier; the equality obtained is an identity". (Issuu.com, 2019).

«Also called "algebraic identities" » (Editors. Lexus. 2008).

Following are the most important notable products with the factors that originate it (factoring).

5.1 Reference 1: Fundamentals of Higher Mathematics.

In this reference, six (6) notable algebraic identities appear. The factors (those that are multiplied) are written on the right side of the equality, while the products (the result to be factored) on the left side:

1. $ab + ac - ad = a(b + c - d)$
2. $acx^2 + (ad + cd)x + bd = (ax + b)(cx + d)$
3. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
4. $a^2 \pm 2ab + b^2 = (a \pm b)^2$
5. $a^2 - b^2 = (a + b)(a - b)$
6. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

(Ayres,Jr.,Ph. D., 1969)

As seen, there is no expression that shows the product of two factors resulting in the sum of two squares $a^2 + b^2$ when multiplied.

5.2 Reference 2: Algebra. Pre-university preparation manual.

(Editores.Lexus.S.A., 2008) mentions the following notable products:

- I. Square of a sum and a difference
 - $(a + b)^2 = a^2 + 2ab + b^2$
 - $(a - b)^2 = a^2 - 2ab + b^2$

In general: $(a \pm b)^2 = a^2 \pm 2ab + b^2$
- II. Product of a sum and a difference
It is equal to the difference of a square

$$(a + b)(a - b) = a^2 - b^2$$
- III. Square of the Trinomial.

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$
- IV. Cube of a sum or difference

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

In general: $(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$
- V. Product of two binomials that have a common term.

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$
- VI. Product of a binomial by a trinomial that gives a sum
Or a difference of cubes.

$$(a + b)(a^2 - ab + b^2) = a^3 + b^3$$

$$(a - b)(a^2 + ab + b^2) = a^3 - b^3$$

In general: $(a \pm b)(a^2 \mp ab + b^2) = a^3 \pm b^3$
- VII. Legendre Identities:

$$(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$$

$$(a + b)^2 - (a - b)^2 = 4ab$$
- VIII. Lagrange Identities:

$$(ax + by)^2 + (bx - ay)^2 = (x^2 + y^2)(a^2 + b^2)$$

$$(ax + by + cz)^2 + (bx - ay)^2 + (cx - az)^2 + (cy - bz)^2$$

$$= (a^2 + b^2 + c^2)(x^2 + y^2 + z^2)$$

5.3 Reference 3: Pre-university Algebra digital book.

(Issuu.com, 2019) presents the notable products that are considered most relevant.

Notable Products or Algebraic Identities.

1. Square Binomial

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$

Note: $(a - b)^2 = (b - a)^2$

in general $(a - b)^{2m} = (b - a)^{2m}$; $m \in \mathbb{Z}$

2. Legendre Identities:

- $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
- $(a + b)^2 - (a - b)^2 = 4ab$

3. Difference of Squares: $(a + b)(a - b) = a^2 - b^2$

4. Binomial cubed

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 \quad \text{ó}$$

$$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

Cauchy's identity

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 \quad \text{ó}$$

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

5. Steven Identities

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

$$(x + a)(x + b)(x + c) = x^3 + (a + b + c)x^2 + (ab + ac + bc)x + abc$$

6. Sum and difference of cubes

$$(a + b)(a^2 - ab + b^2) = a^3 + b^3$$

$$(a - b)(a^2 + ab + b^2) = a^3 - b^3$$

7. Trinomial squared

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

8. Trinomial cubed

$$(a + b + c)^3 = a^3 + b^3 + c^3 + 3a^2b + 3a^2c + 3b^2c + 3b^2a + 3c^2a + 3c^2b + 6abc \quad \text{ó}$$

$$(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(a + c)(b + c)$$

5.4 Reference 4: Higher Algebra Schaum. Special Products.

(Spiegel & Moyer, 2006). About these Special Products, the following have been written:

Special products.

“Here are some of the products that are frequently presented in mathematics and the student should become familiar with them as soon as possible. The proofs of these results can be obtained through multiplication”.

- I. Product of a monomial and a binomial:

$$a(c + d) = ac + ad$$

- II. Product of the sum and the difference of two terms:

$$(a + b)(a - b) = a^2 - b^2$$

- III. Squared of a binomial:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

- IV. Product of two Binomials:

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

$$(ax + b)(cx + d) = acx^2 + (ad + cd)x + bd$$

$$(a + b)(c + d) = ac + bc + ad + bd$$

- V. Cube of a Binomial:

$$\bullet (a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$\bullet (a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

- VI. Squared of a Trinomial:

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

5.5 Reference 5: Products with response of the form $a^n \pm b^n$

(Spiegel & Moyer, 2006), present products that generate a binomial in which each of its terms has an exponent "n".

Following are the factors that are multiplied and the resulting binomial.

It can be verified by means of multiplication that:

$$(a - b)(a + b) = a^2 - b^2$$

$$(a - b)(a^2 + ab + b^2) = a^3 - b^3$$

$$(a - b)(a^3 + a^2b + ab^2 + b^3) = a^4 - b^4$$

$$(a - b)(a^4 + a^3b + a^2b^2 + ab^3 + b^4) = a^5 - b^5$$

The rule is clear. These expressions can be summarized as:

$$(a - b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + ab^{n-2} + b^{n-1}) = a^n - b^n$$

Where "n" is any positive integer (1, 2, 3,4...)

$$(a + b)(a^2 - ab + b^2) = a^3 + b^3$$

$$(a + b)(a^4 - a^3b + a^2b^2 - ab^3 + b^4) = a^5 + b^5$$

$$(a + b)(a^6 - a^5b + a^4b^2 - a^3b^3 + a^2b^4 - ab^5 + b^6) = a^7 + b^7, \text{ etc.}$$

The rule is clear. These expressions can be summarized as:

$$(a + b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \dots - ab^{n-2} + b^{n-1}) = a^n + b^n$$

Where "n" is any positive odd integer (1, 3, 5,7...),

Given the restrictions for the value of "n", there is no way to find factors that when multiplied, result in a binomial that is the sum of two $a^2 + b^2$.

5.6 Reference 6: From Basic Algebra. Theory and Practice.

(González López, 2011), from the topic Notable products states the following:

1. Square binomial (Perfect square trinomial)

$$(x + y)(x + y) = (x + y)^2 = x^2 + 2xy + y^2$$

$$(x - y)(x - y) = (x - y)^2 = x^2 - 2xy + y^2$$

$$(ax + by)(ax + by) = (ax + by)^2 = a^2x^2 + 2abxy + b^2y^2$$

$$(ax - by)(ax - by) = (ax - by)^2 = a^2x^2 - 2abxy + b^2y^2$$

2. Trinomial Squared

$$(x + y + z)(x + y + z) = (x + y + z)^2$$

$$= x^2 + y^2 + z^2 + 2xy + 2xz + 2yz$$

$$(ax + by + cz)(ax + by + cz) = (ax + by + cz)^2$$

$$= a^2x^2 + b^2y^2 + c^2z^2 + 2abxy + 2acxz + 2bcyz$$

3. Binomial Cube

$$(x + y)(x + y)(x + y) = (x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x - y)(x - y)(x - y) = (x - y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$$

$$(ax + by)(ax + by)(ax + by) = (ax + by)^3$$

$$= x^3 + 3x^2y + 3xy^2 + b^3y^3$$

$$(ax - by)(ax - by)(ax - by) = (ax - by)^3$$

$$= x^3 - 3x^2y + 3xy^2 - b^3y^3$$

4. Trinomial Cubed

$$(x + y + z)(x + y + z)(x + y + z) = (x + y + z)^3$$

$$= x^3 + y^3 + z^3 + 3(x + y)(y + z)(x + z)$$

$$(x + y + z)^3 = x^3 + y^3 + z^3 + 3(x + y + z)(xy + yz + xz) - 3xyz$$

$$(x + y + z)^3 = x^3 + y^3 + z^3 + 3(x + y + z)(xy + yz + xz) - 3xyz$$

5. Product of a sum and its difference (Difference of squares)

$$(x^n + y^n)(x^n - y^n) = (x^n)^2 - (y^n)^2 = x^{2n} - y^{2n}$$

$$(x + y)(x - y) = x^2 - y^2$$

$$(ax + by)(ax - by) = a^2x^2 - b^2y^2$$

6. Product of a binomial by a trinomial that gives a sum or difference of cubes.

$$(x^n + y^n)(x^{2m} - x^m y^n + y^{2n}) = x^{3m} + y^{3n}$$

$$(x^n - y^n)(x^{2m} + x^m y^n + y^{2n}) = x^{3m} - y^{3n}$$

$$(x + y)(x^2 - xy + y^2) = x^3 + y^3$$

$$(x - y)(x^2 + xy + y^2) = x^3 - y^3$$

$$(ax + by)(a^2 x^2 - abxy + b^2 y^2) = (ax)^3 + (by)^3 = a^3 x^3 + b^3 y^3$$

$$(ax - by)(a^2 x^2 + abxy + b^2 y^2) = (ax)^3 - (by)^3 = a^3 x^3 - b^3 y^3$$

7. Argand Identity

$$(x^{2m} + x^m y^n + y^{2n})(x^{2m} - x^m y^n + y^{2n}) = (x^{4m} + x^{2m} y^{2n} + y^{4n})$$

m y n: número par

$$(x^{2m} + x^m y^n + y^{2n})(x^{2m} - x^m y^n + y^{2n}) = (x^{4m} + x^{2m} y^{2n} + y^{4n})$$

$$(x^{2k} + x^k + 1)(x^{2k} - x^k + 1) = (x^{4k} + x^{2k} + 1)$$

$$(x^2 + xy + y^2)(x^2 - xy + y^2) = (x^4 + x^2 y^2 + y^4)$$

8. Product of a binomial with a common term

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

$$(x + a)(x + b)(x + c) = x^3 + (a + b + c)x^2 + (ab + ac + bc)x + abc$$

9. Product of a binomials of the form $(ax + b)(cx + d)$

$$(ax + b)(cx + d) = acx^2 + (ad + cd)x + bd$$

10. Legendre Identities

$$(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$$

$$(x + y)^2 - (x - y)^2 = 4xy$$

$$(ax + by)^2 + (ax - by)^2 = 2(a^2 x^2 + b^2 y^2)$$

$$(ax + by)^2 - (ax - by)^2 = 4abxy$$

11. Lagrange Identities

$$(ax + by)^2 + (ay - bx)^2 = (a^2 + b^2)(x^2 + y^2)$$

$$(ax + by + cz)^2 + (ay - bx)^2 + (az - cx)^2 + (bz - cy)^2 \\ = (a^2 + b^2 + c^2)(x^2 + y^2 + z^2)$$

12. Complementary Identities

1. Conditionals:

Sí $a + b + c = 0$ se verifica que:

- $a^2 + b^2 + c^2 = -2(ab + bc + ac)$
- $(ab + bc + ca)^2 = a^2b^2 + b^2c^2 + c^2a^2$
- $a^3 + b^3 + c^3 = 3abc$
- $a^4 + b^4 + c^4 = 2(a^2b^2 + a^2c^2 + b^2c^2)$
- $(a^2 + b^2 + c^2)^2 = 2(a^4 + b^4 + c^4)$
- $a^5 + b^5 + c^5 = -5abc(ab + bc + ac)$

2. $(a + b)^4 - (a - b)^4 = 8ab(a^2 + b^2)$

3. Gauss Identities

$$(a + b + c)(a^2 + b^2 + c^2 - ab - ac - bc) = a^3 + b^3 + c^3$$

$$(a + b + c)(ab + bc + ca) = (a + b)(b + c)(c + a) + abc$$

5.7 Reference 7: Manual for Engineers and Technicians.

(McNeese & Hoag, 1959) mentions, on page 157, the following notable products:

1. $(a \pm b)^2 = a^2 \pm 2ab + b^2$
2. $(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$
3. $(a \pm b)^4 = a^4 \pm 4a^3b + 6a^2b^2 \pm 4ab^3 + b^4$
4. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$
5. $a^2 - b^2 = (a + b)(a - b)$
6. $a^2 + b^2 = (a + bi)(a - bi) *$
7. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
8. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
9. $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$
10. $a^n - b^n = (a + b)(a^{n-1} - a^{n-2}b + \dots - b^{n-1})$ where n is even;
11. $a^n + b^n = (a + b)(a^{n-1} - a^{n-2}b + \dots + b^{n-1})$ where n is odd

*This factorization is in complex numbers, where $i^2 = -1$. In the set of real numbers, $a^2 + b^2$ is irreducible, that is, it cannot be factored.

Chapter 6: “What is said about factoring $a^2 + b^2$?”

In this chapter, we are going to review what some literature sources say about factoring the sum of two squares.

6.1 What is said in Euclid's Book of Elements.

(Gonzalez Urbaneja, 2017), states that, «In the first ten propositions, Euclides establishes the geometric equivalence of the main algebraic identities very common in school practice. » «Proposition II.7. If a line is divided in any way, the square of the entire line and that of one of its parts, taken together, are equal to twice the rectangle comprised by the entire line and the part considered, plus the square of the other part.

$$\text{II.7. } (a + b)^2 + a^2 = 2(a + b) a + b^2, \text{ o } a^2 + b^2 = 2a.b + (a - b)^2»$$

(Navarro Loidi, 2002) says, «Euclides presents all these propositions as geometric questions. For example, the statement of proposition II.9 is: “9 if a straight line is cut into equal and unequal parts; the squares of the unequal segments of the whole (line) are twice the square of the middle plus the square of the line between the points of this section. "[Puertas, 1991.v.1, p. 279] »

The previous statement is interpreted as the sum of two squares:

$$a^2 + b^2 = 2 \left[\left(\frac{a+b}{2} \right)^2 + \left(\frac{a+b}{2} - b \right)^2 \right].$$

You can see that none of these sums of squares is written as a product of polynomial factors.

6.2 Analysis of the Elements. A walk-through geometry.

(Navarro Loidi, 2002) referring to the Second Book of Euclid says, «The second book begins with the definitions of rectangle and "gnomon", which is a kind of L, obtained by removing another similar smaller rectangle at a vertex. The first eleven initial propositions try to relate the area of some squares or rectangles that have given segments on their sides with the surface of other quadrilaterals that have additions or subtractions of said segments on their sides. Propositions 12 and 13 equal the property of the sides of a triangle that is now known as "the cosine theorem." [...].

The first eleven propositions of this Book II could be considered algebraic properties, if instead of segments in it, one spoke of quantities [...] the propositions could be indicated as:

$$\text{II.1 } a(b+c+d) = ab+ac+ad$$

$$\text{II.2 } (a+b)a + (a+b)b = (a+b)^2$$

$$\text{II.3 } (a+b)a = a^2+ab$$

$$\text{II.4 } (a+b)^2 = a^2+b^2+2ab$$

$$\text{II.5 } ab + \left(\frac{a+b}{2} - b\right)^2 = \left(\frac{a+b}{2}\right)^2$$

$$\text{II.6 } ab + \left(\frac{a+b}{2} - b\right)^2 = \left(\frac{a+b}{2}\right)^2$$

$$\text{II.7 } (a+b)^2 + a^2 = 2(a+b)a + b^2$$

$$\text{II.8 } 4(a+b)a + b^2 = [(a+b)+a]^2$$

$$\text{II.9 } a^2 + b^2 = 2\left[\left(\frac{a+b}{2}\right)^2 + \left(\frac{a+b}{2} - b\right)^2\right]$$

$$\text{II.10 } (2a+b)^2 + b^2 = 2[a^2 + (a+b)^2]$$

$$\text{II.11 Problem } a(a-x) = x^2$$

$$\text{II.12 y 13 Cosine theorem of trigonometry}$$

$$\text{II.14 Problem } x^2 = ab$$

6.3 Factor, factorization and the irreducible $a^2 + b^2$ polynomial.

(Swokowski, 1988), says:

«If a polynomial is written as a product of other polynomials, each polynomial in the product is a factor of the original polynomial. The process of expressing a polynomial as a product is called **factorization**. For example, since $x^2 - 9 = (x + 3)(x - 3)$, we see that $x + 3$ and $x - 3$ are factors of $x^2 - 9$.

Factorization plays an important role in many mathematical applications, since it allows the study of complicated expressions to be reduced to the study of several simple expressions. For example, important properties of the polynomial $x^2 - 9$ can be determined by doing an analysis of the factors $x + 3$ and $x - 3$.

An integer $a > 1$ is prime if it cannot be written as a product of two positive integers. In the same way, if S denotes a set of numbers, a polynomial with coefficients on S is said to be prime, or irreducible on S , if it cannot be written as a product of two polynomials of positive degree with coefficients on S . A polynomial can be irreducible on one set S , but not on another. For example, $x^2 - 2$, is irreducible over rational numbers since it cannot be expressed as a product of two polynomials that have rational coefficients.

If the factors can have real coefficients, then $x^2 - 2$, would not be prime and we would have: $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$.

Similarly, $x^2 + 1$ is irreducible over real numbers, [...] but not over complex numbers. [...]. It can be shown that every polynomial $ax + b$ of degree 1 is irreducible. Before factoring polynomials, it is necessary to specify the system from which the coefficients of the factors are to be chosen. »

6.4 The importance of factoring polynomials.

(Krick, 1995) tells us the following about the importance of factoring and solving polynomials:

«Constantly, when posing in mathematical terms problems of different areas (economics, physics, engineering, biology, etc.), the following question appears: trying to determine the zeros of certain functions, that is, values for which the function cancels out. » «After linear functions, polynomial functions (in 1 variable) are the simplest. Studying the zeros (roots) of polynomial functions is of great interest for at least the following two reasons: «You cannot claim to be able to solve the problem for more general functions if you cannot solve it in this simpler case. Many times, it is possible to translate in some way the original problem of finding zeros of any function to that of calculating the roots of certain polynomials (which "approximate" the original function). »

««Generally, for applications, one works with real functions, and it is about finding real zeros. Furthermore, due to the structure of the numbers with which computers work, functions usually have rational coefficients and the zeros that we will be able to calculate will be rational numbers that sufficiently approximate a true solution to the problem.

“In section 2.3. Factorization of polynomials, it is said: « $X^2 + 1$ is irreducible in $\mathbf{Q}[X]$ and $\mathbf{R}[X]$ because if it were not, it would be the product of two polynomials of degree 1, and, therefore, it would have roots in $\mathbf{Q}$ or $\mathbf{R}$ (As we saw before, every polynomial of degree 1 has a root). »

Q: is the set of rational numbers and **R**: is the set of real numbers.

6.5 When is a polynomial irreducible?

Regarding irreducible polynomials, we obtain the following information that says, «A non-constant polynomial $f(x) \in F(x)$ is irreducible over a field F if $f(x)$ cannot be expressed as a product of two polynomials $g(x)$ and $h(x)$ in $F(x)$. Here the degrees of $g(x)$ and $h(x)$ are both less than the degree of $f(x)$. Irreducible polynomials function as the "prime numbers" of rings of polynomials ».

«Example 17.11. The polynomial $x^2 - 2 \in \mathbb{Q}[x]$ is irreducible since it cannot be factored over rational numbers. Similarly, $x^2 + 1$ is **irreducible over the real numbers**. (Abstract.ups.edu, 2021) »

6.6 Why is the binomila $x^2 + 1$ said to be irreducible?

The following information was also obtained: «A polynomial $P(x)$ is said to be **IRREDUCIBLE** when it cannot be decomposed into a product of others of lesser degree than itself. Otherwise, it is said to be **REDUCIBLE**. »

««Examples

$P(x) = x^2 - 9$ is reducible since $P(x) = (x - 3)(x + 3)$

$P(x) = 4x^2 - 25$ is reducible since $P(x) = 4x^2 - 25 = (2x - 5)(2x + 5)$

$P(x) = x^2 + 1$ is **irreducible** since there are no first-degree polynomials $(ax \pm b)(cx + d)$ such that $P(x) = x^2 + 1 = (ax \pm b)(cx + d)$

$P(x) = x^3 - 1$ is reducible since $P(x) = x^3 - 1 = (x - 1)(x^2 + x + 1)$ »

(Studylib.es, 2021)

6.7 Factorization, prime factor, prime polynomial.

(Matematicasn.blogspot.com, 2021) gives us the following information about the prime polynomial that says:

«**Factorization.** It is a process of successive transformations of a polynomial in an indicated multiplication of irreducible polynomials in a numerical field. Generally, the set to be worked on is that of Rational numbers, unless otherwise indicated. »

«**Prime Factor:** A prime polynomial is called a polynomial of non-zero absolute degree that cannot be decomposed as an indicated multiplication of two or more polynomials of non-zero degrees. That is, a prime polynomial cannot be factored. A prime or irreducible factor is that prime polynomial that appears as a factor in an indicated multiplication.

Example

$$\text{At } x^2 - 1 = (x + 1)(x - 1)$$

The factor " $x + 1$ " is prime, because it can no longer be factored.

The most common prime factors are:

- First Grade:
x; y; $x + y$; $x - y$; $3x + 7$; etc.
- Second Grade:
 $x^2 + 1$; $x^2 + a^2$; $x^2 + x + 1$; $x^2 - x + 1$; $x^2 + xy + y^2$; $x^2 - xy + y^2$, etc»

6.8 What the Book digital- algebra pre-university saids.

(Issuu.com, 2019), on the subject of the prime factor says the following:

«Prime factor

That polynomial cannot be decomposed into other polynomials.

Example:

$x^2 - y^2 \rightarrow$ is not prime (can be decomposed).

$x^2 + y^2 \rightarrow$ is prime (cannot be decomposed).

Only non-prime polynomials can be factored »

6.9 What Geolay says about Factoring.

(Geolay.com, 2021), says:

«Decomposition of factors (Factorization)

Form $a^n \pm b^n$

$$1. a^2 - b^2 = (a + b)(a - b)$$

$$2. a^2 + b^2 = \text{Irreducible in } \mathbf{R}$$

TIPO $a^2 - b^2$

$$a) x^2 - 1^2 = x^2 - 1^2 = (x + 1)(x - 1)$$

$$b) 4x^2 - 16 = (2x)^2 - 4^2 = (2x - 4)(2x + 4)$$

TIPO $a^2 + b^2$

$$a) x^2 + 1 \rightarrow \text{Cannot factor in } \mathbf{R}$$

$$b) x^2 + 25 \rightarrow \text{Cannot factor in } \mathbf{R}$$

R: It is the symbol for the set of real numbers.

6.10 It is said that The Sum of Squares cannot be factored.

From the Important Facts section, it is said:

«Difference of two squares: $a^2 - b^2 = (a + b)(a - b)$ »

Observation: The sum of two squares $a^2 + b^2$ cannot be factored by means of the real numbers. » (Allen R., 2007)

6.11 The sum of squares is always prime.

«Factor $x^4 - 1$ »

$$x^4 - 1 = (x^2)^2 - 1^2 = (x^2 - 1)(x^2 + 1)$$

$$= ((x)^2 - (1)^2)(x^2 + 1)$$

$$= (x - 1)(x + 1)(x^2 + 1) »$$

« The answers to this exercise depend on the fact that 1, raised to any power, is always 1. »

« Note: Never forget that this formula is for the difference of squares; the sum of squares is always prime (that is, it cannot be factored). This is the case of $(x^2 + 1)$. » (Purplemath.com, 2021).

6.12 What is said in Algebra I. By Fernando Chamizo Lorente.

(Chamizo Lorente, 2002), says that: «Compared with $a^2 + b^2 = p$, the equation $a^2 - b^2 = p$ is very simple [...]. The difference between these two examples is based on the fact that $a^2 + b^2$ cannot be factored and $a^2 - b^2$, yes, which suggests creating rings that extend that of the integers and in which the given equation can be carried out, »

6.13 In the binomial $a^n + b^n$, if "n" is even, it cannot be factored.

(Aulafacil.com, 2021) tells us that:

«The sum or difference of two equal powers.

$$\frac{a^n + b^n}{a + b} = a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \dots + ab^{n-2} - b^{n-1}$$

only if "n" is odd

When clearing it, we have:

$$a^n + b^n = (a + b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \dots + ab^{n-2} - b^{n-1})$$

Where "n" must not be even, this means that $a^2 + b^2$ cannot be factored into the set of real numbers. »

(Issuu.com, 2019) from the de $a^n \pm b^n$, Factor section says that:

$a^n + b^n$, has $(a + b)$ as a factor if and only if n is an odd positive integer. Then:

$$a^n + b^n = (a + b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \dots - ab^{n-2} + b^{n-1})$$

6.14 The sum $a^n + b^n$ can only be factored if "n" is odd.

(Wapedia.mobi, 2021) (Wapedia.mobi, 2021) says, «First of all, it must be said that not every polynomial can be factored using real numbers. If complex numbers are considered, it can. There are factoring methods, for some special cases.

Sum of nth powers, yes and only if "n" is odd,

$$a^n + b^n = (a + b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \dots + b^{n-1})$$

It means that: $a^2 + b^2$, is not considered for this case.

6.15 The binomial $A^2 + B^2$ can be factored if there are common factors.

From (Barnett & Thomas J., 1986), the following information is obtained:

« If we multiply $(A - B)$ y $(A + B)$, we obtain: $= A^2 - B^2$
 $(A - B)(A + B) = A^2 - B^2$ which is the difference of two squares. Written in reverse order this result, we obtain a very useful factoring formula. If we try to factor the sum of two squares $A^2 + B^2$, we observe that it cannot be factored using integer coefficients, unless A and B have common factors”. »

«Sum of two Squares:

1. $A^2 + B^2$ cannot be factored using integer coefficients unless A and B Have common factors.

Difference of two Squares:

2. $A^2 - B^2 = (A - B)(A + B)$

Sum of Cubes:

3. $A^3 + B^3 = (A + B)(A^2 - AB + B^2)$

Difference of Cubes:

4. $A^3 - B^3 = (A - B)(A^2 + AB + B^2)$

Example No. 24

$$m^2 + n^2 = \text{It cannot be factored using integer coefficients. »}$$

The sum of two squares cannot be factored into the set of real numbers.

6.16 Factorization of $a^2 + b^2$ with the identity of Sophie Germain.

(Penalva Cutanda & Martinez Moncayo, 2021) says:

«Identity of Sophie Germain" states "Marie-Sophie Germain (1776-1831), French mathematician.

«Identity of Sophie Germain

$$x^4 + 4y^4 = (x^2 + 2y^2 + 2xy)(x^2 + 2y^2 - 2xy)»$$

(Baldor D., 2008), in the case of factoring a sum of squares as a special case, it tells us:

SPECIAL CASE

FACTORING THE SUM OF TWO SQUARES

«In general, a *sum of two squares* does **not have a decomposition into rational factors**, that is, factors in which there is no root; but there are sums of squares that, adding and subtracting the same quantity, can be taken to the previous case and decomposed. Example:

«Factor $a^4 + 4b^4$

The square root of a^4 is a^2 ; that of $4b^4$ is $2b^2$. For this expression to be a perfect square trinomial, its second term must be $2(a^2)(2b^2) = 4a^2b^2$. Then, as in the previous cases, we add and subtract $4a^2b^2$ from the expression $a^4 + 4b^4$ and we will have:»

$$\begin{array}{r}
 a^4 \qquad \qquad + 4b^4 \\
 + 4a^2b^2 \qquad - 4a^2b^2 \\
 \hline
 a^4 + 4a^2b^2 + 4b^4 - 4a^2b^2 = (a^4 + 4a^2b^2 + 4b^4) - 4a^2b^2 \\
 = (a^2 + 2b^2)^2 - 4a^2b^2 \\
 = (a^2 + 2b^2 + 2ab)(a^2 + 2b^2 - 2ab) \\
 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2) R»
 \end{array}$$

6.17 Factorization of Special Sums of Squares.

Other examples of factoring a sum of squares, which we transform into a perfect square trinomial and then write it as a difference of squares, are shown by the following examples:

(Mendieta, 2001), the sum of two squares is factored:

«Factorize the sum of two squares

$$\begin{aligned}
 1) \quad x^4 + 64y^4 &= x^4 + 64y^4 + 16x^2y^2 - 16x^2y^2 \\
 &= (x^4 + 16x^2y^2 + 64y^4) - 16x^2y^2 \\
 &= (x^2 + 8y^2)^2 - 16x^2y^2 \\
 &= [(x^2 + 8y^2) + 4xy][(x^2 + 8y^2) - 4xy] \\
 &= (x^2 + 4xy + 8y^2)(x^2 - 4xy + 8y^2)
 \end{aligned}$$

$$\begin{aligned}
 2) \quad 4x^8 + y^8 &= 4x^8 + y^8 + 4x^4y^4 - 4x^4y^4 \\
 &= (4x^8 + 4x^4y^4 + y^8) - 4x^4y^4 \\
 &= (2x^4 + y^4)^2 - 4x^4y^4 \\
 &= [(2x^4 + y^4) + 2x^2y^2][(2x^4 + y^4) - 2x^2y^2] \\
 &= (2x^4 + 2x^2y^2 + y^4)(2x^4 - 2x^2y^2 + y^4)
 \end{aligned}$$

$$\begin{aligned}
 3) \quad a^4 + 324b^4 &= a^4 + 324b^4 + 36a^2b^2 - 36a^2b^2 \\
 &= (a^4 + 36a^2b^2 + 324b^4) - 36a^2b^2 \\
 &= (a^2 + 18b^2)^2 - 36a^2b^2 \\
 &= [(a^2 + 18b^2) + 6ab][(a^2 + 18b^2) - 6ab] \\
 &= (a^2 + 6ab + 18b^2)(a^2 - 6ab + 18b^2) \gg
 \end{aligned}$$

6.18 Complex number and the Sum of Squares.

(Mendoza Walsh, 2003) tells us:

«With the later development of mathematics, the need arose to consider square roots of negative numbers, which is not possible in the set of real numbers, and which gave rise to imaginary numbers whose base is:

$$\sqrt{-1} = i \rightarrow i^2 = -1$$

The union of the real and imaginary numbers form a larger set called the set of complex numbers, denoted by C . As we progress in the study of mathematics and its applications, we will see the importance of this numerical set.

$$C = \{a + bi \mid a, b \in \mathbb{R} \wedge i^2 = -1; \mathbb{R} \subset C\}$$

When we carry out some operations or when we solve equations, it is very important to keep in mind the numerical set in which we are working. For example, the equation $2x^2 - 3 = 0$ has no solution in the set of integers, but in the set of rationales. The expression $x^2 + 4$ **cannot be factored into the set of real numbers**, but it can be factored into the set of complex numbers. »

In complex numbers: $a^2 + b^2 = (a + bi)(a - bi) ; i^2 = -1$.

To illustrate the above, factor $x^2 + 4$

Solution: $x^2 + 4 = x^2 + (2)^2$ here, $a = x$ and $b = 2, i^2 = -1$

$$x^2 + 2^2 = x^2 - 2^2 i^2 = (x + 2i)(x - 2i)$$

6.19 Consultation question on the factorization of $a^2 + b^2$.

Currently, to find out what is done to factor $a^2 + b^2$, and as part of the research on factoring this sum of squares, the following was elaborated for key informants:

Question addressed to people dedicated to mathematics

Dear Sir/Madam:

Granting attention to your experience in mathematics (algebra, geometric algebra, theory of numbers, etc.), you are hereby being requested to respond to a question whose answer will be of much use for research being done since August, 2011, geared towards improving the teaching of mathematics in Nicaragua, Central America. (Thanks for your kind collaboration). Please reply to Bienvenido Hernaldo Acevedo González, bienvenidoheraldoa@gmail.com or bienv1@yahoo.com

Do you know the factorization of $a^2 + b^2 = c^2$ in which only “a” and “b” are included, void of intervention by any other mathematical element such as: **Radicals** ($\sqrt{2ab}$), **Cos π** , **Complex Numbers** ($i^2 = -1$), **¿etc.?**

a. Yes b. No

NOTICE: A no-reply status to this question within a reasonable period of time (one month) will lead us to assume that there is little attention given to, or no knowledge of the factorization of binomial given.

Thanks for your kind collaboration

Bienvenido Hernaldo Acevedo González

Phone: 87233764

Actual: +505 8418 3408

As part of the research on factoring the sum of squares, $a^2 + b^2$, the previous question was presented in the Spanish language for Spanish-speaking key informants:

Pregunta a personas dedicadas a la matemática

Estimado(a) Sr(a): Atendiendo a su experiencia con la Matemática (Álgebra, Álgebra Geométrica, Teoría de Números, etc) se le solicita responder a la siguiente pregunta, cuya respuesta será de mucha utilidad para una investigación que se realiza a partir de agosto del 2011, para mejorar la Enseñanza de la Matemática en Nicaragua América Central. (Gracias por su colaboración). Responder a: Bienvenido Hernaldo Acevedo González, correo bienv1@yahoo.com o

bienvenidohernaldoa@gmail.com

Fecha: _____

¿Conoce la factorización de $a^2 + b^2 = c^2$ en la que solo se incluyen " a " y " b ", sin intervención de ningún otro elemento matemático como: **Radicales** ($\sqrt{2ab}$), **Cos π** , **Números Complejos** ($i^2 = -1$), ¿etc.?

1. Si b. No

NOTA: De no responder el formato de ésta pregunta dentro de un período de tiempo razonable (un mes), se asumirá que el entrevistado, no tiene interés en el tema o desconoce la factorización del binomio dado.

Gracias por su amable colaboración

Bienvenido Hernaldo Acevedo González

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6.20 Mathematicians consulted on factoring $a^2 + b^2 = c^2$.

The following list of key informants was obtained from <https://sites.google.com/site/joeharris60/registered-participants>, and <https://www.math.uic.edu/conferences/joe60/registration>, where a Celebration of Geometric Algebra is being made known... Lecture in Honor of Joe Harris on his 60th birthday. Harvard University Cambridge Massachusetts August 25-28, 2011. The Lecture took place in Lecture Room C of the Harvard University Science Center.

On this occasion, prestigious mathematicians from around the world met to participate in a series of conferences on Geometric Algebra. The question on the previous page was e-mailed to each of them. Below is the list:

1. Izzet Coskun / uic; coskun@math.uic.edu ; Ph.D. 2004. Harvard University. Associate Professor, University of Illinois at Chicago 2010 – present. I am an Algebraic geometer.
2. Aise Johan de Jong / Columbia; dejong@math.columbia.edu : Ph.D. Katholieke Universiteit Nijmegen 1992. Professor of Columbia University. Algebraic Geometry.
3. David Eisenbud / Berkeley: de@msri.org ; Ph.D. University of Chicago 1970. Professor of Mathematics, UC Berkeley. Algebra and Number Theory.
4. Gavril Farkas/ Humboldt u berlin: farkas@math.hu-berlin.de ; Ph.D., Universiteit van Amsterdam, 2000: Professor of Mathematics Humboldt Universität zu Berlin. My field of work is algebraic geometry.

5. William Fulton / Michigan: wfulton@umich.edu ; Ph.D., Princeton University, 1966. Professor of Mathematics University of Michigan Ann Arbor. Specializing in Algebraic Geometry.
6. Tom Graber / Caltech: graber@caltech.edu ; Ph.D. Professor of Mathematics California Institute of Technology.
7. Phillip Griffiths / ias; pg@ias.edu ;Ph.D. from Princeton University in 1962. He is co-author, with Joe Harris, of *Principles of Algebraic Geometry*. Institute for Advanced Study, Einstein Drive, Princeton, New Jersey 08540 USA.
8. Benedict Gross / Harvard; gross@math.harvard.edu ; Ph.D. Harvard University, 1978. He is known for his work in number theory, particularly the Gross–Zagier theorem on L-functions of elliptic curves.
9. Brendan Hassett / rice; hassett@rice.edu ; Ph.D., 1996(supervised by Joe Harris) Yale College, Professor of Mathematics Chair, Department of Mathematics Rice University. Research interests: Algebraic geometry and number theory.
10. Robert Lazarsfeld / Michigan; rlaz@umich.edu ; Ph.D., Brown University, 1980 Department of Mathematics University of Michigan Ann Arbor, MI 48109.
11. James McKernan / mit; mckernan@math.mit.edu ; Ph.D. from Harvard in 1991, under the direction of Joseph Harris. Professor McKernan's research interests are in algebraic geometry. Massachusetts Institute of Technology.

12. Kieran o'grady / rome I; ogrady@mat.uniroma1.it ; Prof. Kieran G. O'Grady. 1986: Ph.D. in Mathematics, Brown University, advisor Joe Harris.
13. David Smyth / Harvard; dsmyth@math.harvard.edu ;
14. Jason Starr /Stony Brook; jstarr@math.sunysb.edu ; Ph.D., Jason Starr Associate Professor, Stony Brook State University of New York.
15. Claire voisin / Jussieu; voisin@math.jussieu.fr she is a French professor of mathematics and director of research at the Institut de mathématiques de Jussieu at the University of Paris VI: Pierre et Marie Curie. She is noted for her work in algebraic geometry.
16. Ph. D Michael Aschbacher; asch@caltech.edu
17. Professor Jan-Erik Roos: jeroos@math.su.se
18. Professor Torsten Ekedahl: teke@math.su.se
19. Christopher Hacon; hacon@math.utah.edu; He received his doctorate from the University of California, Los Angeles in 1998, under supervision of Robert Lazarsfeld. His research interests include algebraic geometry.
20. Wei Ho (Columbia University); who@math.columbia.edu ; Ph.D., Princeton University, 2009, Algebraic geometry.
21. Mikhail Kapranov (Yale University); kapranov@math.toronto.edu ; Ph. D. 1988 (Steklov Institute, Moscow) (Professor, Mossman Chair).

22. Sam Payne (Yale University); spayne@stanford.edu ; Acting Assistant Professor, Stanford Ph.D.: University of Michigan, Ann Arbor. April, 2006.University.
23. Zhiwei Yun (MIT); zyun@math.mit.edu; June 2009: Ph.D. at Princeton University. My advisor was Robert MacPherson. Geometric Representation Theory.
24. Ann-Charlotte Hell; ann-charlotte.hell@musakad.se
25. Ravi Vakil / Stanford; vakil@math.edu ; I am a Professor of Mathematics at Stanford University. I will post notes on the Foundation of algebraic geometry.
26. **Dan Abramovich** / brown: abrmovic@math.brown.edu Ph.D., Harvard University, 1991. Professor of Mathematics Brown University. *Algebraic and Arithmetic Geometry*.
27. Enrico Arbarello / rome I: ea@mat.uniroma1.it ; Ph.D.: Columbia University, 1973 Professor of Geometry, Università di Roma "La Sapienza". *Algebraic curves and their moduli*.
28. Lucia Caporaso / Rome III; caporaso@mat.uniroma3.it ; Ph.D. in Mathematics - Harvard University, 1993; Professor of Mathematics/Geometry, University Roma TRE (since 2001). *Algebraic and Arithmetic Geometry*.

Following is a table-summary being sent you of the previous list with e-mail addresses, names, date and time when e-mails were sent with the question about the factorization of the sum of two squares.

6.21 Tabulation of data: Mails, Names and surnames and date.

Everyone on the following list was e-mailed the question about factoring the sum of squares.

de **Bienvenido Hernaldo Acevedo González** <bienvenidohernaldoa@gmail.com>:

para:

fecha:

coskun@math.uic.edu;	Ph. D Izzet Coskun	27 ago 2011 4:44
dejong@math.columbia.edu	Ph. D Aise Johan de Jong	27 ago 2011 4:46
de@msri.org	Ph. D David Eisenbud	27 ago 2011 4:48
farkas@math.hu-berlin.de	Ph. D Gavril Farkas	27 ago 2011 4:50
wfulton@umich.edu	Ph. D William Fulton	27 ago 2011 4:52
graber@caltech.edu	Ph. D Tom Graber	27 ago 2011 4:54
pg@ias.edu	Ph. D Phillip Griffiths	27 ago 2011 4:55
gross@math.harvard.edu	Ph. D Benedict Gross:	27 ago 2011 4:57
hassett@rice.edu	Ph. D Brendan Hassett	27 ago 2011 5:06
rlaz@umich.edu	Ph. D Robert Lazarsfeld	27 ago 2011 5:09
mckernan@math.mit.edu	Ph. D James Mckernan	27 ago 2011 5:10
ogradey@mat.uniroma1.it	Ph. D Kieran o'grady	27 ago 2011 5:12
dsmyth@math.harvard.edu	Ph. D David Smyth	27 ago 2011 5:16
jsarr@math.sunysb.edu	Ph. D Jason Starr	27 ago 2011 5:17
voisin@math.jussieu.fr	Ph. D Claire voisin	27 ago 2011 5:22
asch@caltech.edu	Ph. D Michael Aschbacher	27 ago 2011 5:26
jeroos@math.su.se	Professor Jan-Erik Roos	27 ago 2011 5:29
teke@math.su.se	Professor Torsten Ekedahl	27 ago 2011 5:30
hacon@math.utah.edu	Ph. D Christopher Hacon	27 ago 2011 5:32
who@math.columbia.edu	Ph. D Wei Ho	27 ago 2011 5:34
kapranov@math.toronto.edu	Ph. D Mikhail Kapranov	27 ago 2011 5:35
spayne@stanford.edu	Ph. D Sam Payne	27 ago 2011 5:47
zyun@math.mit.edu	Ph. D Zhiwei Yun	27 ago 2011 5:48
ann-charlotte.hell@musakad.se	Ph. D Ann-Charlotte Hell	27 ago 2011 5:50
vakil@math.edu	Ph. D Ravi Vakil	27 ago 2011 5:54
abrmovic@math.brown.edu	Ph. D Dan Abramovich	27 ago 2011 5:57
ea@mat.uniroma1.it	Ph. D Enrico Arbarello	27 ago 2011 6:01
caporaso@mat.uniroma3.it	Ph. D Lucia Caporaso	27 ago 2011 6:03

6.22 Dialogue with a key informant, about the binomial $a^2 + b^2 = c^2$.

Dialogue with the Ph.D. Dan Abramovich about factoring the polynomial $a^2 + b^2 = c^2$. Here is the sustained communication:

Bienvenido Acevedo writes and asks:

Ph. D Dan Abramovich abrmovic@math.brown.edu 27 ago. 2011

Dear Ph. D Dan Abramovich:

Granting attention to your experience in mathematics (algebra, geometric algebra, theory of numbers, etc.), you are hereby being requested to respond to a question whose answer will be of much use for research being done since August, 2011, geared towards improving the teaching of mathematics in Nicaragua, Central America. (Thanks for your kind collaboration). Please reply to Bienvenido Hernaldo Acevedo González, or bienvenidohernaldoa@gmail.com or bienv1@yahoo.com

Do you know the factorization of $a^2 + b^2 = c^2$ in which only “a” and “b” are included, void of intervention by any other mathematical element such as: **Radicals** ($\sqrt{2ab}$), **Cos π** , **Complex Numbers** ($i^2 = -1$), **¿etc.?**

a. Yes b. No

NOTICE: A no-reply status to this question within a reasonable period of time (one month) will lead us to assume that there is little attention given to, or no knowledge of the factorization of binomial given.

Thanks for your kind collaboration

Bienvenido Hernaldo Acevedo González

Phone: 87233764

Here is a free translation of the previous text:

Estimado Ph. D Dan Abramovich:

Atendiendo a su experiencia en matemáticas (álgebra, álgebra geométrica, teoría de números, etc.), por la presente se le solicita que responda a una pregunta cuya respuesta será de mucha utilidad para una investigación que se está realizando desde agosto de 2011, orientada a mejorar la enseñanza de las matemáticas en Nicaragua, Centroamérica. (Gracias por su amable colaboración). Por favor respuesta a Bienvenido Hernaldo Acevedo González, o bienvenidohernaldoa@gmail.com o bienv1@yahoo.com

¿Conoce la factorización de $a^2 + b^2 = c^2$ en la que sólo " a " y " b " están incluidos, sin intervención de ningún otro elemento matemático tales como: Radicales ($\sqrt{2ab}$), Cos π , Números complejos ($i^2 = -1$), etc.?

a. Si b. No

NOTA: De no responder el formato de ésta pregunta dentro de un período de tiempo razonable (un mes), se asumirá que el entrevistado, no tiene interés en el tema o desconoce la factorización del binomio dado.

Gracias por su amable colaboración.

Bienvenido Hernaldo Acevedo González
Teléfono: 87233764

Dan Abramovich responds:

From: **Dan Abramovich** abrmovic@math.brown.edu
To: Bienvenido Hernaldo Acevedo González
<bienvenidohernaldoa@gmail.com>
Date: 27 ago. 2011 16:53
Subject: Re: Respond to a Question
Submitted by: math.brown.edu

Dan Abramovich

27 ago. 2011 16:53

Yes, you want to read about the rational parametrization for instance in
<http://mathnow.wordpress.com/2009/11/06/a-rational-parameterization-of-the-unit-circle/>

da

Here is a free translation:

Sí, ud así lo desea puede leer sobre la parametrización racional, en el siguiente link:

<http://mathnow.wordpress.com/2009/11/06/a-rational-parameterization-of-the-unit-circle/>

da

Bienvenido Acevedo writes and responds:

From: **Bienvenido Hernaldo Acevedo**

González bienvenidohernaldoa@gmail.com

To: Dan Abramovich <abrmovic@math.brown.edu>

Date: 28 ago. 2011 6:56

Subject: Re: Respond to a Question

Sent by: gmail.com:

This message is important because you have marked it as such.

Este mensaje es importante porque lo has marcado como tal.

I read the document that you recomend me, but I do not find any algebraic expression that factorize $a.a + b.b = c.c$

Thank you Dr. Dan

Cordially yours

Bienvenido Acevedo

Here a free translation:

Leí el documento que me recomendó, pero no encuentro ninguna expresión algebraica que factorize $a.a + b.b = c.c$

Gracias Dr. Dan

Cordialmente de Ud.

Bienvenido Acevedo

Dan Abramovich writes and responds:

From: **Dan Abramovich** <abrmovic@math.brown.edu>

To: Bienvenido Hernaldo Acevedo González
<bienvenidoheraldoa@gmail.com>

Date: 28 ago. 2011 13:26

Subject: Re: Respond to a Question

Submitted by: math.brown.edu

Hello –

The polynomial $a^2 + b^2 = c^2$ is irreducible, there is no factorization. The best you can do is a rational parametrization of the solutions, which is what that document gives.

D.A.

Here a free translation:

Hola –

El polinomio $a^2 + b^2 = c^2$ es irreducible, no hay factorización. Lo mejor que puede hacer es una parametrización racional de las soluciones, que es lo que da el documento.

DA

Bienvenido Acevedo writes and says goodbye:

Bienvenido Hernaldo Acevedo González bienvenidoheraldoa@gmail.com

28 ago 2011 21:12

For -Dan

Hello Dr. Dan:

Im very much obliged to you, for your kind collaboration, thank for your answer to my question.

Gracias por responder mi pregunta, le quedo muy agradecido

6.23 Screenshots of the Key Informant Dialogue.

Bienvenido Hernaldo Acevedo González <bienvenidohernaldc
abrmovic@math.brown.edu

27 ago 2011 5:57

Respond to a Question

gmail.com

Este mensaje es importante porque lo has marcado como tal.

On Sat, 27 Aug 2011, Bienvenido Hernaldo Acevedo González wrote:

Dear Ph.D Dan Abramovich:

Granting attention to your experience in mathematics (algebra, geometric algebra, theory of numbers, etc.), you are hereby being requested to respond to a question whose answer will be of much use for a research being done since August, 2011, geared towards improving the teaching of mathematics in Nicaragua, Central America. (Thanks for your kind collaboration). Please reply to Bienvenido Hernaldo Acevedo González, or bienvenidohernaldoa@gmail.com or bienv1@yahoo.com

Do you know the factorization of $a^2 + b^2 = c^2$ in which only a and b are included, void of intervention by any other mathematical element such as: Radicals ($\sqrt{2ab}$), $\cos \pi$, Complex Numbers ($i^2 = -1$), etc.?

a. Yes b. No

NOTICE: A no-reply status to this question within a reasonable period of time (one month) will lead us to assume that there is little attention given to, or no knowledge of the factorization of binomial given.

Thanks for your kind collaboration

Bienvenido Hernaldo Acevedo González
Phone : 87233764

The Dr. Abramovich responds:

El 27 de agosto de 2011 16:53, Dan Abramovich <abrmovic@math.brown.edu> escribió:

Yes, you want to read about the rational parametrization for instance in

<http://mathnow.wordpress.com/2009/11/06/a-rational-parameterization-of-the-unit-circle/> <<http://mathnow.wordpress.com/2009/11/06/a-rational-parameterization-of-the-unit-circle/>>

d.a.

On Sun, 28 Aug 2011, Bienvenido Hernaldo Acevedo González wrote:

> I read the document that you recomend me, but I do not find any algebraic
> expression that factorize
> $a.a + b.b = c.c$
>
> Thank you Dr.Dan
>
> Cordialy yours
> Bienvenido Acevedo

Dan Abramovich <abrmovic@math.brown.edu>

para mí ▼

Hello -

The polynomial $a^2+b^2-c^2$ is irreducible, there is no factorization.
The best you can do is a rational parametrization of the solutions, which is what that document gives.

D.A.

Bienvenido Hernaldo Acevedo González <bienvenidohernaldoa@gmail.com>

para Dan ▼

Hello Dr. Dan:

Im very much obliged to you, for your kind collaboration, thank for your answer to my question.

From the previous dialogue, it is concluded that: the polynomial $a^2 + b^2 = c^2$ is irreducible on the set of real numbers, and as Dr. Dan Abramovich says, «there is no factorization. »

In what follows, an equality will be presented in which the sum of two squares is expressed as the product of two factors f_1 and f_2 . The factor f_1 is the sum of the sides of each square and f_2 is the side of the smaller square plus a fraction of the side of the second square.

Chapter 7: Formula “*bienve₁*” that factors $a^2 + b^2$ into R ,

7.1 The beginning of the search for factors of $a^2 + b^2$

In 2001, I participated in a Postgraduate Course in Mathematics Didactics, from July 9 to 20, and August 20 to 31, in the city of León, department of Nicaragua. It was taught to math teachers across the country, 20 years ago. One of the topics was Notable Products and Factoring. The ten best-known and most widely used cases in secondary education in Nicaragua were factored. I asked the facilitators about the factorization of a sum of squares. I told them that if a difference of squares had factorization, why did the sum of squares not have, if the difference of squares comes from a sum, as you can see here: $a^2 + b^2 = c^2$, that is $b^2 = c^2 - a^2 = (c + a)(c - a)$. They told me that it could be factored, but using the complex numbers that have the imaginary unit, $i^2 = -1$. With the following formula:

$$a^2 + b^2 = (a + bi)(a - bi) ; i^2 = -1.$$

Since then, I began to investigate, pray and work to discover the factors other than c^2 , with rational coefficients of the real numbers. It was until October 25, 2010, that Jehovah, the Creator of the Universe, heard my prayers. In a dream between 5:00 - 6:00 a.m., I saw the formula as on a screen. This formula is the one that is presented as the objective of this book. It was my turn to demonstrate the formula and I do not know if it is right or wrong, whether it works or not, let those who know judge it.

I wrote down the formula and dedicated myself to testing with numbers and it was always fulfilled, hence I decided to publish this mathematical contribution in 2021, 20 years later, a promise that I made in the early morning of February 25, 2021, when I was in agony. I am a COVID-19 survivor.

7.2 Factorization of $a^2 + b^2 = c^2$, in complex numbers.

The binomial, $a^2 + b^2 = c^2$ with factors $(a + bi)(a - bi)$, has no factoring formula in the set of real numbers ($\mathbb{R}$), and it is said that in this numerical field, the binomial is *irreducible*. In this case the factors $(a + bi)$ and $(a - bi)$ belong to the set of complex numbers ($\mathbb{C}$). In this system the equality, $i^2 = -1$, is a special number, called an *imaginary unit*.

Let us illustrate the use of : $i^2 = -1$, to obtain the sum of two squares as a result.

$$\begin{array}{c} i \\ \boxed{i^2} \\ i \end{array} = -1$$

Fig.41: Imaginary Square

$$(a + bi)(a - bi) = a^2 + abi - abi - b^2i^2 = a^2 - b^2i^2$$

$$\text{Since } i^2 = -1, \text{ so } a^2 - b^2i^2 = a^2 - b^2(-1) = a^2 + b^2$$

Now, what we want is to factor, $a^2 + b^2 = c^2$, so that the rational factors of c^2 belong to the set of real numbers, *which are obtained naturally without artifacts that transform it into another algebraic expression with known factorization. No square roots, no trigonometric relationships, no complex or transcendent numbers, etc. The rational factors of c^2 must only contain the values of a and b conveniently combined, with the four operations of basic arithmetic, they must belong to the set of real numbers, in such a way that the coefficients of the factors can be integers, or fractions (rational) or any real number.*

7.3 The rational factors of $a^2 + b^2$, in real numbers.

We are going to determine the rational factors f_1 and f_2 of $a^2 + b^2$, with rational coefficients, these factors contain sum of terms. To obtain these rational factors, let's start with the following graph:

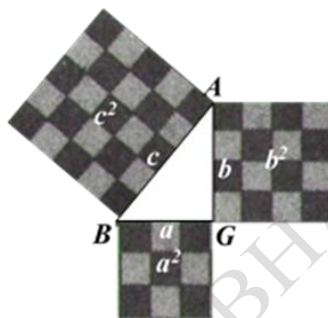


Fig.42; Squares of Pythagoras' Theorem

We draw a horizontal line **MN**, which passes through **BG**.

1. We turn c^2 , anticlockwise until we touch MN.
2. We rotate b^2 vertically around **MN**.
3. We draw a broken white line, **CA**, with this the small rectangle **CFGA** is formed, with base **b**, and height, **(b - a)**.
4. We identify, each of the useful elements and the graph looks like this:

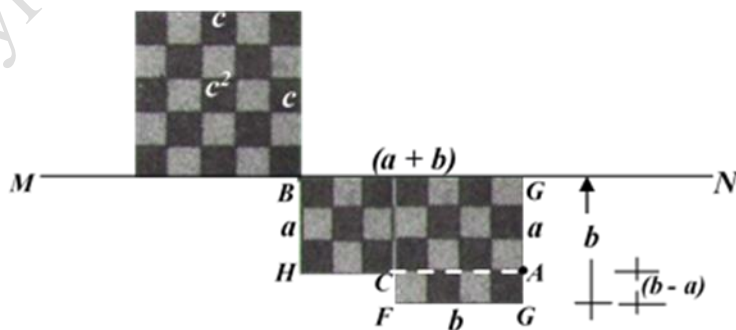


Fig.43; Squares aligned to segment MN

From the previous graph, the following data is obtained:

BG = HA = $(a + b)$; likewise, **BH = AG = a** ;

CA = FG = b ; and **CF = AG = $(b - a)$**

In addition, the formation of the following two rectangles is observed:

1. The rectangle **ABDE**, its base is: **$(a + b)$** and its height is: **a**
2. The rectangle **CFGH**, its base is: **b** and its height is: **$(b - a)$**

Separating the two previous rectangles, the graph looks to us like this:

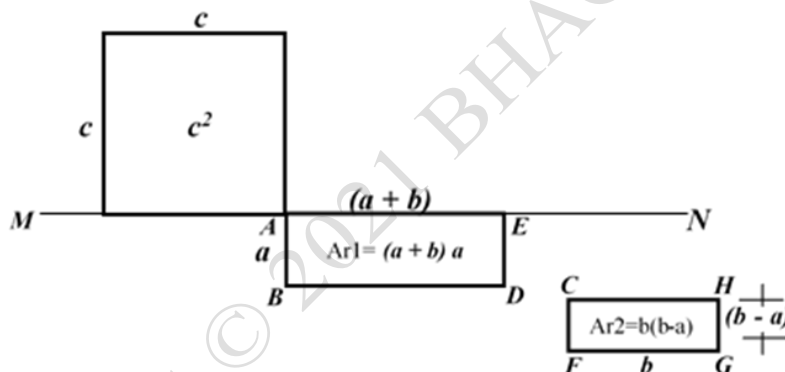


Fig.44; Square and Rectangles obtained in line with MN:

The area of the rectangle **ABDE** is: *base $\times$ height* = **$(a + b) a$**

The area of the rectangle **CFGH** is: *base $\times$ height* = **$b (b - a)$** ; we must distribute this area over the entire length, $(a + b)$, that is, plot it, so that the sought rectangle will have a small increase in its height. Let's call the increment delta (Δ), that is, $a + \Delta$ of such that, the new rectangle would be:

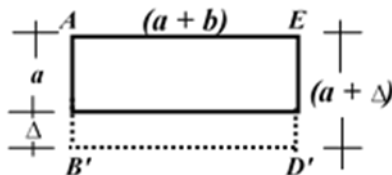


Fig.45; Rectangle AB'D'E with increment Δ with MN

7.4 Determination of the increase “ Δ ” in the height of $AB'D'E'$

To determine the value of delta (Δ), the increase in height of a rectangle $AB'D'E'$, we do the following:

1. The vertex of c^2 , we will identify it as **H**.
2. In the given graph, we mark a point **K**, between **MB**, from which it results, the segment $KB = BI = EV = a + \Delta$.
3. With non-continuous lines, we form the 90° **KHV** triangle.

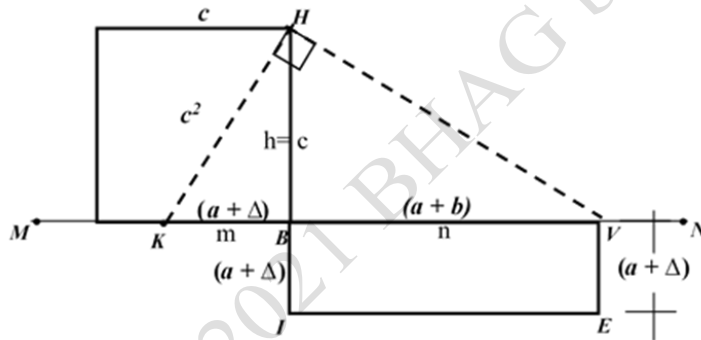


Fig.46; Rectangle $AB'D'E'$ with increment $(a + \Delta)$

Applying in the right triangle **KHV** the Euclid Height Theorem, we find the increment " Δ "

$$h^2 = c^2 = m \cdot n \quad \text{donde } m = (a + \Delta) \text{ y } n = (a + b); \text{ entonces:}$$

$$h^2 = c^2 = (a + \Delta) (a + b)$$

$$c^2 = (a + \Delta) (a + b); \text{ multiplying results:}$$

$$c^2 = a^2 + ab + \Delta a + \Delta b; \text{ factoring:}$$

$$c^2 = a(a + b) + \Delta(a + b); \text{ As: } c^2 = a^2 + b^2; \text{ then:}$$

$$c^2 = a^2 + b^2 = a^2 + ab + \Delta(a + b) =, \text{ if } a^2 \text{ is eliminated it remains:}$$

$$ab + \Delta(a + b) = b^2; \text{ passing } ab \text{ next to } b^2, \text{ it results:}$$

$$\Delta(a + b) = b^2 - ab; \text{ factoring for } b, \text{ results:}$$

$$\Delta(a + b) = b(b - a)$$

Solving for Δ , finally, we have to: $\Delta = \frac{(b-a)}{(a+b)} b$

Since $c^2 = a^2 + b^2 = (a+b)(a+\Delta)$.

Re-writing the previous expression and substituting Δ results in:

$$c^2 = (a+b) \left[a + \left(\frac{b-a}{a+b} \right) b \right]$$

In the graph below, we designate the square as **BHAG**, whose area is equal to the area of the rectangle **BIEN**.

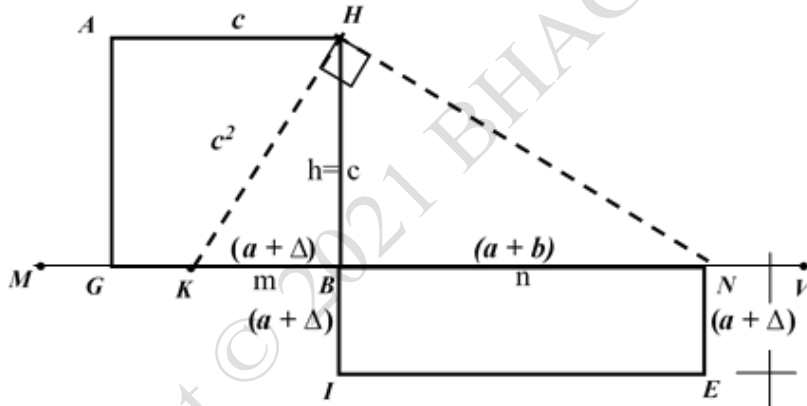


Fig.47; Working KHN Rectangle Triangle

Therefore, the factorization of the sum of two squares, expressed as the product of two rational factors with factors in which there is a sum by a sum, is as follows:

"bienve₁":

$$a^2 + b^2 = c^2 = (a+b) \left[a + \left(\frac{b-a}{a+b} \right) b \right]$$

We will call it the formula "bienve₁", which contains the product of two distinct rational factors. Each factor contains a sum, and one of them has a rational coefficient, both rational factors belong to the real numbers.

7.5 Checking “bienve₁” in the factorization of $a^2 + b^2$

Below is the image of a page in excel, where the formula “bienve₁” is verified. It is shown that:

“bienve₁”:
$$a^2 + b^2 = c^2 = (a + b) \left[a + \left(\frac{b - a}{a + b} \right) b \right]$$

The formula “bienve₁” is fulfilled:

Los Factores Racionales de una Suma de Cuadrados					bienve ₁					
Terna Pitagórica							T.Pitagoras	f ₁ y f ₂ al multiplicarse resultan en c ²		
Rectángulo		a	b	c				f ₁	f ₂	f ₁ *f ₂
n	m	m ² - n ²	2.m.n	m ² + n ²	a ²	b ²	c ² = a ² + b ²	(a + b)	$\left[a + \left(\frac{b - a}{a + b} \right) b \right]$	c ²
1	2	3	4	5	9	16	25	7	3,57143	25
2	3	5	12	13	25	144	169	17	9,94118	169
3	4	7	24	25	49	576	625	31	20,16129	625
4	5	9	40	41	81	1.600	1.681	49	34,30612	1.681
5	6	11	60	61	121	3.600	3.721	71	52,40845	3.721
6	7	13	84	85	169	7.056	7.225	97	74,48454	7.225
7	8	15	112	113	225	12.544	12.769	127	100,54331	12.769
8	9	17	144	145	289	20.736	21.025	161	130,59006	21.025
9	10	19	180	181	361	32.400	32.761	199	164,62814	32.761
10	11	21	220	221	441	48.400	48.841	241	202,65975	48.841
11	12	23	264	265	529	69.696	70.225	287	244,68641	70.225
12	13	25	312	313	625	97.344	97.969	337	290,70920	97.969
13	14	27	364	365	729	132.496	133.225	391	340,72890	133.225
14	15	29	420	421	841	176.400	177.241	449	394,74610	177.241
15	16	31	480	481	961	230.400	231.361	511	452,76125	231.361
16	17	33	544	545	1.089	295.936	297.025	577	514,77470	297.025
17	18	35	612	613	1.225	374.544	375.769	647	580,78671	375.769
18	19	37	684	685	1.369	467.856	469.225	721	650,79750	469.225
19	20	39	760	761	1.521	577.600	579.121	799	724,80726	579.121
20	21	41	840	841	1.681	705.600	707.281	881	802,81612	707.281
21	22	43	924	925	1.849	853.776	855.625	967	884,82420	855.625
22	23	45	1.012	1.013	2.025	1.024.144	1.026.169	1.057	970,83160	1.026.169
23	24	47	1.104	1.105	2.209	1.218.816	1.221.025	1.151	1.060,83840	1.221.025
24	25	49	1.200	1.201	2.401	1.440.000	1.442.401	1.249	1.154,84468	1.442.401
25	26	51	1.300	1.301	2.601	1.690.000	1.692.601	1.351	1.252,85048	1.692.601
26	27	53	1.404	1.405	2.809	1.971.216	1.974.025	1.457	1.354,85587	1.974.025
27	28	55	1.512	1.513	3.025	2.286.144	2.289.169	1.567	1.460,86088	2.289.169
28	29	57	1.624	1.625	3.249	2.637.376	2.640.625	1.681	1.570,86556	2.640.625
29	30	59	1.740	1.741	3.481	3.027.600	3.031.081	1.799	1.684,86993	3.031.081
30	31	61	1.860	1.861	3.721	3.459.600	3.463.321	1.921	1.802,87402	3.463.321
31	32	63	1.984	1.985	3.969	3.936.256	3.940.225	2.047	1.924,87787	3.940.225
32	33	65	2.112	2.113	4.225	4.460.544	4.464.769	2.177	2.050,88149	4.464.769

Image of 32 Pythagorean Triples

The formula "bienve₁" is fulfilled:

33	34	67	2.244	2.245	4.489	5.035.536	5.040.025	2.311	2.180,88490	5.040.025
34	35	69	2.380	2.381	4.761	5.664.400	5.669.161	2.449	2.314,88812	5.669.161
35	36	71	2.520	2.521	5.041	6.350.400	6.355.441	2.591	2.452,89116	6.355.441
36	37	73	2.664	2.665	5.329	7.096.896	7.102.225	2.737	2.594,89404	7.102.225
37	38	75	2.812	2.813	5.625	7.907.344	7.912.969	2.887	2.740,89678	7.912.969
38	39	77	2.964	2.965	5.929	8.785.296	8.791.225	3.041	2.890,89938	8.791.225
39	40	79	3.120	3.121	6.241	9.734.400	9.740.641	3.199	3.044,90184	9.740.641
40	41	81	3.280	3.281	6.561	10.758.400	10.764.961	3.361	3.202,90420	10.764.961
41	42	83	3.444	3.445	6.889	11.861.136	11.868.025	3.527	3.364,90644	11.868.025
42	43	85	3.612	3.613	7.225	13.046.544	13.053.769	3.697	3.530,90857	13.053.769
43	44	87	3.784	3.785	7.569	14.318.656	14.326.225	3.871	3.700,91062	14.326.225
44	45	89	3.960	3.961	7.921	15.681.600	15.689.521	4.049	3.874,91257	15.689.521
45	46	91	4.140	4.141	8.281	17.139.600	17.147.881	4.231	4.052,91444	17.147.881
46	47	93	4.324	4.325	8.649	18.696.976	18.705.625	4.417	4.234,91623	18.705.625
47	48	95	4.512	4.513	9.025	20.358.144	20.367.169	4.607	4.420,91795	20.367.169
48	49	97	4.704	4.705	9.409	22.127.616	22.137.025	4.801	4.610,91960	22.137.025
49	50	99	4.900	4.901	9.801	24.010.000	24.019.801	4.999	4.800,92118	24.019.801
50	51	101	5.100	5.101	10.201	26.010.000	26.020.201	5.201	5.002,92271	26.020.201
51	52	103	5.304	5.305	10.609	28.132.416	28.143.025	5.407	5.204,92417	28.143.025
52	53	105	5.512	5.513	11.025	30.382.144	30.393.169	5.617	5.410,92558	30.393.169
53	54	107	5.724	5.725	11.449	32.764.176	32.775.625	5.831	5.620,92694	32.775.625
54	55	109	5.940	5.941	11.881	35.283.600	35.295.481	6.049	5.834,92825	35.295.481
55	56	111	6.160	6.161	12.321	37.945.600	37.957.921	6.271	6.052,92952	37.957.921
56	57	113	6.384	6.385	12.769	40.755.456	40.768.225	6.497	6.274,93074	40.768.225
57	58	115	6.612	6.613	13.225	43.718.544	43.731.769	6.727	6.500,93192	43.731.769
58	59	117	6.844	6.845	13.689	46.840.336	46.854.025	6.961	6.730,93306	46.854.025
59	60	119	7.080	7.081	14.161	50.126.400	50.140.561	7.199	6.964,93416	50.140.561
60	61	121	7.320	7.321	14.641	53.582.400	53.597.041	7.441	7.202,93522	53.597.041
61	62	123	7.564	7.565	15.129	57.214.096	57.229.225	7.687	7.444,93626	57.229.225
62	63	125	7.812	7.813	15.625	61.027.344	61.042.969	7.937	7.690,93726	61.042.969
63	64	127	8.064	8.065	16.129	65.028.096	65.044.225	8.191	7.940,93822	65.044.225
64	65	129	8.320	8.321	16.641	69.222.400	69.239.041	8.449	8.194,93916	69.239.041
65	66	131	8.580	8.581	17.161	73.616.400	73.633.561	8.711	8.452,94008	73.633.561
66	67	133	8.844	8.845	17.689	78.216.336	78.234.025	8.977	8.714,94096	78.234.025
67	68	135	9.112	9.113	18.225	83.028.544	83.046.769	9.247	8.980,94182	83.046.769
68	69	137	9.384	9.385	18.769	88.059.456	88.078.225	9.521	9.250,94265	88.078.225
69	70	139	9.660	9.661	19.321	93.315.600	93.334.921	9.799	9.524,94346	93.334.921
70	71	141	9.940	9.941	19.881	98.803.600	98.823.481	10.081	9.802,94425	98.823.481
71	72	143	10.224	10.225	20.449	104.530.176	104.550.625	10.367	10.084,94502	104.550.625
72	73	145	10.512	10.513	21.025	110.502.144	110.523.169	10.657	10.370,94576	110.523.169
73	74	147	10.804	10.805	21.609	116.726.416	116.748.025	10.951	10.660,94649	116.748.025
74	75	149	11.100	11.101	22.201	123.210.000	123.232.201	11.249	10.954,94720	123.232.201
75	76	151	11.400	11.401	22.801	129.960.000	129.982.801	11.551	11.252,94788	129.982.801
76	77	153	11.704	11.705	23.409	136.983.616	137.007.025	11.857	11.554,94855	137.007.025
77	78	155	12.012	12.013	24.025	144.288.144	144.312.169	12.167	11.860,94921	144.312.169
78	79	157	12.324	12.325	24.649	151.880.976	151.905.625	12.481	12.170,94984	151.905.625
79	80	159	12.640	12.641	25.281	159.769.600	159.794.881	12.799	12.484,95046	159.794.881
80	81	161	12.960	12.961	25.921	167.961.600	167.987.521	13.121	12.802,95107	167.987.521
81	82	163	13.284	13.285	26.569	176.464.656	176.491.225	13.447	13.124,95166	176.491.225
82	83	165	13.612	13.613	27.225	185.286.544	185.313.769	13.777	13.450,95224	185.313.769

Image of 50 Pythagorean Triples

Chapter 8: Formula “*bienvez*” to factor $a^2 + b^2$ in $\mathbb{R}$

8.1 A little about Greek geometric algebra.

(Ruiz Felipe, 2009), tells us: “Book II of the Elements (one of the shortest, with 14 propositions,) is constructed as a geometric algebra analogous in its time to symbolic algebra. The initial propositions deal with transformations of areas. The fourteenth statement explains how to find a square whose area is equal to that of a given rectilinear figure. The first eleven propositions in this book could be considered algebraic properties, if they were quantities instead of segments. However, geometric solutions precede algebraic solutions. [...]. The Second Book of Euclid's Elements is called "geometric algebra" »

(Gonzalez Urbaneja, 2017), with respect to Greek geometric algebra, mentions: «After the appearance of immeasurable magnitudes, the Greeks could not admit the existence of irrational numbers. In this way, the numerical treatment of lengths, areas and volumes was impossible. Added to this operational limitation was a poor numbering system that used the letters of the Greek alphabet themselves to represent whole numbers, which meant considerable difficulty in performing operations. These conditions prevented assigning to geometric figures numbers that measured their lengths, areas and volumes. Therefore, it was not feasible to subject them to algebraic manipulations, as it is done with numbers, so that it was necessary to operate directly with the figures that were treated as magnitudes. Thus appears the so-called Geometric Algebra of the Greeks developed by the Pythagorean School, as a kind of geometric algorithm. ”

(Sapiens, 2002 PASA), states that: «In ancient Greece (around the 4th century BC), mathematicians performed most of our arithmetic calculations (that is, with numbers) using geometric constructions. This classical "geometric algebra" allows in some cases to give a simple interpretation to some of our general expressions »For example, « In the figure on the left appears a square with side c and, within it, another square with side a . The area of the dark part is expressed algebraically as $c^2 - a^2$ »

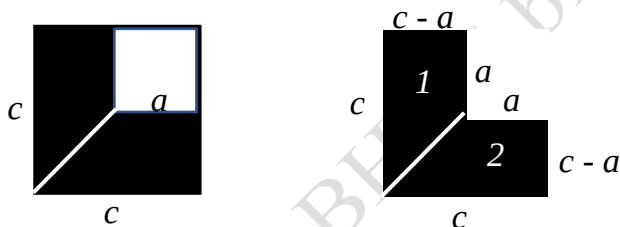


Fig.48: To the square of side c , we subtract the one of side a

On the right, what is left after subtracting $a^2 - b^2$. The figure is divided into two sectors, 1 and 2 which, placed horizontally, look like this:

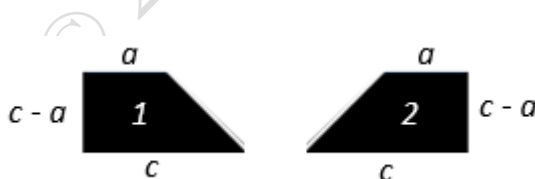


Fig.49: Trapezoids placed horizontally

Flipping part 2 vertically, it remains:



Fig.50: Trapezoid 2, turned vertically.

Putting parts 1 and 3 together results in the following rectangle whose base is $(c + a)$ and height $(c - a)$. If, Ar: Area of the rectangle.

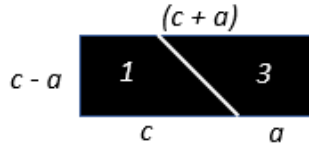


Fig.51: Joining trapezoids 1 and 3

Then: $Ar = (c + a)(c - a) = c^2 - a^2$

By the Pythagorean Theorem, we know that $a^2 + b^2 = c^2$

This implies that: $b^2 = c^2 - a^2 = (c + a)(c - a) = Ar$

The square with side b , obtained from the Pythagorean triangle, is equivalent to the area of the rectangle with sides $(c + a)$ and $(c - a)$, as seen in the following graph:

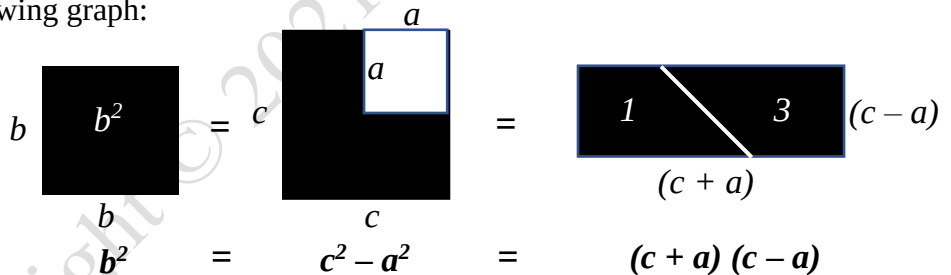


Fig.52: Subtract a square from another square

In conclusion:

1. From a difference of two squares, square results, and its area is equivalent to the area of a rectangle and vice versa. This is:

$$c^2 - a^2 = b^2 = (c + a)(c - a).$$
2. The sum of two squares is a square, and the area of this square must equal the area of a rectangle. As has been shown graphically and analytically, $Ar = (c + a)(c - a)$

Let us see now, the quadrature of a rectangle and the geometric algebra.

8.2 Geometric algebra in the quadrature of a rectangle.

(Gonzalez Urbaneja, 2017) mentions the following: «The most important part of Book II of Euclid's Elements deals with the foundations of the Geometric Algebra of the Greeks that operated directly with the figures that were treated as magnitudes. The numbers are treated as line segments and the operations between them are carried out by means of geometric constructions. The study of the relationships between rectangles or squares of the same height built on the sum or difference of two segments, allows the geometric solution of quadratic equations. »

(Gonzalez Urbaneja, 2017) states, «Proposition II.14. Construct a square equivalent to a given rectilinear figure. [*Quadrature of polygonal figures*] ».

8.3 Squaring a Rectangle.

It is very important for the purposes of this book to review from Greek geometric algebra, how to obtain the area of a rectangle so that it be equivalent to the area of a square. It is about determining the base and the height of the rectangle, which, when multiplied, turns out to be the area of a square.

To review, we refer to (Gonzalez Urbaneja, 2017) who says, «Book II ends with one of the most important problems of Euclid's Elements and with all Greek geometry in relation to the squares of geometric figures. »

The quadrature of a plane figure is the construction, with a ruler and compass, of a square with the same surface as the initial figure.

The problem that fascinated Greek culture for its obsession with reducing the complex to the simple (as simple is the symmetrical regularity of the square), in a natural world governed by reason, perfection, beauty and order-has an enormous interest from a practical point of view. This is because the determination of the area of an irregular figure is not easy at all, while the squaring of a figure reduces the calculation of its area to that of a square. » "The first step in these problems is the squaring of the rectangle that appears in this Proposition II.14 of Euclid's Elements." I will try to explain what is said about the squaring of the rectangle, and how it should be understood. We will do it one-step at a time.

«Be it that **BCDE** be any rectangle. You must build, with a ruler and a compass, a square whose area is that of the rectangle **BCDE**. »

1. Let us draw a rectangle **BCDE**:

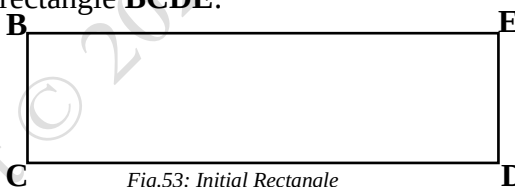


Fig.53: Initial Rectangle

«With the ruler the line **BE** is prolonged, and the compass is used to mark the segment **EF** of equal length **ED**.»

We center at **E**, and with radius **ED**, we draw the part of a circle, from **D**, until it cuts the extension **BE**, the cut point is **F**. This guarantees that: **ED = EF**

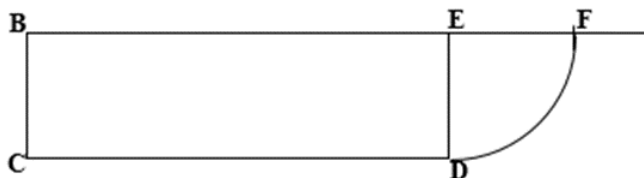


Fig.54: Rectangle with $ED = EF$

We draw with dashed lines, a square of side **BF**. We draw the diagonals of this square. From the point where the diagonals intersect, we draw with dashed line until it intersects the segment **BF**, there we mark the center **G**. With radius **R = GB** or **GF**, I draw a semicircle, from **B** to **F**.

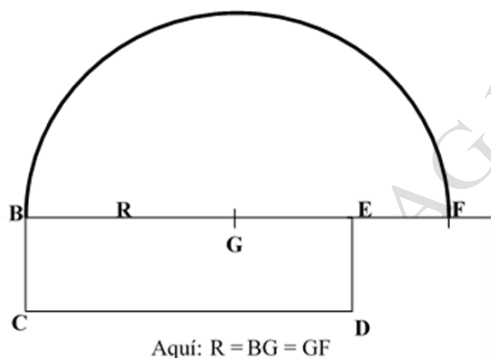
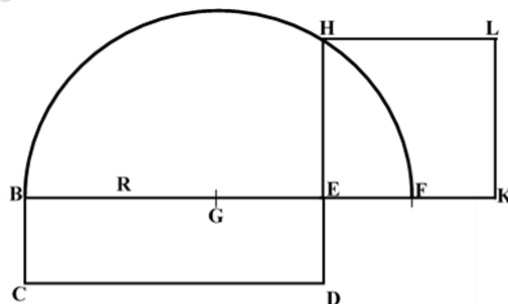


Fig.55: Semicircle with Diameter = BF and radius BG = GF

«From **E** I draw the line **EH** perpendicular to **BF**. At **H**, it is cut to the semicircle. The segment **EH** is the side of the square **EKLH** sought. »

2. We extend the **DE** line, until we cut the semicircle. At the point of intersection, we mark **H**. The line **EH** is the side of the square we are looking for, and then we form the square **EKLH**.



Sabemos que $BG = GH = GF = R = \text{Radio del círculo}$

Fig.56: Extensión of ED until H is cut in semicircle

In the previous figure, let us plot the radius **GH**; the right triangle, **GEH**, is formed, which fulfills the Pythagorean Theorem. We conveniently name the sides of the triangle **GE = a**; **EH = b**; and **BG = GH = GF = c**, with this, all the elements of the figure are identified, which will allow the arithmetic-algebraic calculations to be made and analytically demonstrate the equality of areas of the rectangle **BCDE**, with the square **EKLH**. The graph looks like this:

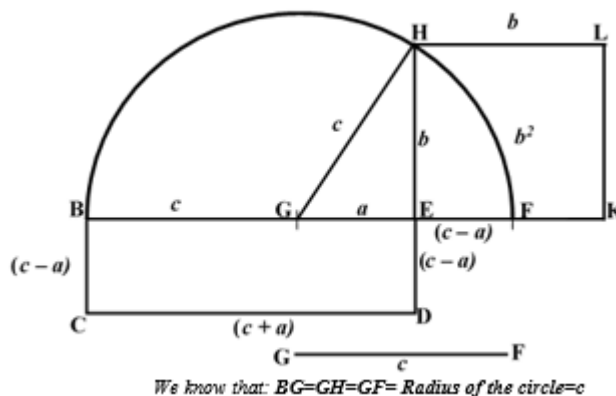


Fig.57: Formation of square EKLH and rectangle BCDE

8.4 Demonstration of the equivalence of areas.

1. In the triangle **GEH**, it is satisfied: $a^2 + b^2 = c^2$,
this means that: $b^2 = c^2 - a^2 = (c + a)(c - a)$
2. If, **GF = c**, and **GE = a**, then: **BC = ED = EF = (c - a)**; **BE = (c + a)**
3. The area of the rectangle **BCDE** = **(BE) . (BC) = (c + a)(c - a)**
4. The area of the square **EKLH**, is: $b^2 = c^2 - a^2 = (c + a)(c - a)$

Then the area of BCDE = To the area of square EKLH

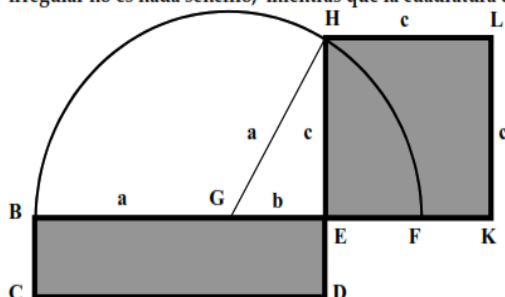
$$(c + a)(c - a) = b^2 = c^2 - a^2 = (c + a)(c - a)$$

What I wanted to demonstrate (W.I.W.T.D)

Here, I present you an image of page 57, of the book Critical Analysis of the Elements of Euclid (Gonzalez Urbaneja, 2017)

Termina el Libro II con uno de los problemas más importantes de *Los Elementos* de Euclides y de toda la Geometría griega, la cuestión de las cuadraturas de las figuras geométricas.

La cuadratura de una figura plana es la construcción, con regla y compás, de un cuadrado con la misma superficie que la figura inicial. El problema –que fascinó a la cultura griega por su obsesión por reducir lo complejo a lo simple (como simple es la regularidad simétrica del cuadrado), en un mundo natural gobernado por la razón, la perfección, la belleza y el orden– tiene un enorme interés desde un punto de vista práctico, ya que la determinación del área de una figura irregular no es nada sencillo, mientras que la cuadratura de una figura reduce el cálculo de su área al de un cuadrado.



El primer paso en estos problemas es la cuadratura es la cuadratura del rectángulo que aparece en esta Proposición II.14 de *Los Elementos* de Euclides:

Sea BCDE un rectángulo cualquiera. Se debe construir, con regla y compás, un cuadrado que tenga por área la del rectángulo BCDE. Con la regla se prolonga la recta BE, y se utiliza el compás para señalar el segmento EF de longitud igual a ED. A continuación se halla el punto medio G de BF y se describe un semicírculo de centro G y diámetro BE. Finalmente desde E se traza la línea EH perpendicular a BF, siendo H el punto de intersección de esta perpendicular con el semicírculo. El segmento EH es el lado del cuadrado EHLK buscado.

En efecto, sean: $a = HG$, $b = EG$, $c = EH$. Ahora se tiene:

- $a^2 = b^2 + c^2$ (Teorema de Pitágoras, I.47).
- $FG = BG = HG = a$, al ser todos radios del semicírculo.
- $EF = FG - EG = a - b$, $BE = BG + GE = a + b$
- Área del rectángulo BCDE = $BE \times ED = BE \times EF = (a+b) \cdot (a-b) = a^2 - b^2 = c^2 = \text{Área del cuadrado EHLK}$.

Así pues, se ha demostrado que el área del rectángulo inicial BCDE es igual al área del cuadrado EHLK, construido con regla y compás. Se ha resuelto, por tanto, el problema de la cuadratura del rectángulo.

Fig.58: Image of Page 57 - Critical Analysis of the Elements

Every document is perfectible, and in this image, there are typographical errors that do not affect what the author wants to say and what Euclid wrote about squaring a rectangle.

The interpretation of this image allows us to find the area of a rectangle equivalent to the area of a square that originates from a Pythagorean triple. The base and height of the rectangle will be the factors of the sum of squares, which produces the Pythagorean triangle. In other words, another way to factor the sum of two squares into the real numbers.

8.5 “Rectangle” a square from a Pythagorean triangle.

In this section, starting from a Pythagorean triangle resulting in a sum of squares, we are going to obtain, from the square on the hypotenuse, a rectangle with area equal to base times height, equivalent to the area of the square that originates it.

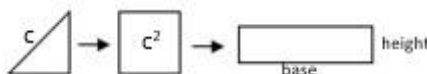


Fig.59: A Pythagorean triangle creates a square, and from this square, a rectangle is obtained.

Pythagorean Triangle $\rightarrow$ The Square on the hypotenuse $\rightarrow$ Rectangle

What you want to do is obtain from the largest square c^2 , a rectangle, with the condition that their respective areas be equal.

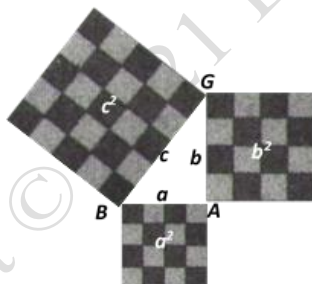


Fig.60: Image of the initial Pythagorean triangle

What I am going to do is, put the larger square in a horizontal position and that the hypotenuse (c) of the triangle coincides with one side of the square, doing this, the figure looks like this:

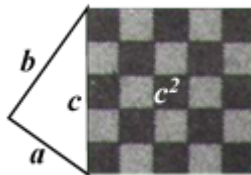


Fig.61: Image of the triangle and square in the hypotenuse

With the situation of the previous figure, applying the procedure of squaring a rectangle but in the opposite direction. As we already have the square, we are going to "get it in rectangular".

We currently have the following triangle and square:

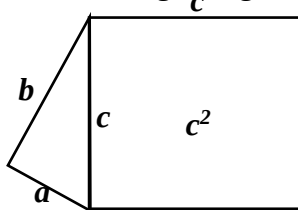


Fig.62: Initial triangle and square

1. Let's name the triangle, **BHM** and the square as **HAGM**

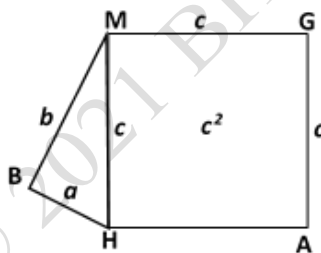


Fig.63: Image of triangle BHM and square HAGM

2. Let's extend **MH** down to **H'**, and **HA** horizontally to the left to **A'**, then, we extend side **b = MB**, downward-left until it intersects line **A'A**, at point **B'**.

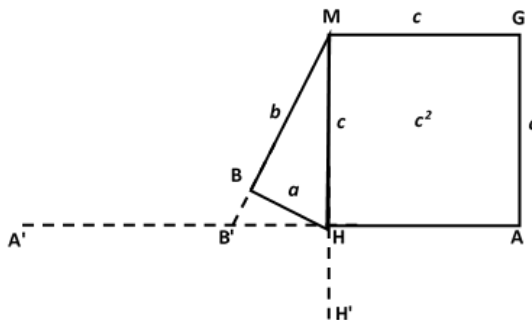


Fig.64: Extension of MH and HA

3. In the figure, two rectangle triangles are formed: the Pythagorean triangle **BHM** whose known sides are a , b and c ; as well as the triangle **B'HM**, with sides c , and a , where a is the height of this triangle. The **B'M** and **B'H** sides are unknown. Until now, the figure looks like this:

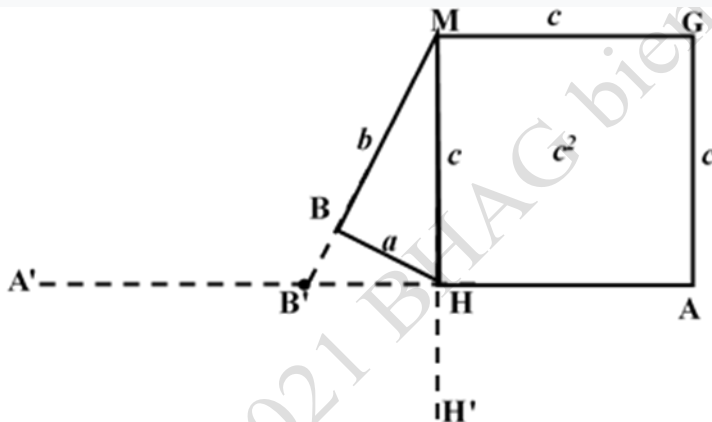


Fig.65: Formation of triangle B'HM

4. We identify the sides of triangle **B'HM** as **B'B = s**; **B'H = t** and **BH = a**, is the height of triangle B'HM. Here, **B'M = R = b + s**. The figure looks like this:

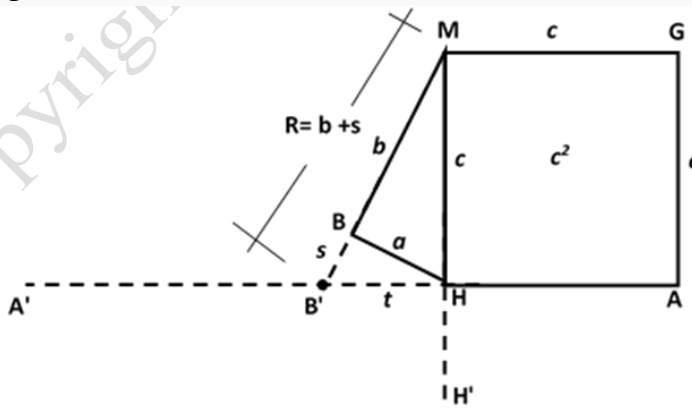


Fig.66: Identification of elements of triangle B'HM

5. We center at B' , and with radius $B'M = B'F = A'B' = R = b + s$; we draw a semicircle that goes from F to A' , passing through M . We already know that HM , is the side c , of the $HAGM$ square. With the above, the figure looks like this:

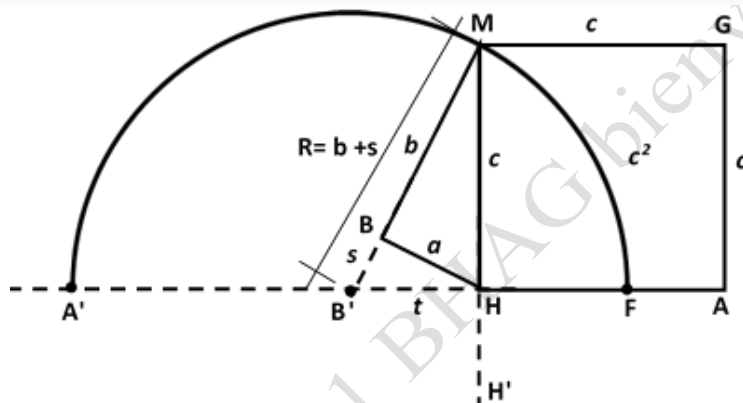


Fig.67: Semicircle $A'F$, radius $R = b + s$

6. It is observed that the radius $R = A'B' = B'F = B'M = b + s$, is the hypotenuse of triangle $B'HM$. The rectangle $A'CHH'$ with base $A'H = b + s + t$, and height $HF = HH' = A'C = b + s - t$. The graph looks like this:

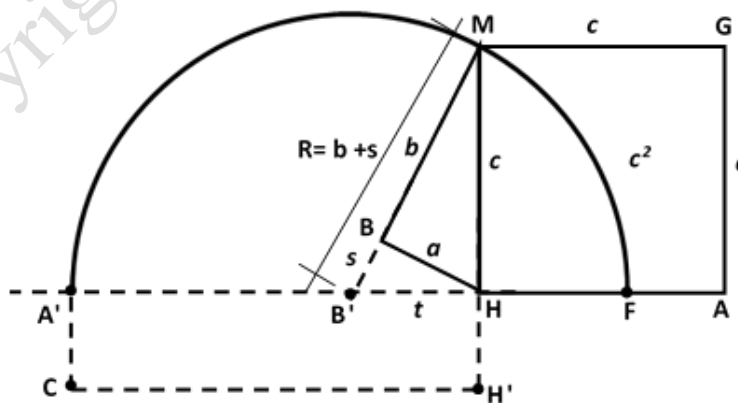


Fig.68: Formation of rectangle $A'CHH'$

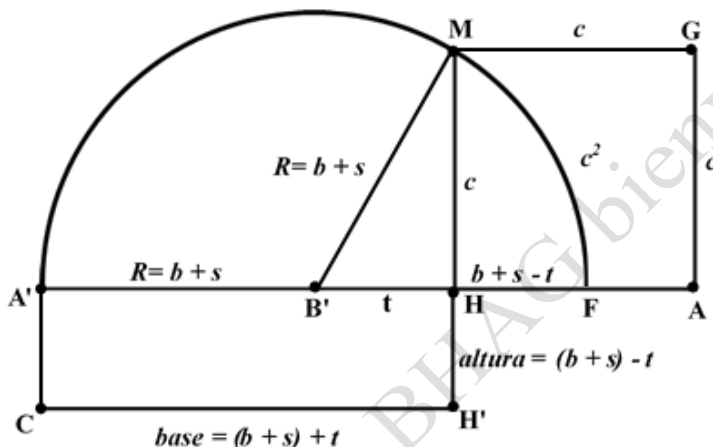


Fig.69: Rectangle A'CHH 'obtained from square HAGM

8. Let's show that the area of square **HAGM** is equal to that of rectangle **A'CHH'**
 - a. From the triangle **B'HM**, we obtain the area of the square **HAGM**. This triangle has sides: t , c y $(b + s)$. Applying the Pythagorean Theorem, it results:

$$t^2 + c^2 = (b + s)^2,$$

$$c^2 = (b + s)^2 - t^2$$

Then, the area of **HAGM** = $c^2 = (b + s + t) (b + s - t)$

- b. The rectangle **A'CHH'** with base = **$(b + s + t)$** ; and height: **$(b + s - t)$** , has for area: **$Ar = (b + s + t) (b + s - t)$**

Therefore: The area of **HAGM** is equal to the area of **A'CHH'**

This is: $c^2 = (b + s + t) (b + s - t) = Ar = (b + s + t) (b + s - t)$

What I wanted to demonstrate (W.I.W.T.D)

Comparison of the triangles **BHAG** and **B'H'A'G'**

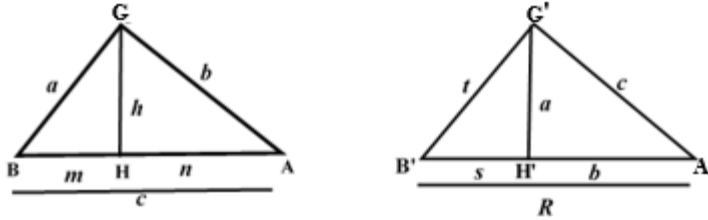


Fig.72: Comparison of triangles BHAG and B'H'A'G'

The following metric relations are obtained:

$$a^2 = c.m$$

$$t^2 = R.s \rightarrow s = \frac{t^2}{R}$$

$$b^2 = c.n$$

$$c^2 = R.b \rightarrow R = \frac{c^2}{b}$$

$$h^2 = m.n$$

$$a^2 = s.b \rightarrow s = \frac{a^2}{b}$$

$$a.b = c.h \rightarrow a = \frac{c.h}{b}$$

$$t.c = R.a \rightarrow t.c = \frac{c^2}{b} a$$

$$t = a.c / b$$

Found values:

$$R = \frac{c^2}{b} = b + s \quad s = \frac{a^2}{b} \quad t = \frac{a.c}{b}$$

$$\text{As: } a^2 + b^2 = c^2 = (R + t)(R - t) = (b + s + t)(b + s - t)$$

$$R + t = (b + s + t) = \frac{c^2 + a.c}{b} : \text{ is the base of rectangle.}$$

$$\text{Likewise: } R - t = (b + s - t) = \frac{c^2 - a.c}{b} : \text{ is the height of rectangle.}$$

If we graph the previous, we have:

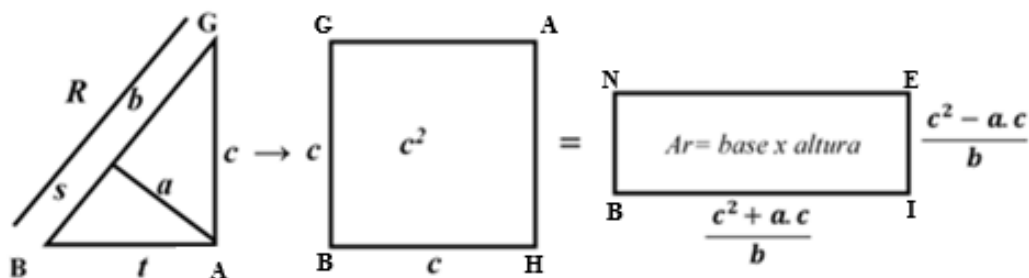


Fig.73: From Triangle **BAG** $\rightarrow$ Square **BHAG** $\rightarrow$ Rectangle **BIEN**

Therefore, another way of factoring the sum of two squares, with factors in which there is an addition and a subtraction is the following:

“bienve₂”:

$$a^2 + b^2 = c^2 = \left(\frac{c^2 + a \cdot c}{b} \right) \left(\frac{c^2 - a \cdot c}{b} \right)$$

We will call it the formula "bienve₂", which contains the product of two distinct rational factors, that belong to the set real numbers. One factor f_1 is a sum and the other f_2 contains a difference.

8.7 Proof examples of the formula “*bienve2*”.

Example No.1: Let n and m be the first two consecutive natural numbers, that is: $n = 1$ and $m = 2$; let's say these two numbers form a rectangle, let's call it a parent rectangle.

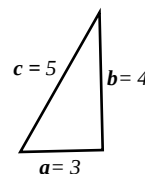
$$\boxed{\text{Ar} = m \cdot n = 2(1)} \quad \begin{matrix} n=1 \\ m=2 \end{matrix}$$

From these two numbers we get another three:

$$a = m^2 - n^2 = (m+n)(m-n) = (2+1)(2-1) = 3 \cdot 1 = 3$$

$$b = 2 \cdot m \cdot n = 2(2)(1) = 4$$

$$c = m^2 + n^2 = 2^2 + 1^2 = 4 + 1 = 5$$



The rectangle triangle 3, 4, 5 is formed:

The Pythagorean Theorem is fulfilled: $a^2 + b^2 = c^2$

$$3^2 + 4^2 = 5^2 = 9 + 16 = 25$$

Applying “*bienve2*”: $a^2 + b^2 = c^2 = \left(\frac{c^2 + a \cdot c}{b}\right) \left(\frac{c^2 - a \cdot c}{b}\right)$

$$9 + 16 = 25 = \left(\frac{5^2 + 5 \cdot 3}{4}\right) \left(\frac{5^2 - 5 \cdot 3}{4}\right)$$

$$9 + 16 = 25 = \left(\frac{25 + 15}{4}\right) \left(\frac{25 - 15}{4}\right)$$

$$9 + 16 = 25 = \left(\frac{40}{4}\right) \left(\frac{10}{4}\right) = 10(2.5) = 25$$

$$\begin{array}{c} 5 \\ \boxed{\text{Ac} = (5)(5) = 25} \\ 5 \end{array} = \begin{array}{c} \boxed{\text{Ar} = (10)(2.50) = 25} \\ \text{Base} = 10 \end{array} \quad \text{Height} = 2.5$$

Rta. The factors are **10** and **2.5**

Conclusion: The proposed formula, for this case, is fulfilled!

Ejemplo No.2: Let n and m be the following two consecutive natural numbers, that is: $n = 2$ and $m = 3$; let's say these two numbers form a rectangle.

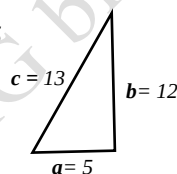
$$\boxed{\text{Ar} = m \cdot n = (3)(2) = 6} \quad \begin{matrix} n=2 \\ m=3 \end{matrix}$$

From these two numbers we get another three: `

$$a = m^2 - n^2 = (m+n)(m-n) = (3+2)(3-2) = 5 \cdot 1 = 5$$

$$b = 2 \cdot m \cdot n = 2(3)(2) = 12$$

$$c = m^2 + n^2 = 3^2 + 2^2 = 9 + 4 = 13$$



The triple or triad is; 5, 12 y 13: The triangle is formed:

The Pythagorean Theorem is fulfilled: $a^2 + b^2 = c^2$

$$5^2 + 12^2 = 13^2 = 25 + 144 = 169$$

Let's factor with: $a^2 + b^2 = c^2 = \left(\frac{c^2 + a \cdot c}{b}\right) \left(\frac{c^2 - a \cdot c}{b}\right)$

$$25 + 144 = 169 = \left(\frac{13^2 + 5 \cdot 13}{12}\right) \left(\frac{13^2 - 5 \cdot 13}{12}\right)$$

$$25 + 144 = 169 = \left(\frac{169 + 65}{12}\right) \left(\frac{169 - 65}{12}\right)$$

$$25 + 144 = 169 = \left(\frac{234}{12}\right) \left(\frac{104}{12}\right) = 19,50 \left(\frac{26}{3}\right) = 169$$

The square and rectangle have the same area:

$$\begin{array}{c} 13 \\ \square \\ \text{Ar} = 169 \\ 13 \end{array} = \begin{array}{c} \boxed{\text{Ar} = (19,50) \left(\frac{26}{3}\right) = 169} \\ \text{Base} = 19,50 \end{array} \quad \text{Height} = \frac{26}{3}$$

Rta. The factors are: 19,50 and 8,666666667

Observation: $\frac{104}{12} = \frac{26}{3} = 8,666666667$

Conclusion: The proposed formula, for this case, is fulfilled!

8.8 Proofs of “bienve₂” in excel for factoring $a^2 + b^2$.

In what follows, I present an image of a page in excel, with Pythagorean triples (triads) factored with the formula “bienve₂”:

It is shown that:

$$\text{“bienve}_2\text{”}: \quad a^2 + b^2 = c^2 = \left(\frac{c^2 + a \cdot c}{b}\right) \left(\frac{c^2 - a \cdot c}{b}\right)$$

The formula “bienve₂” is fulfilled:

Los Factores Racionales de una Suma de Cuadrados									bienve ₂	
Terna Pitagórica							T.Pitágoras	f ₃ y f ₄ al multiplicarse resultan en c ²		
Rectángulo		a	b	c				f ₃	f ₄	f ₃ *f ₄
n	m	m ² - n ²	2.m.n	m ² + n ²	a ²	b ²	c ² = a ² + b ²	[(c ² + a.c)/b]	[(c ² - a.c)/b]	c ²
1	2	3	4	5	9	16	25	10,0000	2,5000	25
2	3	5	12	13	25	144	169	19,5000	8,6667	169
3	4	7	24	25	49	576	625	33,3333	18,7500	625
4	5	9	40	41	81	1.600	1.681	51,2500	32,8000	1.681
5	6	11	60	61	121	3.600	3.721	73,2000	50,8333	3.721
6	7	13	84	85	169	7.056	7.225	99,1667	72,8571	7.225
7	8	15	112	113	225	12.544	12.769	129,1429	98,8750	12.769
8	9	17	144	145	289	20.736	21.025	163,1250	128,8889	21.025
9	10	19	180	181	361	32.400	32.761	201,1111	162,9000	32.761
10	11	21	220	221	441	48.400	48.841	243,1000	200,9091	48.841
11	12	23	264	265	529	69.696	70.225	289,0909	242,9167	70.225
12	13	25	312	313	625	97.344	97.969	339,0833	288,9231	97.969
13	14	27	364	365	729	132.496	133.225	393,0769	338,9286	133.225
14	15	29	420	421	841	176.400	177.241	451,0714	392,9333	177.241
15	16	31	480	481	961	230.400	231.361	513,0667	450,9375	231.361
16	17	33	544	545	1.089	295.936	297.025	579,0625	512,9412	297.025
17	18	35	612	613	1.225	374.544	375.769	649,0588	578,9444	375.769
18	19	37	684	685	1.369	467.856	469.225	723,0556	648,9474	469.225
19	20	39	760	761	1.521	577.600	579.121	801,0526	722,9500	579.121
20	21	41	840	841	1.681	705.600	707.281	883,0500	800,9524	707.281
21	22	43	924	925	1.849	853.776	855.625	969,0476	882,9545	855.625
22	23	45	1.012	1.013	2.025	1.024.144	1.026.169	1.059,0455	968,9565	1.026.169
23	24	47	1.104	1.105	2.209	1.218.816	1.221.025	1.153,0435	1.058,9583	1.221.025
24	25	49	1.200	1.201	2.401	1.440.000	1.442.401	1.251,0417	1.152,9600	1.442.401
25	26	51	1.300	1.301	2.601	1.690.000	1.692.601	1.353,0400	1.250,9615	1.692.601
26	27	53	1.404	1.405	2.809	1.971.216	1.974.025	1.459,0385	1.352,9630	1.974.025
27	28	55	1.512	1.513	3.025	2.286.144	2.289.169	1.569,0370	1.458,9643	2.289.169
28	29	57	1.624	1.625	3.249	2.637.376	2.640.625	1.683,0357	1.568,9655	2.640.625
29	30	59	1.740	1.741	3.481	3.027.600	3.031.081	1.801,0345	1.682,9667	3.031.081
30	31	61	1.860	1.861	3.721	3.459.600	3.463.321	1.923,0333	1.800,9677	3.463.321
31	32	63	1.984	1.985	3.969	3.936.256	3.940.225	2.049,0323	1.922,9688	3.940.225
32	33	65	2.112	2.113	4.225	4.460.544	4.464.769	2.179,0313	2.048,9697	4.464.769

Image of 32 Pythagorean Triples

The formula "bienve2" is fulfilled:

32	33	65	2.112	2.113	4.225	4.460.544	4.464.769	2.179,0313	2.048,9697	4.464.769
33	34	67	2.244	2.245	4.489	5.035.536	5.040.025	2.313,0303	2.178,9706	5.040.025
34	35	69	2.380	2.381	4.761	5.664.400	5.669.161	2.451,0294	2.312,9714	5.669.161
35	36	71	2.520	2.521	5.041	6.350.400	6.355.441	2.593,0286	2.450,9722	6.355.441
36	37	73	2.664	2.665	5.329	7.096.896	7.102.225	2.739,0278	2.592,9730	7.102.225
37	38	75	2.812	2.813	5.625	7.907.344	7.912.969	2.889,0270	2.738,9737	7.912.969
38	39	77	2.964	2.965	5.929	8.785.296	8.791.225	3.043,0263	2.888,9744	8.791.225
39	40	79	3.120	3.121	6.241	9.734.400	9.740.641	3.201,0256	3.042,9750	9.740.641
40	41	81	3.280	3.281	6.561	10.758.400	10.764.961	3.363,0250	3.200,9756	10.764.961
41	42	83	3.444	3.445	6.889	11.861.136	11.868.025	3.529,0244	3.362,9762	11.868.025
42	43	85	3.612	3.613	7.225	13.046.544	13.053.769	3.699,0238	3.528,9767	13.053.769
43	44	87	3.784	3.785	7.569	14.318.656	14.326.225	3.873,0233	3.698,9773	14.326.225
44	45	89	3.960	3.961	7.921	15.681.600	15.689.521	4.051,0227	3.872,9778	15.689.521
45	46	91	4.140	4.141	8.281	17.139.600	17.147.881	4.233,0222	4.050,9783	17.147.881
46	47	93	4.324	4.325	8.649	18.696.976	18.705.625	4.419,0217	4.232,9787	18.705.625
47	48	95	4.512	4.513	9.025	20.358.144	20.367.169	4.609,0213	4.418,9792	20.367.169
48	49	97	4.704	4.705	9.409	22.127.616	22.137.025	4.803,0208	4.608,9796	22.137.025
49	50	99	4.900	4.901	9.801	24.010.000	24.019.801	5.001,0204	4.802,9800	24.019.801
50	51	101	5.100	5.101	10.201	26.010.000	26.020.201	5.203,0200	5.000,9804	26.020.201
51	52	103	5.304	5.305	10.609	28.132.416	28.143.025	5.409,0196	5.202,9808	28.143.025
52	53	105	5.512	5.513	11.025	30.382.144	30.393.169	5.619,0192	5.408,9811	30.393.169
53	54	107	5.724	5.725	11.449	32.764.176	32.775.625	5.833,0189	5.618,9815	32.775.625
54	55	109	5.940	5.941	11.881	35.283.600	35.295.481	6.051,0185	5.832,9818	35.295.481
55	56	111	6.160	6.161	12.321	37.945.600	37.957.921	6.273,0182	6.050,9821	37.957.921
56	57	113	6.384	6.385	12.769	40.755.456	40.768.225	6.499,0179	6.272,9825	40.768.225
57	58	115	6.612	6.613	13.225	43.718.544	43.731.769	6.729,0175	6.498,9828	43.731.769
58	59	117	6.844	6.845	13.689	46.840.336	46.854.025	6.963,0172	6.728,9831	46.854.025
59	60	119	7.080	7.081	14.161	50.126.400	50.140.561	7.201,0169	6.962,9833	50.140.561
60	61	121	7.320	7.321	14.641	53.582.400	53.597.041	7.443,0167	7.200,9836	53.597.041
61	62	123	7.564	7.565	15.129	57.214.096	57.229.225	7.689,0164	7.442,9839	57.229.225
62	63	125	7.812	7.813	15.625	61.027.344	61.042.969	7.939,0161	7.688,9841	61.042.969
63	64	127	8.064	8.065	16.129	65.028.096	65.044.225	8.193,0159	7.938,9844	65.044.225
64	65	129	8.320	8.321	16.641	69.222.400	69.239.041	8.451,0156	8.192,9846	69.239.041
65	66	131	8.580	8.581	17.161	73.616.400	73.633.561	8.713,0154	8.450,9848	73.633.561
66	67	133	8.844	8.845	17.689	78.216.336	78.234.025	8.979,0152	8.712,9851	78.234.025
67	68	135	9.112	9.113	18.225	83.028.544	83.046.769	9.249,0149	8.978,9853	83.046.769
68	69	137	9.384	9.385	18.769	88.059.456	88.078.225	9.523,0147	9.248,9855	88.078.225
69	70	139	9.660	9.661	19.321	93.315.600	93.334.921	9.801,0145	9.522,9857	93.334.921
70	71	141	9.940	9.941	19.881	98.803.600	98.823.481	10.083,0143	9.800,9859	98.823.481
71	72	143	10.224	10.225	20.449	104.530.176	104.550.625	10.369,0141	10.082,9861	104.550.625
72	73	145	10.512	10.513	21.025	110.502.144	110.523.169	10.659,0139	10.368,9863	110.523.169
73	74	147	10.804	10.805	21.609	116.726.416	116.748.025	10.953,0137	10.658,9865	116.748.025
74	75	149	11.100	11.101	22.201	123.210.000	123.232.201	11.251,0135	10.952,9867	123.232.201
75	76	151	11.400	11.401	22.801	129.960.000	129.982.801	11.553,0133	11.250,9868	129.982.801
76	77	153	11.704	11.705	23.409	136.983.616	137.007.025	11.859,0132	11.552,9870	137.007.025
77	78	155	12.012	12.013	24.025	144.288.144	144.312.169	12.169,0130	11.858,9872	144.312.169
78	79	157	12.324	12.325	24.649	151.880.976	151.905.625	12.483,0128	12.168,9873	151.905.625
79	80	159	12.640	12.641	25.281	159.769.600	159.794.881	12.801,0127	12.482,9875	159.794.881
80	81	161	12.960	12.961	25.921	167.961.600	167.987.521	13.123,0125	12.800,9877	167.987.521
81	82	163	13.284	13.285	26.569	176.464.656	176.491.225	13.449,0123	13.122,9878	176.491.225
82	83	165	13.612	13.613	27.225	185.286.544	185.313.769	13.779,0122	13.448,9880	185.313.769

Image of 50 Pythagorean Triples

8.9 Final words of the experience in writing this book.

In Nicaragua, I do not know if in other countries, there is a saying which says that to have lived fully, the following three things must have been done: 1) Plant a tree, 2) Procreate a son / daughter, at least, and 3) Write a book.

In my case, when I was a child at the age of 8 years old, I planted a "custard apple" tree that produces a fruit similar to "noni", but sweeter and larger.

Regarding children, this plural includes males and females. I had four different women give birth to six children, in this order: Roxanita, Germán, Diana, Idaly, Hernaldito, and finally Napoleón.

As for writing a book, I have been writing it since October 25, 2010, and I never finished it. It was until this year, February 25, 2021, while in agony and facing death, I asked Jehovah to keep me alive even for a year more. I made Him the promise to write this book for His glory and honor. Inasmuch, he provided me with the formula that is presented here.

I started writing again without interruption, following my discharge from the Nuevo Amanecer Hospital in Bilwi, on Thursday, April 4 of this year (2021). My schedule was from 3:00 A.M. to 12:00 noon. In the afternoon from 3:00 P.M. until 6:00 P.M. I go to bed at 8:00 P.M.

These are the last two pages that I write, I hope to have it ready, printed, even just one copy, on August 20, 2021, within 5 days from today.

Writing this book has been passionate, pleasant, exciting, and now I understand the writers, such as: Miguel de Cervantes, Gabriel García Márquez, Rubén Darío, etc., who have dedicated their lives to the profession of writing. How fortunate!

Some among other reasons why this book exists:

1. It is a promise that I had to fulfill to the glory of the Creator of the Universe, Jehovah God.
2. I had pending the need to write a book.
3. Collect and present information from authors dedicated to mathematics who like to write on the subject. Most of this book is not entirely unpublished. It is only a compilation of information found in documents written by others.
4. Present two ways to factor a sum of two squares. If they work or not, it does not matter. For me, the most important thing was and is, to make them known.

(Paenza, 2007), says: « [...], once the solution, the problem itself, is known, it is very easy. Of course, it is very easy once you know, (or are told) how to solve it, [...]. Many times, one looks from the wrong place, [...]. That is why Dr. Paenza says, you have to: «think differently ... from another angle, not be restricted or trapped only in what you see, (you have to look to the side where others are not looking) » If one does not exit the dimension in which he is, finding solution to a problem is impossible. » [...]. No matter how inaccessible it may seem, it is possible that there is another way of looking at it that makes it simple»

END

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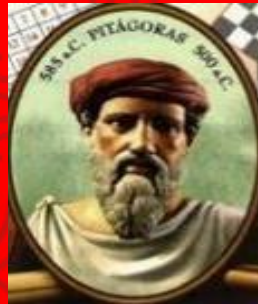
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